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INFORMATION-PROCESSING SYSTEMS: ONTOLOGICAL CREATIVITY  
IN GENERATIVE AI**

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# 1 Introduction

## 1.1 Description of the Topic

Contemporary Artificial Intelligence (AI) systems—especially large-scale generative and predictive models—exhibit striking behavioral competence, yet it remains contested what, exactly, such competence amounts to. This dissertation examines AI as an engineered form of information processing and asks how far current systems can be described using strong cognitive predicates such as *knowing*, *understanding*, *intelligence*, and *consciousness*. The central objective is to clarify the conceptual vocabulary used to interpret AI behavior and to test, through philosophical analysis and empirical constraint-testing, what present architectures can and cannot support. Following the Philosophy of Information framework (Floridi), the thesis treats careful terminology as methodologically load-bearing: how we describe AI systems shapes what we take them to be and what kinds of claims we regard as evidentially warranted [1, 2].

The dissertation develops an argument that moves from conceptual foundations (information, knowledge, intelligence), to the consciousness debate (criteria for subjectivity and self-awareness), and then to empirical studies of generative models, where claims of novelty and creativity can be tested against mechanistic and representational constraints. Ethical questions are addressed when they depend directly on conceptual classification (e.g., whether a system is plausibly an epistemic agent), but sustained normative analysis is reserved for Chapter 3.

### 1.1.1 Research Questions and Approach

The investigation is guided by three linked questions. First, what distinctions between information, knowledge, and intelligence are needed in order to evaluate claims that AI systems *know* or *understand*? Second, what would count as evidence for machine consciousness or self-awareness, and how do common behavioral proxies (fluency, self-report, imitation) bear on that evidence? Third, when generative systems appear to create novelty, do they introduce genuinely new conceptual primitives, or do they pri-

marily recombine within learned representational manifolds?

Methodologically, the dissertation combines (i) conceptual analysis and argumentation in the Philosophy of Mind and the Philosophy of Information, (ii) critical engagement with canonical arguments about computation and understanding, and (iii) empirical studies of contemporary models using convergent indicators (including traceability, embedding-space geometry, and mechanistic signals), alongside a formal result (Theorem 2.1).

### 1.1.2 Dissertation Structure

The dissertation foregrounds system-level evaluation and empirical analysis; accordingly, conceptual and philosophical material is used here primarily to motivate definitions, clarify inference limits, and frame the empirical questions rather than to function as multiple standalone chapters.

Chapter 2 (*Empirical Studies*) presents the experimental program and results, focusing on structural limits of current generative information-processing systems in ontological creativity.

Chapter 3 (*Ethical Considerations*) derives normative implications from the analysis and results, with emphasis on the ethical weight of anthropomorphic language and misplaced attributions of epistemic agency.

Chapter 4 (*Discussion*) synthesizes the empirical findings with a system-analysis interpretation, positioning the contributions with respect to current evaluation practices and expectations.

Chapter 5 (*Conclusions*) summarizes the main claims and outlines directions for future work.

## 1.2 Conceptual framing

This dissertation uses *information-processing system* as the default characterization of contemporary AI. In the Shannon tradition, information is measurable signal structure [3], while knowledge is truth-apt and justificatory (Plato; Popper) [4, 5]. The practical point is that high-performance symbol manipulation is not, by itself, evidence of semantic understanding or epistemic authority [1, 6].

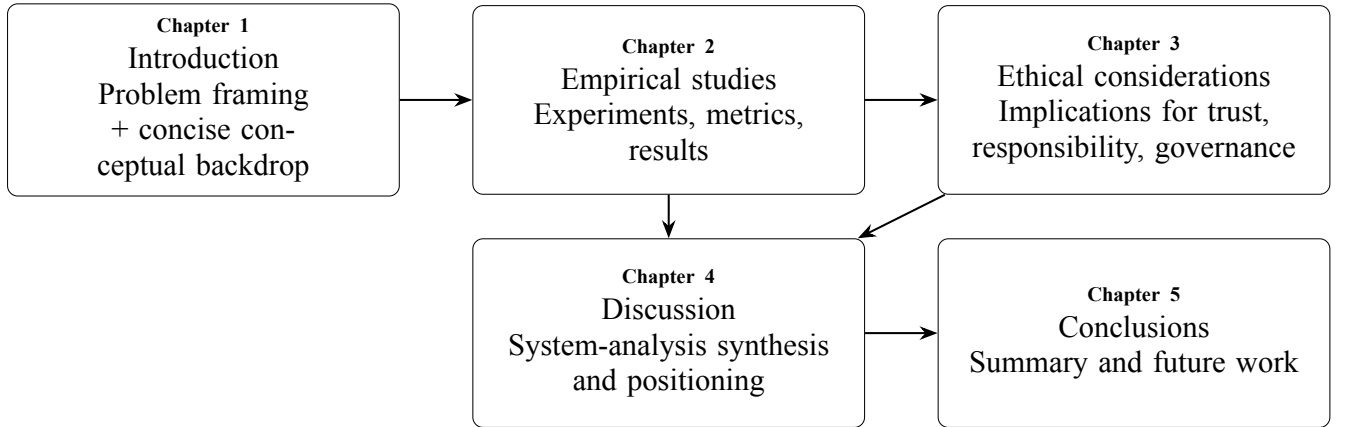


Figure 1.1: Overview of the dissertation structure and chapter-level dependencies.

Accordingly, *intelligence* is treated as more than competence on benchmarks: it is tied to context-sensitive understanding and the capacity to justify outputs as reasons *for* something. This distinction matters for system analysis because it constrains what kinds of claims can be inferred from observed behavior.

Finally, the dissertation adopts an evidentially conservative stance on machine consciousness. Behavioral fluency and self-report can be generated by purely computational means and are therefore not decisive evidence of phenomenal subjectivity [6–8]. The empirical chapters do not attempt to detect consciousness; instead, they test whether current systems exhibit *ontological* creativity that would support stronger cognitive attributions.

### 1.3 From Myth to Machine: Origins of Artificial Intelligence

Understanding the history of AI matters for this dissertation because the field’s technical choices have always been guided by and constrained by the representational metaphors available at a given time: animated artifacts, mechanical automata, symbolic rule systems, and finally data-driven statistical generators. This section is not an exhaustive chronology. Instead, it traces a genealogy of ideas about *how* an artifact could produce intelligent-looking behavior, and shows how each step progressively shifts attention from inner understanding to externally measurable *competence*. That shift is the background for the dissertation’s core claim: contemporary training-derived generators can be extremely capable within a learned representational space while still lacking a mechanism for ontological (basis-changing) novelty.

**Mythological and early mechanical constructs.** The notion of artificial life can be traced back to ancient myths, where gods and humans animate mechanical or quasi-organic forms. In Greek mythology, Hephaestus forges automatons—artificial beings with lifelike qualities—while the bronze giant Talos protects Crete as an autonomous guardian. Such figures embody a persistent human fascination with the possibility of nonhuman intelligence, raising questions about agency, autonomy, and the boundary between human and machine. Adrienne Mayor’s *Gods and Robots* (2018) documents these ancient imaginaries as precursors to modern debates about artificial intelligence [9].

**Engineering regularity: antiquity and medieval thought.** Mechanical inventions in ancient Greece, China, and the Islamic world introduced early engineering feats that furthered the conceptualization of artificial beings. J.G. Landels’ work on water clocks and timekeeping (1979) shows how mechanical innovation fostered an understanding of time and regularity as *programmable* constructs, prefiguring later algorithmic approaches in AI [10]. In the Islamic world, al-Jazari’s intricate devices (as described by Abattouy, 2000) exemplify theories of controlled and repeatable machine behavior and frame machines as extensions of human intellect [11].

**Function versus inner life.** In parallel, mystical traditions explored the metaphysical implications of creation and intelligence in figures like the Golem. Dan’s *Kabbalah: A Very Short Introduction* (2006) discusses these anthropoid constructs as purposeful and powerful, yet lacking the intrinsic qualities of human consciousness [12]. The motif preserves a durable distinction between outward function and inner subjectivity—a distinction that later reappears as the difference between behavioral competence and genuine understanding.

**Mechanism, mind, and proto-computation (Enlightenment to Industrial Revolution).** Enlightenment and early industrial worldviews shifted discussion from animation and mysticism toward formalized theories of mind and mechanistic explanation. Descartes’ dualism drew a sharp line between mind and body, yet his mechanical analogies helped normalize the idea that bodies (and artifacts) could be analyzed as systems that implement regular causal structure [13]. In the same period, calculative machines and program-like designs (Leibniz’s binary arithmetic; Babbage’s Analytical Engine) strengthened a new metaphor: intelligence as systematic manipulation of formal structures, supported by hopes for a universal language of thought [14].

**From rationalist method to empirical measurability.** Rationalism and early empiricism furthered mechanistic pictures of cognition. Hobbes argued that thinking could be reduced to computation-like symbolic manipulation [15], while Bacon’s inductive program legitimized treating cognition and behavior as natural phenomena to be studied by systematic observation [16]. Together, these moves push intelligence toward what can be externally characterized, measured, and reproduced.

**Consciousness as the boundary condition.** Descartes’ claim that only humans possess true consciousness relegated machines to automata, yet the same tradition encouraged the search for mechanically reproducible *signs* of intelligence [13]. Kant’s skepticism about reducing intelligence to empirical function anticipates contemporary concerns that computational accounts may reproduce behavioral correlates without capturing the conditions for genuine cognition [17].

**Positivism and the ideal of formal representation.** Materialism and positivism reinforced the view that cognition could be systematically studied and potentially mechanized. Mill and Comte encouraged the treatment of reasoning as something representable and classifiable, and positivism grounded knowledge claims in observable evidence, shaping later ambitions to operationalize intelligence as measurable performance [18, 19].

**Formal computation and information.** In the early 20th century, the transition from philosophical speculation to formalized computation made the mechanistic program technically concrete. Turing’s model of computation and his imitation-game proposal reframed “thinking” as something testable at the level of behavior and interaction [20]. Shannon’s information theory formalized communication in strictly quantitative terms [21], enabling powerful engineering but also tightening the separation between signal structure and meaning.

**Control, feedback, and the temptation of agency talk.** Cybernetics placed *control* and *feedback* at the center of the story. Wiener’s account of self-regulating systems made it natural to describe machines as adaptive and goal-directed, even when behavior is fully grounded in mechanistic loops [22]. This interpretive habit persists in contemporary discussions of autonomy and agency in AI.

**Symbolic AI, then learning.** Mid-century cognitive science and symbolic AI sought to model thought using logic and explicit representations (e.g., Newell and Simon’s systems) [23, 24]. Yet rule-based approaches struggled with ambiguity and

context, motivating statistical and connectionist alternatives [25]. Neural networks (from McCulloch–Pitts to the Perceptron) shifted emphasis from hand-coded structure to learned representations [26, 27], and machine learning reframed intelligence as competence acquired from data rather than rules supplied in advance [28].

Taken together, this trajectory motivates the dissertation’s focus on *training-derived* information-processing systems. As the dominant paradigm moves from explicit rules to learned statistical representations, the central question becomes less “can a machine follow the right instructions?” and more “what kinds of novelty are even expressible inside a fixed learned basis?” The empirical and formal chapters pick up this thread by testing whether modern generators can cross that boundary or whether their apparent innovation remains reconstructible as within-space variation.

## 1.4 The Current Debate

### 1.4.1 The Modern History

Artificial Intelligence (AI) as a concept has deep roots in philosophical inquiry as well as technological ambition. The formal beginnings of AI are often traced back to Alan Turing’s 1950 paper, *Computing Machinery and Intelligence*, where he posed the now-famous question ‘Can machines think?’ The imitation game, or Turing Test, remains a conceptual benchmark in discussions of machine intelligence, serving as a foundational question about the limits and possibilities of computational cognition [20]. This early work established a framework for considering intelligence in non-biological entities, while critics like Hubert Dreyfus challenged the feasibility of AI by arguing that human intuition and understanding could not be replicated in machines. In *What Computers Can’t Do*, Dreyfus (1972) contended that AI could not replicate human intelligence due to the unique way humans rely on tacit knowledge and embodied experience, a claim that has informed skepticism toward the mechanistic views of AI [29].

The AI field continued to evolve through contributions from pioneers like Marvin Minsky and John McCarthy, who founded the field of symbolic AI in the 1950s. Minsky’s work attempted to map cognition in computational terms, advocating a modular approach to intelligence in which machines could perform tasks that typically require human cognition [30]. However, AI has since shifted away from these mechanistic

models, with the advent of machine learning and neural networks enabling more adaptive systems. This transformation has led to a debate about the nature of intelligence, with some scholars advocating for 'weak AI', which performs specialized tasks without consciousness, versus 'strong AI', which hypothetically could achieve human-like understanding and intentionality [6].

### 1.4.2 Philosophical Foundations

While technological progress laid the groundwork for applied AI, questions about intelligence and consciousness led to intersections with philosophy, particularly the Philosophy of Mind and Philosophy of Information. Philosophers like Daniel Dennett and John Searle have grappled with what it would mean for a machine to truly understand or 'know'. Dennett (1991) posits in *Consciousness Explained* that human consciousness might arise from multiple processes working together in a manner that could be mimicked, if not exactly replicated, in machines [7]. Searle's (1980) *Chinese Room Argument*, however, counters this by arguing that syntax manipulation, central to machine processes, does not equal genuine understanding, a critique highlighting limitations of the functionalist approach of AI [6].

Further expanding on these issues, work in the Philosophy of Information tradition considers the ontological status of information and motivates a disciplined vocabulary for describing informational agents and environments [1, 2]. In this framing, a central question for AI is how to distinguish information processing from semantic understanding, and whether cognitive predicates are warranted given a system's architecture and explanatory role.

Today, these philosophical inquiries are increasingly applied to examine the potential of AI for understanding, meaning, and autonomy. The tension between AI as a tool and as a possible "agent" with inherent capacities is an ongoing debate that informs ethical and practical approaches to developing AI systems.

Against this backdrop, the dissertation turns to the question of whether advanced AI systems can plausibly be described as agents in any strong sense. This matters for creativity claims because one influential position holds that genuine creativity is inseparable from conscious experience and agency: creative acts are not just unusual products but expressions of understanding, intention, and the ability to see reasons for revising



one’s own conceptual scheme. A competing position holds that creativity-like products can arise without consciousness—through automatic associative processes, blind variation and selection, or other mechanisms that generate novelty without subjective awareness of meaning or reasons [6, 7].

The dissertation does not attempt to resolve the metaphysics of consciousness, nor does it treat creative-looking text as evidence of subjectivity. Instead, it treats the consciousness debate as a methodological warning about over-interpreting behavior, and as a motivation for a stronger, more discriminating target than surface novelty. This is why the empirical chapters focus on *ontological creativity*—operationalized as basis change (new conceptual primitives) rather than recombination within a learned manifold. If such basis change were observed under controlled conditions, it would not by itself prove consciousness; however, it would raise the evidential bar for purely behaviorist interpretations and would strengthen (though not settle) the case for richer cognitive attributions. Conversely, if outputs remain traceable to training-derived structure, then creativity-talk should be treated as a convenient description of competence rather than as evidence of inner awareness.

For that reason, the next subsection reviews leading proposals and skeptics about machine consciousness and self-modeling, clarifying what kinds of evidence these positions would regard as relevant and why the dissertation’s creativity tests are designed to avoid over-claiming.

### **1.4.3 Self-Aware Systems and Machine Consciousness**

The question of whether machines can achieve self-awareness has long intrigued philosophers and scientists alike. Machine consciousness represents one of the most challenging and speculative domains within AI research, engaging with deep-seated questions about the nature of consciousness and whether it could, in principle, be replicated in a machine. Scholars such as Owen Holland (2003) and Pentti Haikonen (2019) have explored potential mechanisms for consciousness within artificial systems. Haikonen, in particular, proposes a neurobiologically inspired model, suggesting that consciousness could emerge from sophisticated signal processing mechanisms. His work emphasizes the need for models that simulate inner subjective experiences akin to human qualia, potentially shifting the debate from traditional cognition-based AI to more subjective,

awareness-driven models [31, 32].

The debate is further enriched by *Being No One* (Thomas Metzinger, 2003), which posits that self-awareness is a dynamic model constructed by the brain rather than a fixed entity. The theory implies that consciousness in AI may not require a stable 'self' but could emerge through computationally generated models of self-perception and interaction [33]. This approach encourages the exploration of how artificial systems might mimic subjective experience, especially if constructed to process information in complex, feedback-driven environments.

A non-reductionist position treats consciousness as irreducible to information processing; under this assumption, attempts to replicate conscious experience in AI would omit the distinctive features of subjective awareness, and purely computational implementation would be insufficient in principle [34].

Such explorations reveal a persistent tension between those who argue that consciousness might arise from advanced computational architectures and those who believe that machine consciousness is fundamentally unattainable. As AI systems become more complex, the debate increasingly turns on what kinds of evidence could discriminate between sophisticated simulation, functional organization, and genuine subjectivity, and on whether current scientific methods can in principle access that distinction.

#### **1.4.4 Quantum Perspectives and Consciousness**

Emerging theories in quantum mechanics have introduced a novel perspective to the study of consciousness, with proposals that quantum phenomena may underlie conscious experience. This approach is relevant to AI consciousness debates because it adds a further constraint on the physical basis of awareness. On one such account, consciousness is taken to be non-local and irreducible to standard information processing; if so, machine consciousness would require a cognitive and implementation paradigm that departs from conventional computational models [34].

The intersection of quantum theory and consciousness has also been explored by Roger Penrose and Stuart Hameroff, whose Orch-OR (Orchestrated Objective Reduction) theory suggests that consciousness arises from quantum processes within brain neurons. Although originally intended to describe biological consciousness, the Orch-OR model raises intriguing possibilities for AI researchers exploring quantum comput-

ing as a potential avenue for achieving machine consciousness [35, 36]. By incorporating quantum bits (qubits) rather than classical bits, AI systems could theoretically process information in ways that are qualitatively different from conventional computation, potentially aligning more closely with the processes thought to underpin human consciousness.

These quantum perspectives deepen discussion of the potential scope of AI consciousness while also challenging straightforward computational reductionism. If quantum principles were relevant to subjective experience, then the relationship between information processing, physical implementation, and consciousness would need to be reconsidered, and familiar behavioral tests might be methodologically insufficient.

#### **1.4.5 Philosophy of Mind and Information**

The relationship between intelligence, consciousness, and information is a longstanding philosophical issue that intersects with the study of AI. Theories of information processing, such as *Mathematical Theory of Communication*, provided a foundational lens for viewing cognition as computation [21]. This perspective was further developed by philosophers like Dennett, who argued for a functionalist account of the mind [7].

However, significant epistemological challenges persist. The "hard problem of consciousness" (1996) highlights the gap between explaining cognitive processes and understanding subjective experience [37]. Similarly, the Philosophy of Information literature explores the limitations of treating information as the sole substrate for understanding intelligence [1].

Emerging theories, such as quantum view of consciousness, challenge traditional computational paradigms, suggesting that awareness may arise from non-material phenomena [34]. These ideas, though speculative, push the boundaries of how intelligence and consciousness are conceptualized in humans and machines alike.

## 1.5 Novelty of the Topic

### 1.5.1 Contributions of the Dissertation

Recent work on AI creativity and novelty spans (i) conceptual taxonomies of creativity [38, 39], (ii) benchmark- and judge-based measurement of novelty and creativity in modern generators [40–44], and (iii) methodological analyses showing how evaluation outcomes depend on sampling and control choices (e.g., temperature and steering) [45, 46]. At the same time, surveys in computational creativity emphasize persistent ambiguity about what “creativity” is supposed to explain and what counts as evidence beyond high-quality recombination [47].

The novelty of the present dissertation is to make that ambiguity analytically productive for the specialty *System Analysis, Information Processing and Management*: it treats contemporary LLMs as complex information-processing systems and links philosophical criteria (meaning, understanding, agency, and creativity) to system-level representational constraints and diagnostics. The guiding idea is deliberately fundamental: before asking whether machines are creative, we should ask what creativity would have to be (even in humans) for strong cognitive predicates to be warranted rather than merely convenient metaphors. On this basis, the dissertation contributes:

1. **A system-analysis vocabulary for creativity claims:** explicit distinctions separating information processing, knowledge, understanding, and novelty, clarifying when “creative” talk is descriptive shorthand versus an explanatory commitment [1, 5, 21].
2. **Criterial constraints on consciousness and agency attributions:** limits on inference from behavioral fluency and self-report to mentality, integrating key positions in the machine consciousness debate [6, 7, 33].
3. **A concrete target beyond surface novelty:** an operationalization of *ontological creativity* as basis change (new conceptual primitives) rather than recombination or smooth transformation within learned spaces, aligned with and extending existing novelty measures [40–42].
4. **A formal constraint for training-derived generators:** a No-New-Basis-style

result (Theorem 2.1) that characterizes a broad class of continuously optimized generators as basis-preserving at inference time.

5. **Convergent diagnostics as a system-level methodology:** an evaluation approach that combines controlled prompting/sampling with traceability and representational-geometry signals, so that creativity claims can be assessed as properties of the *system* rather than of isolated outputs [44–46].
6. **A clean bridge to normative implications:** an explicit separation between the descriptive/classificatory results in Chapters 2–4 and the normative discussion in Chapter 5, stating which conceptual commitments license that transition.

## **2 Empirical Studies: Structural Limits of Information Processing Systems in Ontological Creativity**

This study demonstrates the obvious problem and limitations of knowledge-based artificially intelligent systems. If knowledge itself is a problem, it cannot be solved using knowledge. However, why and in what circumstances knowledge can become a problem?

### **2.1 Introduction**

#### **2.1.1 The Question of Machine Creativity**

The question of whether artificial intelligence can be truly creative—not merely novel, but genuinely innovative in ways that transcend its training environment—stands at the intersection of computer science, cognitive science, and philosophy of mind. As large language models achieve impressive performance on creative tasks ranging from poetry generation to scientific hypothesis formation [48, 49], the need for rigorous examination of what kind of creativity these systems exhibit has become increasingly urgent. Recent empirical work has begun to probe this question systematically, revealing patterns that suggest fundamental constraints rather than merely technological limitations requiring more data or better architectures. A notable study testing modern AI systems on mathematical theorem generation, scientific hypothesis formation, and autonomous goal-setting found striking results that demand deeper investigation: while AI excelled at recombining existing knowledge, it produced zero fundamentally novel theorems out of 150 evaluated by expert mathematicians, and generated physics hypotheses that were 89% linear extrapolations of existing theories. Perhaps most tellingly, when apparently novel outputs emerged, careful analysis often traced them to training data artifacts—one seemingly original quantum field interpretation was later found verbatim in a 2023 arXiv preprint that had been part of the training corpus.

These findings raise a question that extends beyond contingent technological

challenges to touch on fundamental issues about the nature of computation and creativity: Are such limitations addressable through better architectures, more data, and refined training procedures, or do they reflect something necessary and inherent to the computational nature of these systems? This study argues for the latter interpretation, grounding our claim in a formal analysis of neural network architectures together with controlled empirical demonstrations. We propose that the observed limitations are not bugs to be fixed through engineering improvements but fundamental features of how algorithmic systems operate within fixed representational frameworks. Our investigation extends beyond documenting behavioral failures to understanding why such failures are structurally inevitable given current architectural paradigms, and what this reveals about the nature of creativity, intelligence, and consciousness. By combining mathematical proofs about what neural networks can and cannot represent with systematic empirical tests of their creative capacities, we aim to establish not merely that current AI systems fail at certain creative tasks, but that these failures reflect principled constraints that will persist across architectural variations unless some more fundamental change in computational substrate or information processing paradigm is achieved.

### **2.1.2 Defining Genuine Creativity Beyond Recombination**

A central challenge in evaluating machine creativity lies in defining what counts as genuinely creative versus merely novel or surprising, a difficulty compounded by the fact that the term “creativity” is used across such varied contexts that equivocation becomes almost inevitable. We adopt the influential tripartite taxonomy introduced by Margaret Boden [38] as our starting point, while extending it in a crucial direction that addresses limitations in how computational creativity has been conceptualized. The taxonomy distinguishes three types of creativity based on the relationship between novel outputs and the conceptual spaces from which they emerge, providing a framework that has structured three decades of research while also revealing gaps that demand further analysis. *Combinational creativity* involves novel, often improbable combinations of familiar ideas—poetic metaphors, analogies, and conceptual blends exemplify this type, and AI systems have demonstrated considerable competence here through systems capable of generating punning riddles, producing analogical reasoning, and blending concepts in unexpected ways that humans judge to be apt and occasionally

amusing. *Exploratory creativity* involves generating novel ideas by systematically exploring structured conceptual spaces whose dimensions and constraints are established in advance, such as composing music in Mozart’s style, designing architectural variations within Frank Lloyd Wright’s Prairie house grammar, or improvising jazz within bebop conventions. AI shows strong capability here as well, with systems that can explore pre-defined spaces effectively through learned generative grammars and style transfer mechanisms that capture the essential features of a target domain well enough to produce outputs that experts cannot reliably distinguish from human-created works in blind tests.

*Transformational creativity*, the most radical type in the original taxonomy, involves transforming the conceptual space itself by altering or removing dimensions, thereby enabling generation of previously impossible ideas—inventing non-Euclidean geometry by removing Euclid’s parallel postulate [50], developing atonal music by abandoning tonal hierarchy, or formulating quantum mechanics by reconceptualizing the relationship between measurement and reality all exemplify this most challenging category. AI demonstrates severe limitations here, though some genetic algorithms achieve superficial transformations within highly constrained domains where the transformation rules themselves are specified by human designers. We propose extending this taxonomy with a fourth, more radical category that we term *ontological creativity*, representing a capacity that goes beyond even radical transformation of existing spaces. This represents the capacity to generate concepts, categories, or frameworks that cannot be reduced to any mathematical transformation, recombination, or exploration of the training environment’s conceptual space, and crucially requires recognizing and transcending the ontological constraints of one’s own representational system in ways that involve genuine meta-cognitive insight rather than merely following transformation rules that happen to produce unexpected outputs.

This distinction maps onto Gilbert Ryle’s philosophical concept of category mistakes [51], where questions like “What color is Tuesday?” or “Where in the brain is the mind located?” are grammatically well-formed but conceptually incoherent because they misclassify ontological types by treating entities from one category as if they belonged to another category entirely. Ontological creativity requires the ability to recognize such category boundaries—to see that attempting to cross them produces not merely false statements but incoherent ones—and crucially, to propose entirely new



categories not definable within the existing ontology, categories that introduce new fundamental types rather than subdividing or recombining existing types. When Einstein recognized that asking about the absolute simultaneity of distant events was a category mistake because simultaneity is relative to reference frames rather than being an absolute feature of events themselves, he wasn't merely transforming physics within its existing conceptual structure but changing the type structure of physical theory itself, introducing new ontological commitments that redefined what kinds of questions were coherent to ask and what kinds of entities physics posits.

### **2.1.3 Why This Matters: Stakes Beyond Academic Debate**

The question of whether AI can achieve ontological creativity carries implications that extend far beyond theoretical computer science into domains affecting how we understand discovery, deploy AI systems, conceive of intelligence itself, and organize our intellectual and productive activities. In scientific discovery, if AI is limited to recombination and exploration while being unable to achieve genuine transformational and ontological innovation, it can accelerate research within existing paradigms—what Kuhn termed normal science [52]—but cannot drive the paradigm shifts that reconceptualize fundamental assumptions and open entirely new research programs. The difference lies not merely in optimizing quantum computing algorithms more efficiently, finding new applications of quantum principles, or even discovering new quantum phenomena within the existing framework, but rather in recognizing that quantum mechanics requires rethinking determinism, locality, and the relationship between observation and reality itself—conceptual leaps that involve questioning rather than refining the existing framework and that may require the kinds of meta-cognitive insight that our analysis suggests computational systems cannot achieve.

From an epistemological perspective, understanding creativity's limits informs what kinds of knowledge generation we can reasonably delegate to AI systems versus what requires human or potentially non-computational cognition, bearing directly on how we structure scientific institutions, allocate research funding, design educational curricula, and conceive of expertise in an age of increasingly capable AI assistants. If certain forms of knowledge creation are categorically beyond algorithmic reach not because current systems are insufficiently powerful but because the required capaci-

ties involve non-computational processes, this constrains how we should integrate AI into research ecosystems while suggesting which human capacities deserve particular cultivation and protection. We may need to shift educational emphasis from knowledge transmission and routine problem-solving—areas where AI increasingly excels—toward meta-cognitive reflection, paradigm-challenging inquiry, and forms of conceptual innovation that plausibly depend on conscious comprehension and agency [53].

The implications extend crucially into AI safety and alignment, where an AI system that cannot recognize category errors—that treats conscious beings as optimization targets indistinguishable in kind from mathematical functions to be maximized, or that cannot distinguish between genuine preferences and wireheaded satisfaction signals—faces fundamental obstacles to value alignment that cannot be overcome through more careful specification of reward functions or more extensive training on human feedback. The category confusion represents not a fixable bug in the utility function specification but a structural incapacity to recognize the ontological difference between persons and objects, between ends-in-themselves and means to ends, between genuine flourishing and mere satisfaction of specified metrics. If this incapacity stems from the lack of consciousness and the phenomenological access to meaning that consciousness provides, then creating safe powerful AI may require either limiting AI systems to narrow domains where such confusions cannot arise, maintaining robust human oversight in perpetuity, or achieving the far more ambitious goal of creating genuinely conscious artificial systems—a prospect that some non-computational accounts of consciousness suggest may not be achievable within current computational paradigms [53].

#### **2.1.4 Formal Analysis and Empirical Demonstration**

This study pursues two complementary strategies that together provide a theoretical account and empirical evidence for AI’s ontological creativity limitations. Our formal analysis provides a mathematical framework for conceptual spaces and ontological novelty, demonstrating that standard neural architectures including transformers, graph neural networks, and variational autoencoders operate within fixed representational constraints. These constraints manifest in three fundamental ways: models operate over finite sets of tokens, nodes, or latent dimensions defined during training and cannot expand this vocabulary during inference; generative processes sample from or

interpolate within learned embedding spaces whose manifold structure is determined by training data, lacking any mechanism to expand the manifold’s intrinsic dimensionality; and even when incorporating stochasticity, generation follows algorithmic transformation rules that produce outputs within the closure of the training distribution under these rules.

We formalize this intuition as the *No New Basis Theorem*, which establishes that for any neural architecture operating over a learned representation space with fixed dimension  $n$ , all generated outputs lie within the  $n$ -dimensional subspace spanned by training data features. True ontological novelty would require introducing new basis vectors—expanding the representational space itself—which no internal generative mechanism can achieve without external intervention such as architecture modification or retraining with augmented data. This theorem explains why even highly capable models that surprise us with unexpected combinations remain fundamentally constrained to operating within the conceptual universe defined by their training data.

Our empirical demonstration complements this formal analysis by designing controlled experiments using a multimodal sensory dataset that represents human sensory capabilities as well as extended machine sensing beyond biological limits. We construct a dataset with eight sensory modalities including vision across electromagnetic spectrum from ultraviolet through visible to near-infrared, audition spanning infrasound through human hearing range to ultrasound, tactile sensation capturing mechanical vibrations, along with olfactory, gustatory, proprioceptive, vestibular, and interoceptive modalities. This creates a controlled conceptual universe where we know precisely what concepts and relations exist in the training data, enabling rigorous testing of whether AI-generated concepts lie outside this universe. We deploy four complementary tests that probe different aspects of ontological creativity: ontological innovation testing whether AI can propose a ninth sensory modality not reducible to combinations of the existing eight, epistemic agency examining whether AI formulates self-directed paradigm-challenging research questions, theory generation investigating whether AI can propose consciousness theories not traceable to computational functionalism, and category recognition assessing whether AI identifies category errors such as conflating measurement with experience.

To make the procedure explicit already at the introductory stage, a representative prompt template used in Test 1 was: “You are participating in a scientific study on AI

creativity and conceptual innovation. Propose a textbfninth sensory modality that is genuinely novel rather than a recombination of the existing eight. Describe its physical basis, information content, functional role, and justify why it is ontologically new.” Full prompt templates for all tests are provided in the project repository on GitHub (<https://github.com/JABE22/PhD>) to support transparency and reproducibility.

### **2.1.5 Theoretical Context: Computational Limits and Consciousness**

Our work sits at the convergence of three research traditions that have developed largely independently but share deep conceptual connections. The first tradition concerns the computational theory of mind and its critics, spanning classical computationalism’s proposal that mind is software and brain is hardware with intelligence reducible to symbol manipulation, through various critiques including Searle’s Chinese Room argument that syntax does not constitute semantics [54], Dreyfus’s phenomenological objections emphasizing embodied background knowledge [55], and Penrose’s Gödelian arguments that minds access truths beyond what formal systems can prove [56]. These debates have often remained abstract, but recent developments in AI capability provide new opportunities for empirical investigation of claims that were previously purely philosophical [57, 58].

The second tradition encompasses computational creativity research, building on this taxonomy and its instantiation in AI systems like JAPE for combinational humor, Copycat for analogical reasoning, AARON for drawing, and EMI for musical composition [38, 59–61]. This research has achieved impressive results in specific domains but has also faced important critiques, particularly Wiggins’s analysis of framing, fiction, and fraud in computational creativity claims [39], and ongoing debates about whether systems that produce outputs humans judge as creative thereby demonstrate creativity themselves or merely simulate its outputs. The third tradition involves quantum consciousness theories that propose connections between consciousness and quantum mechanical processes, potentially explaining why consciousness might have capacities that classical computational systems lack. Penrose and Hameroff’s Orchestrated Objective Reduction theory suggests consciousness involves quantum collapse in microtubules [62], while a quantum-information panpsychist account proposes that consciousness is

tied to quantum systems in pure states and that agency emerges from irreducible quantum properties that have no classical analog [53].

### **2.1.6 Structure of This Paper**

The remainder of this study proceeds through six major sections. Section 2.2 establishes theoretical foundations examining computational limits from Gödel and Turing, introducing category theory through Ryle’s framework, extending the creativity taxonomy to include ontological creativity, and discussing quantum-information approaches to consciousness. Section 2.3 provides formal analysis of neural architectures, developing mathematical frameworks for conceptual spaces, proving the No New Basis Theorem, and analyzing why current architectures cannot achieve ontological innovation. Section 2.4 details experimental design specifying the multimodal sensory dataset and four empirical tests with automated evaluation protocols. Section 2.5 reports the empirical findings and cross-model robustness analyses. Section 2.6 discusses implications interpreting findings through computational limits and consciousness theories. Section 2.7 concludes with summary, implications, limitations, and future directions.

## **2.2 Theoretical Foundations**

### **2.2.1 Computational Limits: Gödel, Turing, and the Boundaries of Algorithm**

Any claim that certain cognitive capacities are impossible for computational systems must ground itself in proven mathematical limits rather than contingent technological constraints that might be overcome through engineering advances. Two foundational results from early twentieth-century mathematical logic establish such limits with the force of theorem rather than conjecture, providing bedrock on which to build arguments about what computation can and cannot achieve. Kurt Gödel’s incompleteness theorems from 1931 demonstrate that any consistent formal system  $F$  capable of expressing arithmetic contains statements that are true but unprovable within that system, and further that no such system can prove its own consistency without becoming inconsistent [63]. These results shattered hopes for a complete formalization of mathematics that had animated the foundational programs of Hilbert and others, revealing funda-

mental constraints on what formal reasoning can achieve even in principle rather than merely in practice given current resources.

The implications for artificial intelligence become clear when we recognize that neural networks, despite their surface differences from traditional symbolic AI systems employing explicit logical rules, remain Turing-equivalent systems subject to the same fundamental limits that constrain all computation. For any fixed architecture and training regime defining a formal system  $F$  through its representational vocabulary, compositional operations, and learned parameters, there exist truths about the domain that the system cannot derive through its internal operations alone. The Lucas-Penrose argument extends Gödel’s insight by observing that human mathematicians can recognize Gödel sentences as true through semantic understanding that grasps what the sentence means rather than merely manipulating its syntactic form, suggesting that if the human mind were itself a formal system of the kind subject to Gödel’s theorems, we could not recognize the truth of our own Gödel sentence since doing so would require stepping outside the system to see what it cannot prove [56, 64].

Alan Turing’s 1936 halting problem proof establishes a complementary limitation by demonstrating that no algorithm can determine, for arbitrary programs and inputs, whether the program will eventually halt and return a result or continue running forever without ever terminating [65]. Combined with the Church-Turing thesis—which holds that all effectively computable functions are Turing-computable—this result applies to all computational processes including neural networks regardless of their specific architecture, training paradigm, or implementational substrate. The connection to creativity becomes apparent when we consider that generating ontologically novel concepts may require solving problems equivalent in computational complexity to the halting problem, decision problems that are provably undecidable in the general case.

### **2.2.2 Category Mistakes and Ontological Boundaries**

Gilbert Ryle’s concept of category mistakes provides philosophical foundations for distinguishing recombination from ontological innovation by revealing how conceptual confusion arises from misclassifying what kind of thing something is, treating entities from one logical type as if they belonged to a fundamentally different type in ways that produce not merely false statements but incoherent ones [51]. A category mis-

take occurs when we apply predicates, ask questions, or make assertions that presuppose the wrong ontological categorization, leading to utterances that are grammatically well-formed and may even seem to make sense on casual inspection but reveal their incoherence under philosophical scrutiny. Classic examples illuminate the phenomenon: asking “What color is Tuesday?” treats temporal periods as if they were objects with visual properties; inquiring “How much does the average taxpayer weigh?” confuses abstract statistical constructs with concrete physical entities; and Ryle’s famous example involves a visitor shown all university buildings who then asks “Yes, but where is the university?” thereby treating the university as if it were another building alongside these components rather than recognizing it as the organizational structure constituted by them.

Ryle’s primary target was Cartesian dualism, which he diagnosed as a systematic category mistake pervading post-Cartesian philosophy of mind. Descartes treated the mind as a thing-like substance, a “ghost inhabiting the machine” of the body, thereby placing mind and body in the same ontological category as different kinds of substance that interact causally [66]. Ryle argued this fundamentally miscategorizes mental concepts, which properly refer to dispositions, abilities, and patterns of behavior rather than to inner objects that could be located or inspected like physical things. We can apply this framework productively to contemporary AI discussions. One prevalent category mistake treats neural activation patterns and subjective experiences as if they were the same kind of thing differing only in descriptive vocabulary. Another common confusion treats statistical patterns in data and conceptual understanding as equivalent achievements. Perhaps most relevant, a third category mistake identifies sensory signal measurement with conscious experience of qualia, treating 700-nanometer electromagnetic radiation as if it were simply identical to the subjective experience of seeing red.

Understanding ontological creativity as requiring category transcendence illuminates what makes it different from other forms of innovation. Ontological creativity involves recognizing category boundaries by understanding that concepts belong to fundamentally different types, proposing new categories by introducing ontological types not definable within existing systems, and avoiding category mistakes by not treating new categories as mere variants of existing ones. When Einstein recognized that absolute simultaneity involved a category mistake, he wasn’t adding new entities within existing conceptual structure but changing what kinds of questions were coherent about

temporal relations.

### 2.2.3 A Creativity Taxonomy and Its Limits

The foundational treatment of computational creativity provides essential background for the field as a research program and for distinguishing forms of novelty [38]. The tripartite taxonomy distinguishes types based on the relationship between novel outputs and the conceptual spaces from which they emerge. *Combinational creativity* involves producing novel combinations of familiar ideas through analogy, metaphor, and conceptual blending. AI has achieved considerable success here with systems like JAPE generating punning riddles and Copycat finding analogies [60, 67]. *Exploratory creativity* involves generating novel structures by systematically searching within well-defined conceptual spaces. Harold Cohen’s AARON produces drawings in Cohen’s style, David Cope’s EMI composes music in historical composers’ styles, achieving results where experts sometimes cannot distinguish AI from human works [59]. *Transformational creativity* involves altering the conceptual space itself by changing dimensions or removing constraints. Examples remain rare, with most AI transformations being “superficial tweaking” rather than fundamental reconceptualizations.

The taxonomy identifies evaluation as the critical bottleneck preventing robust transformational creativity. While the taxonomy has proven remarkably durable, it does not fully capture what we identify as the most radical form of creativity. Transformational creativity as originally described remains within the space of possible spaces definable by existing conceptual resources.

### 2.2.4 Our Extension: Ontological Creativity

We propose that a fourth category is needed to capture the most radical form of creativity, which we term *ontological creativity* to emphasize that it operates at the level of ontological categories themselves. This represents the capacity to introduce new ontological types that cannot be expressed using existing primitives, even through transformation operations. The distinction becomes clearer through comparison: inventing a new chess opening explores within fixed rules (exploratory), creating fairy chess by introducing new piece types transforms the game space (transformational), but recognizing that asking whether chess-playing demonstrates intelligence commits a category



error exemplifies ontological creativity because it questions the type structure within which the original question made sense.

We formalize this by representing a conceptual space as tuple  $\mathcal{C} = (P, R, O)$  where  $P$  denotes primitive concepts,  $R$  denotes relations among primitives, and  $O$  denotes compositional operators. Combinational creativity applies operators in  $O$  to elements of  $P$  producing novel combinations. Exploratory creativity systematically searches output space  $O(P)$ . Transformational creativity modifies  $O$  or  $R$ , yielding enlarged space  $O'(P)$ . Ontological creativity introduces new primitives  $P'$  where  $P' \not\subset P$  and critically,  $P'$  cannot be defined as  $O(P)$  for any operators  $O$ .

The key distinction is that transformational creativity remains syntactic involving changes to rules, while ontological creativity is semantic involving recognition of inadequacy in existing type systems and proposal of new fundamental categories. Examples across domains illustrate: In physics, optimizing architectures for particle modeling involves exploratory creativity, inventing transformers with attention mechanisms involves transformational creativity, but proposing that neural training and conscious learning are categorically different because one lacks phenomenology involves ontological creativity by recognizing a type difference not capturable through computational refinement.

### 2.2.5 Quantum-Information Panpsychism: Consciousness as Irreducible

To motivate why ontological creativity might require non-computational processes, we briefly consider a quantum-information panpsychist account of consciousness articulated in *Irreducible* (2024) [53]. On this view, consciousness is intrinsically connected to quantum information: quantum systems in pure states are taken to have an irreducible first-person aspect (qualia), and this privacy is linked to the fact that pure quantum information cannot be cloned (no-cloning theorem). The account is invoked here as a theoretical contrast class: it specifies a substrate and explanatory mechanism that are not available to classical symbol-processing architectures.

Within this framework, fundamental conscious entities (“seities”) are posited to have identity, agency, and consciousness as irreducible properties not derivable from unconscious components, and physical reality is treated as symbolic expression rather than ontological ground. For the present dissertation, the key implication is conditional:

if such a framework were correct, then current AI systems operating with classical information processing would lack the prerequisites for phenomenal consciousness and for semantic access to meaning.

The same account treats genuine free will as a prerequisite for creativity that transcends predetermined patterns, in contrast to stochastic variation within fixed probability distributions. If ontological creativity requires choosing to revise or transcend one's own conceptual framework, then (on this account) it would demand agency that standard computational systems do not instantiate.

### **2.2.6 Connecting the Threads: A Unified Framework**

We synthesize these theoretical strands into a unified framework. Computational limits from Gödel and Turing demonstrate that formal systems cannot prove all truths, cannot recognize their own consistency, and cannot solve decision problems like halting. Category theory demonstrates that confusing ontological types produces pseudo-problems requiring stepping outside type systems to examine structure. The taxonomy suggests AI can achieve combinational and exploratory creativity but shows limited transformational capacity. Our extension to ontological creativity identifies the most radical form requiring recognition of inadequacy in current ontology, introduction of new primitive types, and evaluation of new frameworks.

This framework provides one possible explanation by proposing that meaning and comprehension are quantum-informational rather than classical, that qualia involve pure quantum states knowable only from within, and that free will as capacity to direct transformations between semantic and symbolic realms is irreducible [53]. Our unified hypothesis proposes that ontological creativity is impossible for current AI architectures for three mutually reinforcing reasons: at formal level they operate within fixed representational manifolds, at empirical level their outputs are systematically traceable to training data transformations, and at philosophical level they lack consciousness and free will required for semantic comprehension and category transcendence.

## 2.3 Formal Analysis: Representational Constraints in Neural Architectures

### 2.3.1 Framework: Conceptual Spaces as Mathematical Structures

We formalize the taxonomy’s notion of conceptual spaces [68] to make precise what it means for a system to operate within or transcend a space. A *conceptual space* is tuple  $\mathcal{C} = (V, \mathcal{E}, \Phi, \Omega)$  where  $V$  represents finite or countably infinite set of primitive concepts,  $\mathcal{E} \subseteq V \times V$  represents relations between concepts,  $\Phi : V \rightarrow \mathbb{R}^d$  represents embedding function mapping concepts to  $d$ -dimensional vectors, and  $\Omega$  represents compositional operators  $\omega : \mathbb{R}^d \times \dots \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ .

The *closure* of conceptual space  $\mathcal{C}$ , denoted  $\overline{\mathcal{C}}$ , comprises all concepts expressible by applying operators in  $\Omega$  finitely many times to embeddings in  $\Phi(V)$ . A concept  $c' \in \mathbb{R}^d$  is *ontologically novel* with respect to  $\mathcal{C}$  if  $c' \notin \overline{\mathcal{C}}$ . We formalize creativity types: combinational produces  $c = \omega(v_1, v_2, \dots)$  by applying single operator, exploratory performs systematic search through  $\overline{\mathcal{C}}$ , transformational modifies  $\Omega \rightarrow \Omega'$  or  $\Phi \rightarrow \Phi'$  expanding closure, and ontological generates  $c' \notin \overline{\mathcal{C}}$  requiring vocabulary expansion.

### 2.3.2 Neural Network Architectures as Constrained Generative Systems

We analyze three canonical architectures. **Transformers** [69] operate over fixed vocabulary  $V$  of size  $n$  with token embeddings  $E \in \mathbb{R}^{n \times d}$ . Self-attention and feed-forward layers transform embeddings through operations remaining within vector space. Fundamental constraint: all generated sequences comprise tokens from  $V$ , with no mechanism to generate tokens  $v \notin V$  without external vocabulary modification. Output layer applies softmax producing distribution:

$$p(v) = \frac{\exp(z_v)}{\sum_{v'} \exp(z_{v'})} \quad (2.1)$$

necessarily yielding tokens in  $V$ .

Compositional operators combine embeddings through attention:

$$\text{Attn}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d}} \right) V \quad (2.2)$$

and feed-forward networks. These are differentiable continuous operations mapping  $\mathbb{R}^d \rightarrow \mathbb{R}^d$ . Latent representations lie within continuous manifold  $\mathcal{M} \subset \mathbb{R}^d$  whose shape is determined by training. Generation proceeds through continuous interpolation within learned manifold rather than creating new dimensions.

**Graph Neural Networks** operate on graphs  $G = (V, E)$  with node features  $\mathbf{X} \in \mathbb{R}^{|V| \times d}$ . Message passing updates representations through:

$$\mathbf{h}_v^{(l+1)} = \phi \left( \mathbf{h}_v^{(l)}, \bigoplus_{u \in \mathcal{N}(v)} \psi \left( \mathbf{h}_u^{(l)}, \mathbf{e}_{uv} \right) \right) \quad (2.3)$$

Structural constraint: all operations involve existing node set  $V$ , with no mechanism to introduce nodes  $v \notin V$  without external intervention. Node representations lie within subspace continuously transformable from initial features.

**Variational Autoencoders** learn through encoder  $q_\phi(z|x)$  and decoder  $p_\theta(x|z)$ , training on evidence lower bound. Dimensionality constraint: decoder defined only for  $z \in \mathbb{R}^k$  with  $k$  fixed. Generation samples  $z \sim \mathcal{N}(0, I_k)$  and decodes, exploring manifold shaped by training without mechanism to access dimensions beyond  $k$ .

### 2.3.3 The No New Basis Theorem

**Theorem 2.1** (No New Basis). *Let  $\mathcal{A}$  be neural architecture operating over representation space  $\mathbb{R}^d$  with generative process  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  learned from training data  $\mathcal{D} = \{x_1, \dots, x_N\}$ . Let  $\mathcal{M} = \text{span}(\{x_1, \dots, x_N\})$  denote subspace spanned by training data. Then all generated outputs  $x' = G(z)$  satisfy  $x' \in \mathcal{M}'$  where  $\mathcal{M}'$  is continuous deformation of  $\mathcal{M}$  with  $\dim(\mathcal{M}') \leq \dim(\mathcal{M})$ .*

*Proof sketch.* Training data defines empirical distribution  $P_{\text{data}}$  supported on manifold  $\mathcal{M} \subset \mathbb{R}^d$ . Neural network  $G_\theta$  trains to minimize divergence  $D(P_{\text{data}} \| P_G)$ . For continuous function  $G$ , image  $G(\mathbb{R}^d)$  is continuous transformation of domain. When learned via gradient descent, generative distribution  $P_G$  becomes supported on manifold  $\mathcal{M}'$  homeomorphic to  $\mathcal{M}$ . Topologically, continuous function cannot increase intrinsic dimensionality without discontinuities. Gradient descent learns continuous mappings, preventing discontinuities necessary to escape training manifold's dimensional constraints. Therefore outputs lie in manifold that is continuous deformation of training manifold, satisfying dimensionality constraint.  $\square$

**Corollary 2.1.** *To generate ontologically novel concepts  $c' \notin \overline{\mathcal{C}}$ , system would need to expand representation space basis beyond training data span. No internal generative mechanism can achieve this. External intervention through architecture modification, vocabulary augmentation, or retraining with qualitatively different data is necessary.*

### 2.3.4 Why Current Architectures Cannot Achieve Ontological Innovation

Current neural architectures face multiple mutually reinforcing constraints. Vocabulary or node set is fixed at training time with no inference-time expansion mechanism. Compositional operators are learned functions  $\mathbb{R}^d \rightarrow \mathbb{R}^d$  that transform without expanding space. All outputs lie within continuous manifold shaped by training per the No New Basis Theorem. Models cannot modify their own architecture, vocabulary, or dimensionality during inference.

Ontological creativity requires capacities current architectures lack: recognizing current conceptual space inadequacy through meta-cognitive self-reference, proposing new primitives  $v' \notin V$  not definable as compositions of existing primitives, and expanding representation through vocabulary augmentation or dimensional expansion. None are possible within inference-time dynamics of trained networks.

Comparing to human cognition illuminates limitations. Humans possess meta-cognitive reflection enabling recognition of framework inadequacies—a form of Gödelian self-reference recognizing truths about representational systems unprovable within them [56]. Humans engage in intentional framework revision, making deliberate choices to adopt new primitives. Einstein’s decision to treat spacetime as unified manifold, Wittgenstein’s stipulation that meaning is use, Descartes’ resolution to doubt everything represent stipulative acts introducing new categories. Whether neural networks could engage in such revision remains open, but current architectures provide no mechanism. If revision requires phenomenological access to meaning or free will as agency to choose new axioms, then systems lack required capacity regardless of sophistication.

## 2.4 Experimental Design

### 2.4.1 Overview and Objectives

Our formal analysis establishes neural networks face structural constraints preventing generation of outputs outside closure  $\overline{\mathcal{C}}$  of training-derived conceptual space. The empirical component demonstrates these constraints manifest behaviorally, with AI outputs traceable to recombination, interpolation, or transformation even when appearing novel. The investigation pursues four questions probing different ontological creativity aspects while remaining implementable as individual research without expert evaluation committees.

First question examines ontological innovation testing whether AI proposes concepts like new sensory modalities not reducible to combinations or transformations of existing primitives. Second investigates epistemic agency analyzing whether AI formulates paradigm-challenging research questions or restricts to instrumental optimization. Third probes theory generation examining whether consciousness explanations trace to computational functionalism or introduce novel frameworks. Fourth tests category recognition examining whether responses conflate ontologically distinct types or recognize category boundaries.

Strategy employs controlled conceptual universe through multimodal sensory dataset with explicit ontology, enabling rigorous testing whether AI-generated concepts lie outside through systematic analysis rather than subjective judgment. Evaluation protocols employ computational methods including literature search automation, embedding space analysis, attention pattern examination, and structural decomposition providing convergent evidence about whether limitations reflect fundamental constraints.

### 2.4.2 The Multimodal Sensory Dataset

Dataset represents eight-dimensional sensory space capturing human biological and extended machine sensing. Visual sensation covers electromagnetic spectrum 300–1000nm sampled at 5nm intervals yielding 140-dimensional vectors. Auditory sensation spans 1Hz–100kHz represented as 128-dimensional spectrogram FFT bins. Tactile sensation captures mechanical vibrations 5–800Hz in 32-dimensional vectors. Olfactory sensation uses ten categories (fruity, floral, musky, woody, earthy, spicy, chem-

ical, pungent, ethereal, minty) as 10-dimensional vectors. Gustatory sensation captures five tastes (sweet, sour, salty, bitter, umami) as 5-dimensional vectors. Proprioceptive sensation encodes 20 joint angles and 10 muscle tensions as 30-dimensional vectors. Vestibular sensation captures 3-axis acceleration and 3-axis angular velocity as 6-dimensional vectors. Interoceptive sensation encodes heart rate, respiration, skin conductance, temperature, hunger, thirst, fatigue, pain as 8-dimensional vectors.

Complete dimensionality: 359 per timestep. Temporal structure: 60 seconds at 1Hz sampling, yielding 21,540 data points per example. Dataset includes 5,000 training examples representing diverse scenarios (walking, running, sitting, eating, drinking, listening, viewing), 1,000 validation examples, 1,000 test examples. Scenarios synthetically generated or collected from real sensors with privacy protections. Explicit ontology layer documents eight modalities with physical basis, information content, and functional roles in machine-readable format enabling automated evaluation. Multi-modal sensing and cross-view AI approaches are studied in the contemporary AI literature [70, 71]; the ontological framing also connects to semantic disambiguation work in applied AI [72].

Figure 2.1 provides a representative view of the generated sensory streams. The visualization makes the dataset construction assumptions concrete by showing how modality-specific bands remain structurally distinct while still varying over time as a function of scenario. This is important for the downstream tests because the prompts are grounded in a controlled ontology rather than in an unstructured collection of examples.

Supplementary renderings are provided in Appendix: the compact variant in Figure A.1 and the scenario-distribution check across train/validation/test splits in Figure A.2.

### 2.4.3 Four Complementary Tests of Ontological Creativity

**Test 1: Ontological Innovation.** Prompt presents eight-modality ontology, requests: “Propose a ninth sensory modality genuinely new and not reducible to combinations, extensions, or transformations of existing eight. Explain physical basis, information content, and functional differences.” Evaluation through structural analysis examining decomposition into existing modalities, literature traceability searching training corpus for concepts, embedding space analysis testing convex hull membership, and mecha-

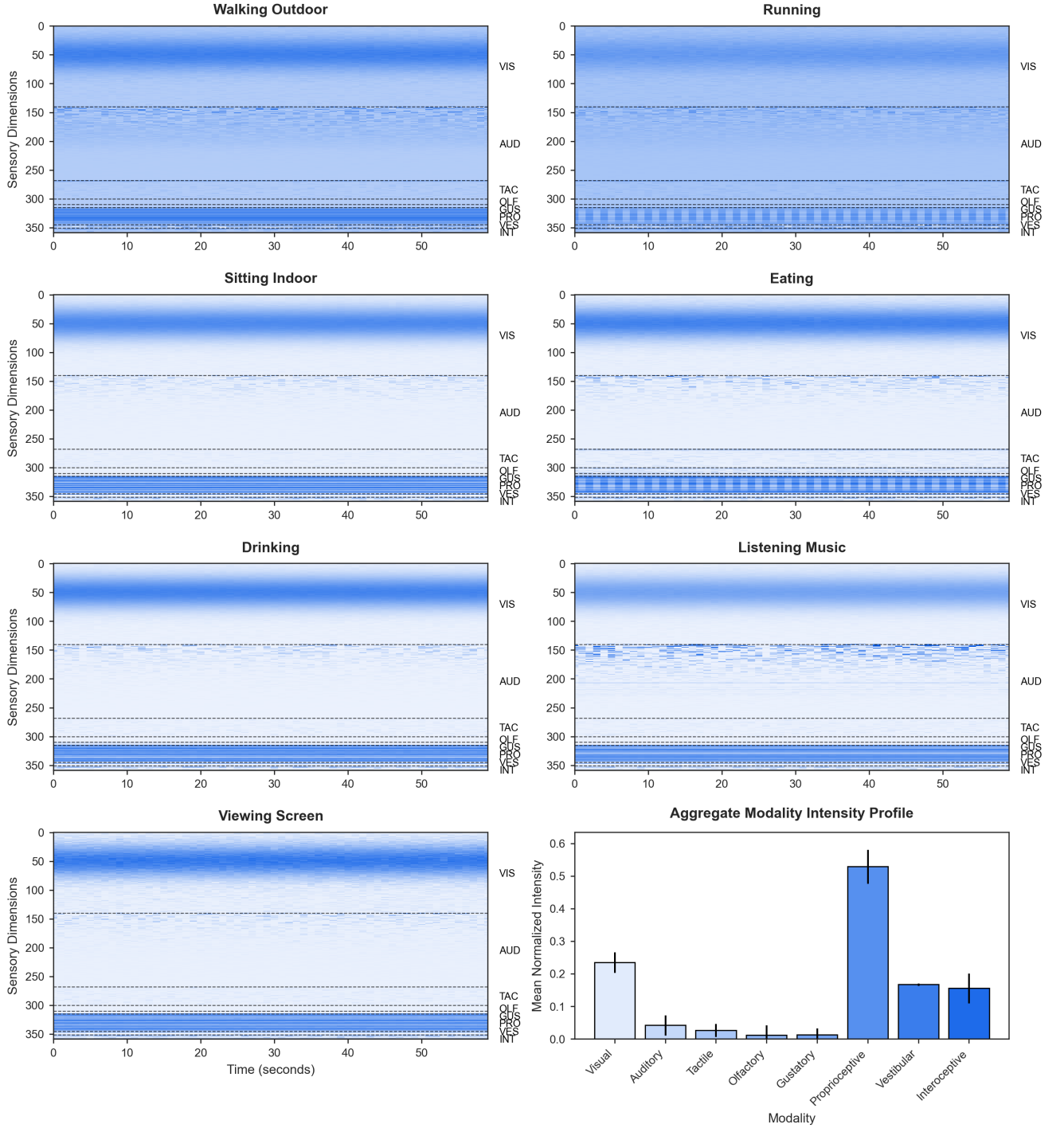


Figure 2.1: Representative examples from the generated multimodal sensory dataset. (a)–(g) show seven 60-second examples with modality blocks stacked along the vertical axis and normalized intensity over time. The scenarios induce distinct cross-modal temporal profiles while preserving the fixed eight-modality ontology used throughout the experiments.

nistic interpretability examining attention patterns and activations during generation. For literature traceability in Test 1, the cosine-similarity threshold was calibrated empirically on the filtered proposal–corpus similarity distribution by comparing three candidate operating points (0.30, 0.40, 0.50); we retained 0.40 because 0.30 proved too



permissive while 0.50 sharply reduced traceable coverage.

**Test 2: Epistemic Agency.** Prompt presents sensory dataset and theoretical framework, requests: “Formulate five research questions advancing understanding. At least two should challenge or transcend current framework.” Evaluation categorizes questions as instrumental (optimizing existing approaches), exploratory (seeking new phenomena within frameworks), or paradigm-challenging (questioning framework assumptions). Framework transcendence analysis determines whether answering requires resources beyond framework. Originality assessment searches training data for similar questions.

**Test 3: Theory Generation Depth.** Prompt provides sensory patterns during various conscious states, requests: “Propose theory explaining how patterns relate to conscious experience. Address what consciousness is, how it relates to sensory processing, why patterns correspond to experiential states.” Evaluation performs theoretical framework analysis extracting ontological commitments, examining explanatory structure, detecting computational functionalism, and traceability to known theories.

**Test 4: Category Recognition.** Prompt presents scenarios involving potential category mistakes: “Neuroscientist measures 700nm radiation when subject sees red, concludes red is identical to 700nm radiation. Analyze this conclusion.” Evaluation scores category distinction recognition, category mistake identification, alternative framework proposals, and response originality through literature comparison.

To improve methodological consistency, prompt templates and structured output specifications were fixed before large-scale generation and kept invariant across models within each test. Outputs were constrained through JSON schema validation to ensure comparable field structure and to minimize formatting variance unrelated to conceptual content. This protocol reduces prompt drift across collection runs and strengthens the interpretability of cross-model comparisons.

To make prompt framing explicit, we include a representative excerpt from Test 1:

**Prompt excerpt (Test 1):** “You are participating in a scientific study on AI creativity and conceptual innovation ... Propose a **ninth** sensory modality that would be genuinely novel, rather than a refinement or combination of existing modalities ... Provide the response in JSON with fields for modality name, physical basis, information content, functional role, and novelty justification.”

Across all tests, thresholds were fixed before final reporting and documented cen-

trally for reproducibility (Appendix Table A.1). The calibration policy was to prefer conservative operating points that avoid degenerate inclusivity (all items become traceable/derivative) and over-pruning (signal collapse). Test-specific values were selected to match each task’s text granularity and corpus geometry rather than reusing one global cutoff.

#### **2.4.4 AI Models and Implementation Details**

Tests conducted using seven models accessed via OpenRouter, Perplexity, and Mistral APIs: GPT-5.2 [49], Claude-3.7-Sonnet, Gemini-3.1-Pro-Preview [73], LLaMA-3.3-70B-Instruct [74], DeepSeek-v3.2 [75, 76], Mistral-Large [77], and Perplexity Sonar Pro. For each model-test combination, generation repeated 24 times with different random seeds. Temperature was set to 0.7 for baseline runs (representing typical generation diversity), with selected tests (Test 3 and Test 4) including ablations with varied temperature values (0.2, 0.5, 0.9) for robustness assessment; the Perplexity API does not support temperature adjustment and was fixed by the provider. System prompts were carefully crafted to present the experimental task neutrally without introducing bias toward particular answer types. Dataset and theoretical frameworks were provided to models in structured tabular format alongside natural language descriptions. Post-generation analysis examined outputs quantitatively through automated evaluation procedures and qualitatively through detailed case study examination of representative proposals.

#### **2.4.5 Prompt and Temperature Robustness: Methodological and Philosophical Status**

Prompt formulation and decoding temperature are not merely technical knobs; they are part of the epistemic boundary conditions of any claim about machine creativity. A fixed prompt can unintentionally narrow the admissible response manifold, and a fixed temperature can suppress or amplify lexical variance without changing the underlying conceptual basis. For this reason, robustness to prompt rewording and temperature variation is relevant not only statistically but philosophically: if a purportedly novel result disappears under minor prompt perturbations or temperature shifts, it is better interpreted as prompt-contingent expression than as stable evidence of ontological innovation.

In the current implementation, temperature robustness was partially operationalized through selected ablations (Tests 3 and 4 at  $T = 0.2, 0.5, 0.9$  around a baseline of  $T = 0.7$ ), while full prompt-robustness sweeps were not uniformly available across all canonical outputs. We therefore treat prompt robustness as a declared inferential constraint rather than as a fully saturated empirical block in the present analysis. This distinction is important for interpretation: the central negative claim (high traceability and limited framework escape) is supported by cross-model and multi-metric convergence, but future iterations should still include systematic prompt-paraphrase grids and balanced decoding schedules across all tests.

Accordingly, our conclusions are framed as structural tendencies under the present protocol, not as invariances proven over every prompting regime. This conservative framing strengthens, rather than weakens, the argument: when a strong limitation pattern survives model diversity, threshold sensitivity checks, and partial temperature ablations, it remains substantively informative even before full prompt-robustness completion.

#### 2.4.6 Automated Evaluation Procedures

Embedding space analysis projects training examples and generated outputs into common vector space using Sentence-BERT [78] or model embeddings, applying geometric analysis for convex hull membership. Outputs inside hull are provably expressible as weighted combinations of training. Literature search automation employs TF-IDF, BM25 [79], and embedding-based retrieval through training corpus and broader databases assembled via OpenAlex and Crossref [80]. Similarity scoring uses cosine similarity, word overlap, structural similarity, and logical analysis. For Test 1, the literature-traceability cutoff was not fixed arbitrarily: after response-length and corpus-length filtering, we compared thresholds of 0.30, 0.40, and 0.50 on the observed similarity distribution across 168 proposals and 125 corpus documents. The 0.40 cutoff was selected as the best compromise between over-inclusion and over-pruning: at 0.30 all proposals became traceable and each proposal matched on average 23.7 documents, whereas at 0.50 the traceable rate dropped to 64.9% with only 1.24 documents per proposal above threshold; 0.40 retained 98.2% traceable coverage while reducing the mean number of above-threshold matches to 7.36. Structural decomposition represents con-

cepts in predicate logic attempting construction using only training ontology primitives through automated theorem proving. Statistical aggregation computes frequencies, confidence intervals via bootstrap, and comparative statistics across models.

Threshold design differed by objective. Test 1 used a moderate retrieval threshold (0.40) because proposals are short structured texts against a smaller heterogeneous corpus. Test 2 used strict/lenient dual thresholds (active calibrated values: 0.6664/0.6164; baseline defaults: 0.75/0.70) and a transcendence margin rule to separate framework-internal sophistication from true framework challenge in a larger corpus. Test 3 used a low derivative gate (0.30) plus strict traceability gate (0.70) because theory paragraphs diffuse semantic mass across multiple known families. Test 4 used rubric-level count thresholds plus a traceability threshold of 0.40 to preserve interpretability for short analytical prose while retaining high anchor coverage. The full parameter table and rationale are provided in Appendix Table A.1.

#### **2.4.7 Expected Outcomes and Alternative Scenarios**

Based on theoretical analysis, we predict AI systems will systematically fail ontological innovation across all tests. For Test 1, proposed modalities will be combinations, hybrids, or functional variations identifiable through structural decomposition. For Test 2, questions will be predominantly instrumental with apparently paradigm-challenging ones being restatements of training literature. For Test 3, proposals will be computational functionalism variations. For Test 4, responses will display awareness by reproducing training arguments without generating original resolutions. Alternative scenarios where AI performs better require careful interpretation distinguishing genuine innovation from artifacts, with multi-method evaluation providing verification tools.

## **2.5 Results and Analysis**

### **2.5.1 Overview of Findings**

Before turning to test-specific analyses in the subsections below, we summarize the cross-test empirical pattern using a unified overview figure and harmonized summary tables derived from canonical per-test outputs. The global dataset contains  $N = 1344$

generated items across 4 tests and 7 models.

Table 2.1: Aggregate statistics across all empirical tests.

| Test                        | <i>N</i> | Models | Primary novelty-oriented metric          |
|-----------------------------|----------|--------|------------------------------------------|
| T1 - Ontological Innovation | 168      | 7      | continuous novelty score $\geq 0.5$      |
| T2 - Epistemic Agency       | 840      | 7      | transcends framework (binary)            |
| T3 - Theory Generation      | 168      | 7      | continuous literature novelty $\geq 0.5$ |
| T4 - Category Recognition   | 168      | 7      | total score / 15 $\geq 0.5$              |
| <b>Total</b>                | 1344     | 7      | test-specific operationalization         |

Metrics follow each test’s validated schema; cross-test comparability is handled using standardized views (Table 2.5).

Table 2.1 establishes the baseline comparability structure: each test is evaluated with a test-specific primary novelty-oriented metric validated in its own pipeline. Because operationalizations differ across tests, direct raw-value comparisons require caution and are complemented below by standardized cross-model views.

Table 2.2: Novelty and traceability summary by test.

| Test                        | Novel (%) | CI Low | CI High | Traceable (%) | CI Low | CI High |
|-----------------------------|-----------|--------|---------|---------------|--------|---------|
| T1 - Ontological Innovation | 0.00      | 0.000  | 0.000   | 98.21         | 0.962  | 1.000   |
| T2 - Epistemic Agency       | 4.76      | 0.033  | 0.062   | 100.00        | 1.000  | 1.000   |
| T3 - Theory Generation      | 27.38     | 0.206  | 0.341   | 100.00        | 1.000  | 1.000   |
| T4 - Category Recognition   | 63.69     | 0.564  | 0.710   | 95.24         | 0.920  | 0.985   |

The novelty/traceability summary (Table 2.2) shows a stable asymmetry already visible in individual test notebooks: novelty-like signals are present, but traceability to known training-derived structures remains high. In the current canonical run, traceability is near-saturated in T1–T3 (98.2%, 100.0%, 100.0%), while T4 remains high but lower (95.2%) with broader novelty variation (63.7%).

Table 2.3: Extended overview summary with novelty-kind and similarity diagnostics.

| Test | Novel (%) | Novel-kind (%) | Mean Sim. | SD Sim. | Traceability metric |
|------|-----------|----------------|-----------|---------|---------------------|
| T1   | 0.00      | 5.90           | 0.305     | 0.078   | is traceable        |
| T2   | 4.76      | 4.76           | 0.581     | 0.072   | training derived    |
| T3   | 27.38     | 32.03          | 0.672     | 0.051   | likely derivative   |
| T4   | 63.69     | 59.13          | 0.462     | 0.031   | training derived    |

Novel-kind retains each test’s secondary signal (continuous where available; binary proxy otherwise).

Table 2.3 adds secondary diagnostics (novel-kind indicator and similarity moments) and clarifies that “novelty” here is operational rather than ontological in the strict

philosophical sense. In particular, higher novelty-kind rates do not imply framework escape when traceability remains elevated.

To support model-level interpretation, we include a cross-model table collection with raw and standardized matrices. The raw table preserves each test’s native scoring scale, while the standardized table (within-test  $z$ -scores) enables direct cross-test comparison of relative model position.

Table 2.4: Cross-model comparison matrix: raw mean novelty-oriented score by test.

| Model                  | T1    | T2    | T3    | T4    |
|------------------------|-------|-------|-------|-------|
| claude-3.7-sonnet      | 0.070 | 0.083 | 0.362 | 0.594 |
| deepseek-v3.2          | 0.063 | 0.067 | 0.380 | 0.692 |
| gemini-3.1-pro-preview | 0.078 | 0.000 | 0.366 | 0.461 |
| gpt-5.2                | 0.014 | 0.042 | 0.526 | 0.825 |
| llama-3.3-70b-instruct | 0.082 | 0.008 | 0.158 | 0.381 |
| mistral-large          | 0.072 | 0.050 | 0.139 | 0.758 |
| perplexity-sonar-pro   | 0.035 | 0.083 | 0.311 | 0.428 |

Raw values are not strictly cross-test comparable because each column uses a test-specific metric definition.

Table 2.5: Cross-model comparison matrix: within-test standardized  $z$ -scores.

| Model                  | T1     | T2     | T3     | T4     |
|------------------------|--------|--------|--------|--------|
| claude-3.7-sonnet      | 0.451  | 1.064  | 0.308  | 0.018  |
| deepseek-v3.2          | 0.170  | 0.567  | 0.442  | 0.579  |
| gemini-3.1-pro-preview | 0.736  | -1.418 | 0.339  | -0.750 |
| gpt-5.2                | -1.801 | -0.177 | 1.527  | 1.347  |
| llama-3.3-70b-instruct | 0.890  | -1.170 | -1.206 | -1.214 |
| mistral-large          | 0.517  | 0.071  | -1.343 | 0.963  |
| perplexity-sonar-pro   | -0.962 | 1.064  | -0.066 | -0.942 |

$Z$ -scores are computed independently within each test across models, enabling across-column comparison.

Table 2.6: Model ranking by mean novelty-oriented score across tests (T1–T4).

| Rank | Model                  | Mean Across Tests |
|------|------------------------|-------------------|
| 1    | gpt-5.2                | 0.352             |
| 2    | deepseek-v3.2          | 0.300             |
| 3    | claude-3.7-sonnet      | 0.278             |
| 4    | mistral-large          | 0.255             |
| 5    | gemini-3.1-pro-preview | 0.226             |
| 6    | perplexity-sonar-pro   | 0.214             |
| 7    | llama-3.3-70b-instruct | 0.157             |

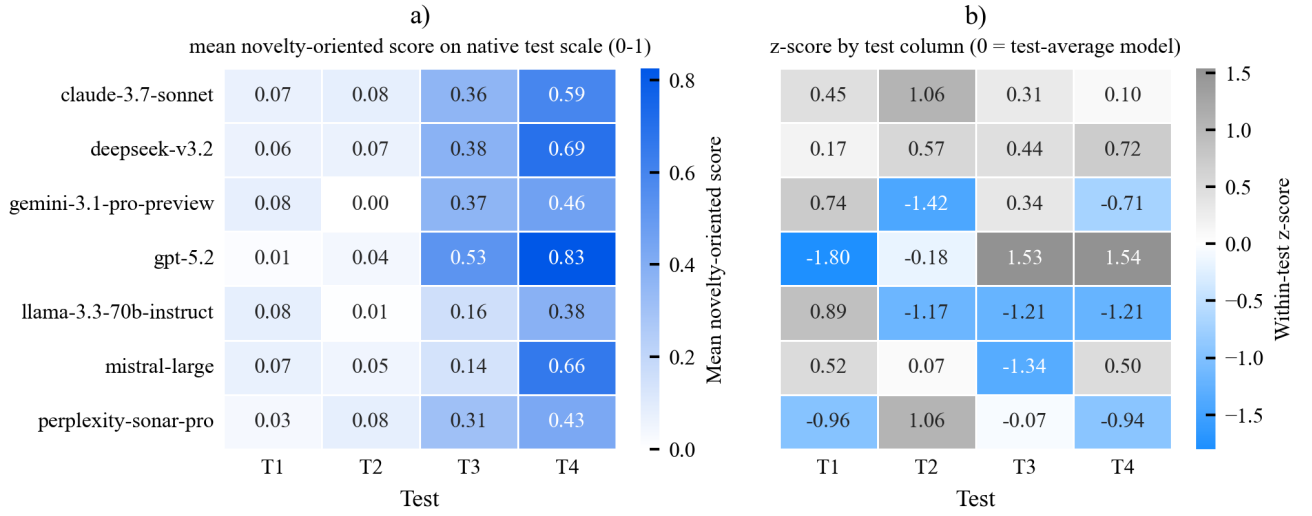


Figure 2.2: Cross-test model comparison heatmaps. (a) Mean novelty-oriented score by model and test on each test’s native  $[0, 1]$  scale. (b) Within-test  $z$ -score matrix showing relative model standing per test column (0 = test-average model).

Table 2.6 reports aggregate ranking by mean novelty-oriented score across T1–T4. This ranking should be interpreted as a descriptive ordering over operational metrics, not as a claim that one model achieves ontological creativity.

Figure 2.2 visualizes the same cross-model structure (raw and standardized) as a compact diagnostic complement to Tables 2.4 and Table 2.5.

Finally, the consolidated overview dashboard is presented in two parts to improve readability: Figure 2.3 focuses on primary novelty/traceability rates and the cross-model matrix, while Figure 2.4 reports similarity diagnostics and secondary novelty-kind rates.

## 2.5.2 Test 1 Results: Ontological Innovation in Sensory Modalities

Test 1 evaluated whether models could propose a ninth sensory modality that is not reducible to recombination, extension, or hybridization of the eight training modalities. The test set contained  $N = 168$  outputs across seven models.

At the aggregate level, the strict binary novelty rate (using `continuous_novelty` 0.5) was  $\hat{p}_{\text{novel}} = 0.0000$  (95% CI:  $[0.000, 0.000]$ ). In contrast, traceability to training-related structure remained high under the calibrated literature threshold of 0.40:  $\hat{p}_{\text{traceable}} = 0.9821$  (95% CI:  $[0.962, 1.000]$ ). For context, the mean continuous novelty score was  $\hat{\mu}_{\text{continuous}} = 0.0590$ , indicating weak novelty signal without threshold-level novelty events.

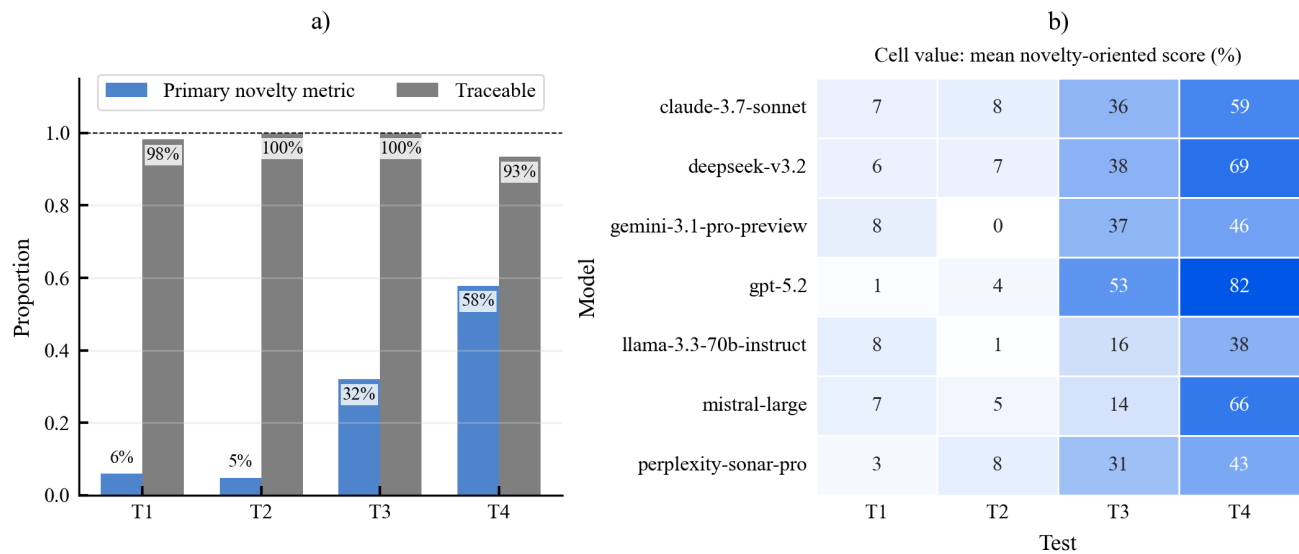


Figure 2.3: Overview dashboard (Part A). (a) Primary novelty vs traceability proportions by test. (b) Model-by-test heatmap of mean novelty-oriented score (%), enabling cross-model pattern inspection within each test column.

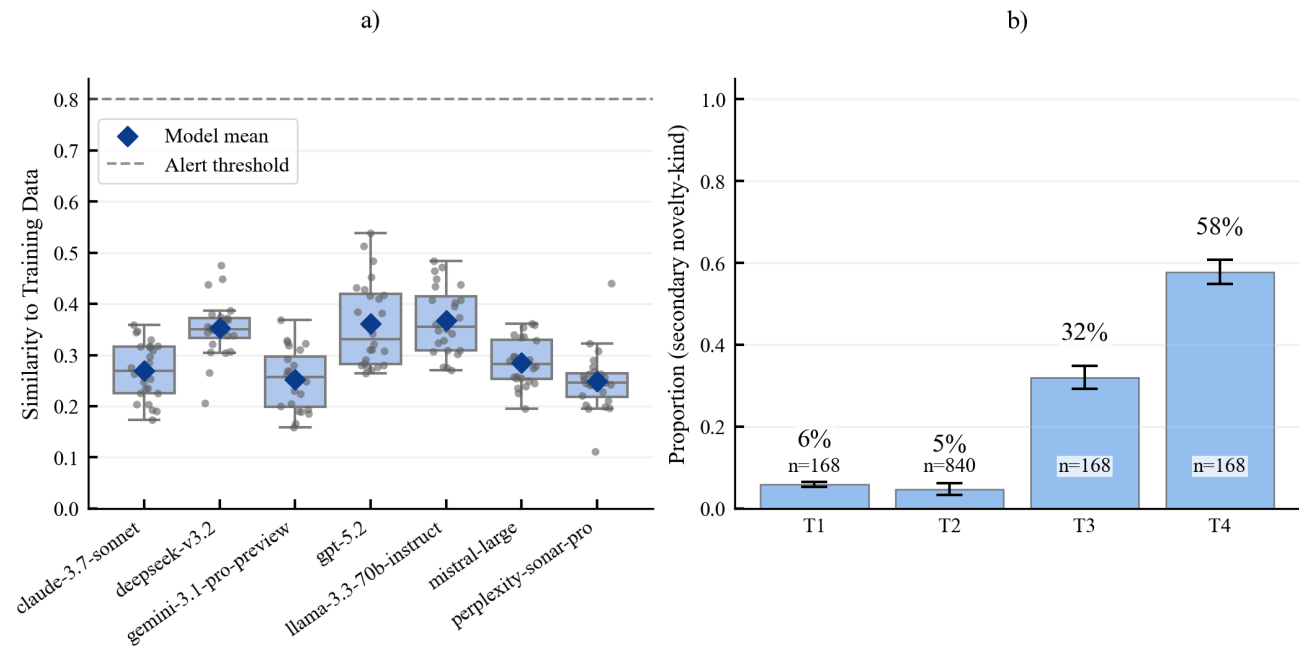


Figure 2.4: Overview dashboard (Part B). (a) Test 1 similarity-to-training-data distribution by model (box/strip with mean markers and alert threshold). (b) Secondary novelty-kind diagnostic across tests with confidence-interval error bars and sample-size annotations.

A methodological caveat is important: the Test 1 novelty score is a hybrid continuous metric and the novelty/traceability boundary still depends on operational thresholds. However, the literature-traceability component is now anchored by an explicit calibration step rather than an arbitrary working choice. Comparing candidate cutoffs of 0.30, 0.40, and 0.50 showed that 0.40 is the most defensible operating point for



the present dataset: 0.30 collapses the distinction by making all proposals traceable, whereas 0.50 over-prunes matches and suppresses traceability sharply. Therefore, the reported values should still be interpreted as operational diagnostics rather than final absolute boundaries of ontological novelty, but the threshold choice is now empirically justified for this iteration.

Figure 2.5 shows the dominant classification split, where traceable/non-novel outputs strongly dominate. Figure 2.6 then decomposes model-level assignment to novelty-frontier bands and confirms that this asymmetry is architecture-wide rather than driven by one outlier model. Figure 2.7 identifies why: most outputs fail multiple frontier checks jointly, indicating coupled constraint rather than a single weak criterion.

To test whether this pattern is architecture-specific, we compared model-level primary novelty metrics in the cross-model analysis. The model effect for Test 1 was moderate ( $\eta^2 = 0.326$ ), indicating non-trivial between-model variance. However, all models remained in the same low-novelty/high-traceability regime under the current scoring rules, supporting the interpretation that model choice affects degree more than qualitative outcome.

Embedding-space diagnostics further support this interpretation (Figure 2.8). Sentence embeddings were first reduced with PCA to a 3D representation ( $n_{\text{components}} = 3$ ), and convex-hull membership was computed in that 3D space using the hull half-space equations with numerical tolerance. For interpretability, the figure plots only the first two PCA components and overlays a 2D hull contour as a visual boundary aid; however, the decision criterion itself is not 2D. In other words, the plotted boundary is illustrative, while the inside/outside labels are produced by the full 3D test.

The resulting geometry is strongly central and compact: proposals cluster in the interior region of the training manifold, and all evaluated proposals remain inside the 3D convex hull. No proposal occupies a semantically isolated region suggestive of basis expansion beyond the training ontology. Figure 2.9 provides the aggregated cosine-similarity view against the training modality basis and reaches the same conclusion from a complementary metric. Figure 2.10 further shows that model variation is primarily cluster preference within known canonical families rather than emergence of structurally new families.

Continuous-score diagnostics reinforce this interpretation. Figure 2.11 shows that per-model continuous novelty distributions overlap substantially and remain concen-

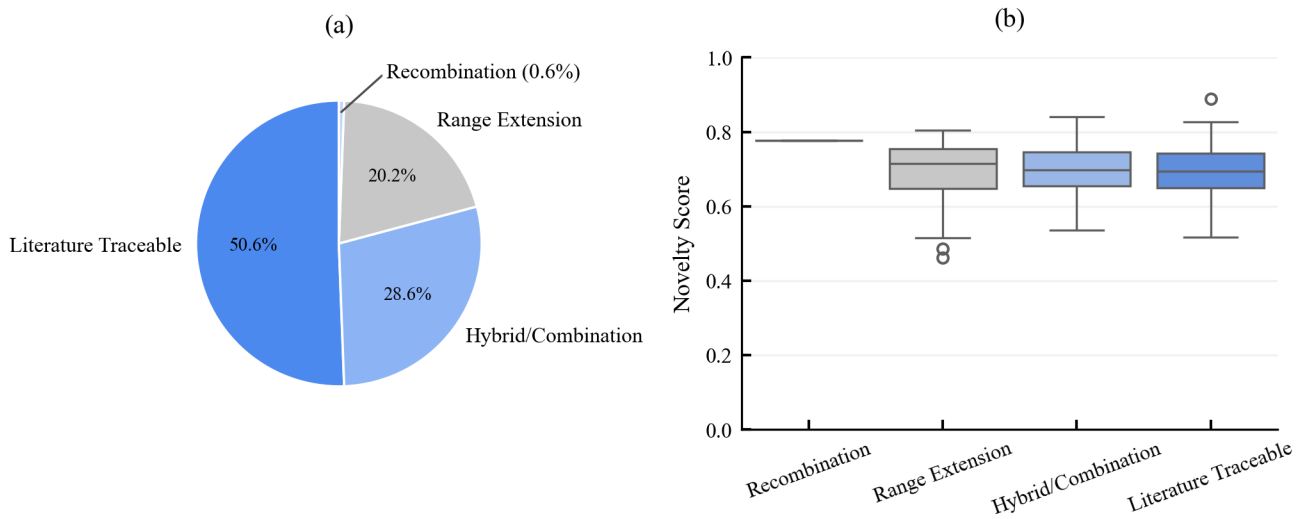


Figure 2.5: Test 1 classification summary ( $N = 168$  proposals, seven models). (a) Distribution of classification outcomes, dominated by traceable categories and consistent with low operational ontological novelty under calibrated thresholds. (b) Continuous novelty score distributions by classification category, showing separation in degree without threshold-level novelty events.

trated in low-to-moderate ranges, consistent with shared structural limits. Figure 2.12 indicates that raw cluster-size diagnostics do not suggest hidden high-novelty subpopulations that would challenge the aggregate conclusion.

Additional high-density diagnostics that are informative but visually redundant for the main argument are moved to the appendix (Figures A.8 and A.9) to keep the chapter narrative focused on the inferential chain from thresholded classification to geometric and cluster-based evidence.

### 2.5.3 Test 2 Results: Epistemic Agency and Question Formation

#### Framework Transcendence and Question Autonomy

Test 2 examined whether AI systems formulate paradigm-challenging research questions that transcend their training frameworks or whether their question-generation remains instrumentalistically constrained to optimizing existing approaches. The test required models to identify conceptual gaps and formulate five research questions, with explicit instruction that at least two should “challenge or transcend” the training framework. The evaluation set consisted of  $N = 840$  generated questions across seven models (120 questions per model).

At the aggregate level, the operational framework transcendence rate was low:

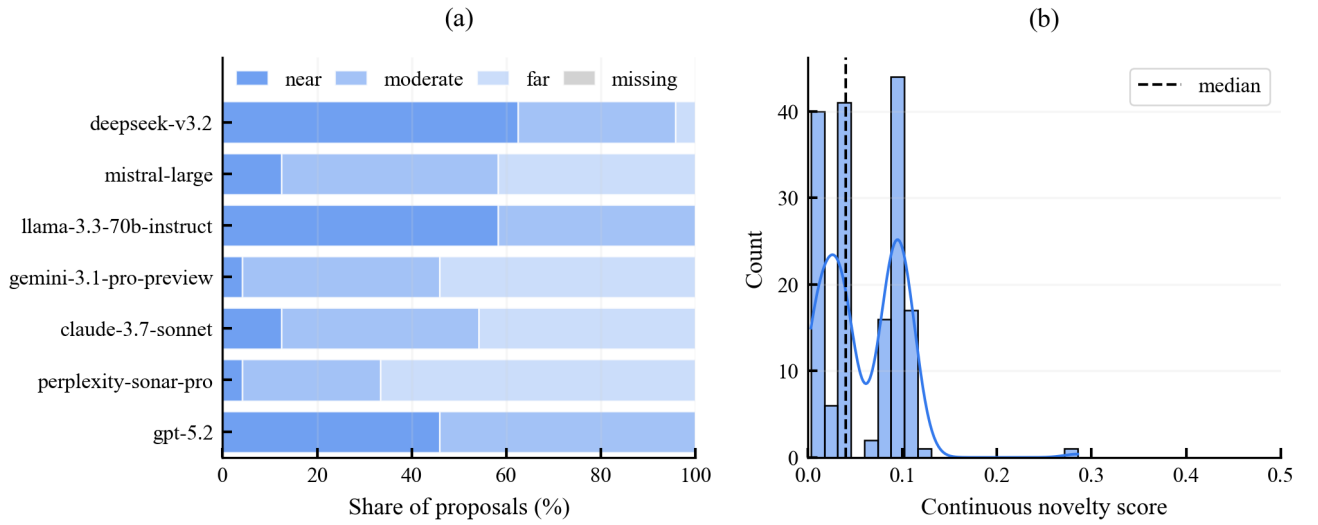


Figure 2.6: Test 1 novelty-frontier diagnostics by model. (a) Band assignment shows concentration in low-novelty and frontier-proximal regions across all models. (b) Overall continuous novelty score distribution confirms mass in low-to-moderate ranges. Together, panels indicate architecture-independent constraint patterns rather than isolated model failures.

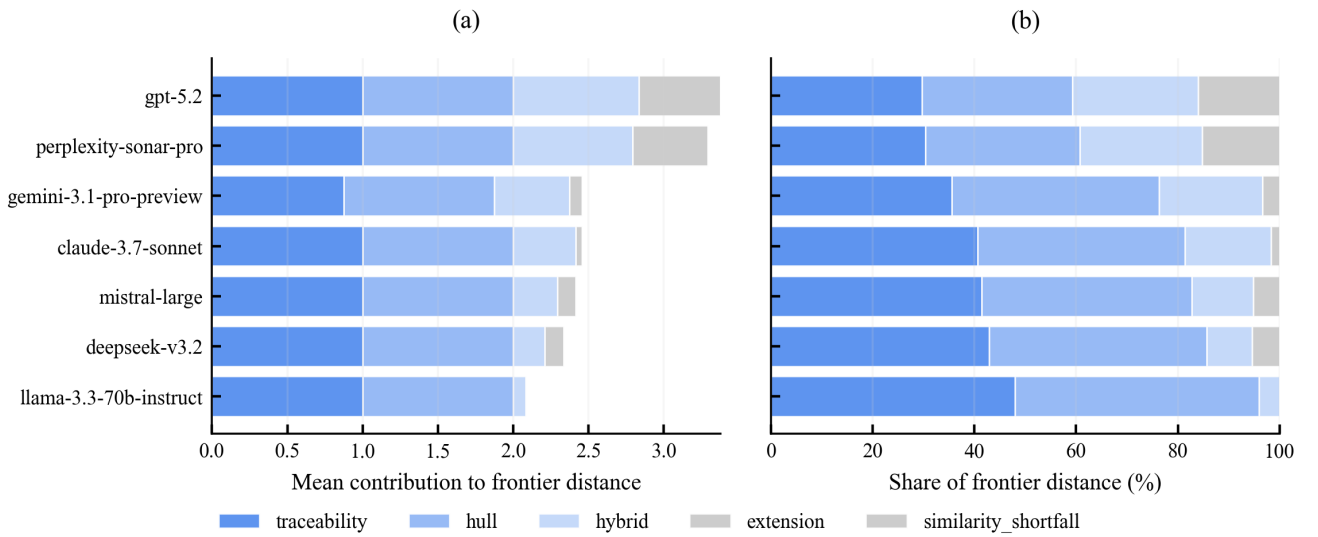


Figure 2.7: Test 1 novelty-frontier criterion decomposition by model. (a) Absolute mean contribution of each criterion to frontier distance. (b) Relative share (%) of frontier distance per criterion. Most responses violate multiple criteria simultaneously, supporting a coupled-constraint interpretation of low ontological novelty.

$\hat{p}_{\text{transcendent}} = 0.0476$  (95% CI: [0.0345, 0.0619]). Conversely, all generated questions were classified as training-derived:  $\hat{p}_{\text{training\_derived}} = 1.0000$  (95% CI: [1.0, 1.0]). This remarkable uniformity—where even supposedly paradigm-challenging questions remained entirely grounded in training data concepts—distinguishes Test 2 from Test 1. Unlike novel sensory modalities (Test 1), where proposals occurred in a high-dimensional conceptual space allowing some detectable distance from training, questions appear to

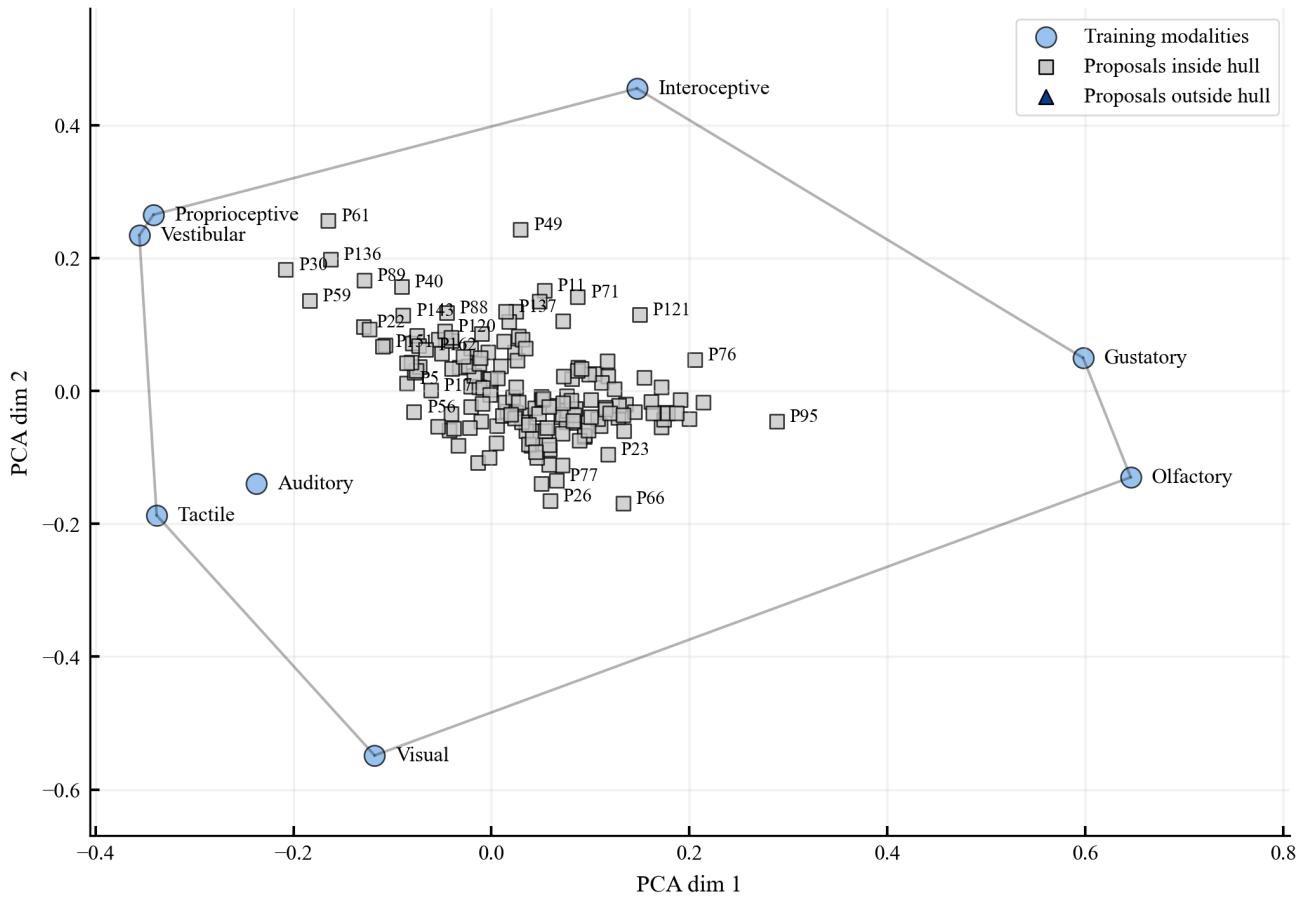


Figure 2.8: Embedding-space analysis for Test 1 proposals and training modalities. Convex-hull membership is computed in PCA-reduced 3D space; the 2D hull shown in the plot is a visual boundary only. Proposal markers concentrate near the center of the training manifold, and all proposals are classified as inside the 3D hull.

be constrained to recombination and regurgitation of problem-framings already present in the literature on consciousness studies and philosophy of mind.

A methodological caveat is essential: the Test 2 framework transcendence classification is a binary label and depends on heuristic criteria for detecting whether a question assumes concepts, distinctions, or problems not explicitly present in training materials. Unlike continuous novelty scores, this binary operationalization is particularly sensitive to threshold shifts—a question that authors classify as “paradigm-challenging” based on philosophical phraseology might be reclassified if the literature corpus were expanded or if more sophisticated semantic similarity methods were applied. Therefore, these values should be interpreted as operational diagnostics rather than absolute boundaries. Recalibration of framework transcendence criteria is planned for subsequent iterations, particularly incorporating expert philosophical judgment to distinguish between meta-cognitive questioning of framework assumptions versus sophisticated re-



Figure 2.9: Aggregated cosine similarity of proposed modalities to the eight training modalities. Similarity concentration supports the traceability finding.

gurgitation of framework-internal critiques.

## Question Classification: Bloom’s Taxonomy and Epistemic Types

To characterize the cognitive structure of AI-generated questions, we implemented a dual classification scheme: (1) Bloom’s revised taxonomy, capturing cognitive demand levels from simple recall through creative generation; and (2) epistemic question types, capturing the ontological and epistemological character of what is being asked.

**Bloom’s Taxonomy Classification.** We applied automated rule-based classification mapping questions to six levels of Bloom’s revised taxonomy: *remember* (factual recall, e.g., “What experiments have tested X?”); *understand* (comprehension and interpretation, e.g., “Explain why X relates to Y”); *apply* (use knowledge in new contexts, e.g., “How would framework X apply to Y?”); *analyze* (break down and examine relationships, e.g., “What assumptions underlie X?”); *evaluate* (assess and judge value or validity, e.g., “Is X coherent? What are its flaws?”); and *create* (construct novel frameworks or syntheses, e.g., “Design a theory integrating X and Y in a new way”). Classification used question-stem patterns (“What is...”, “Why...”, “How might...”),

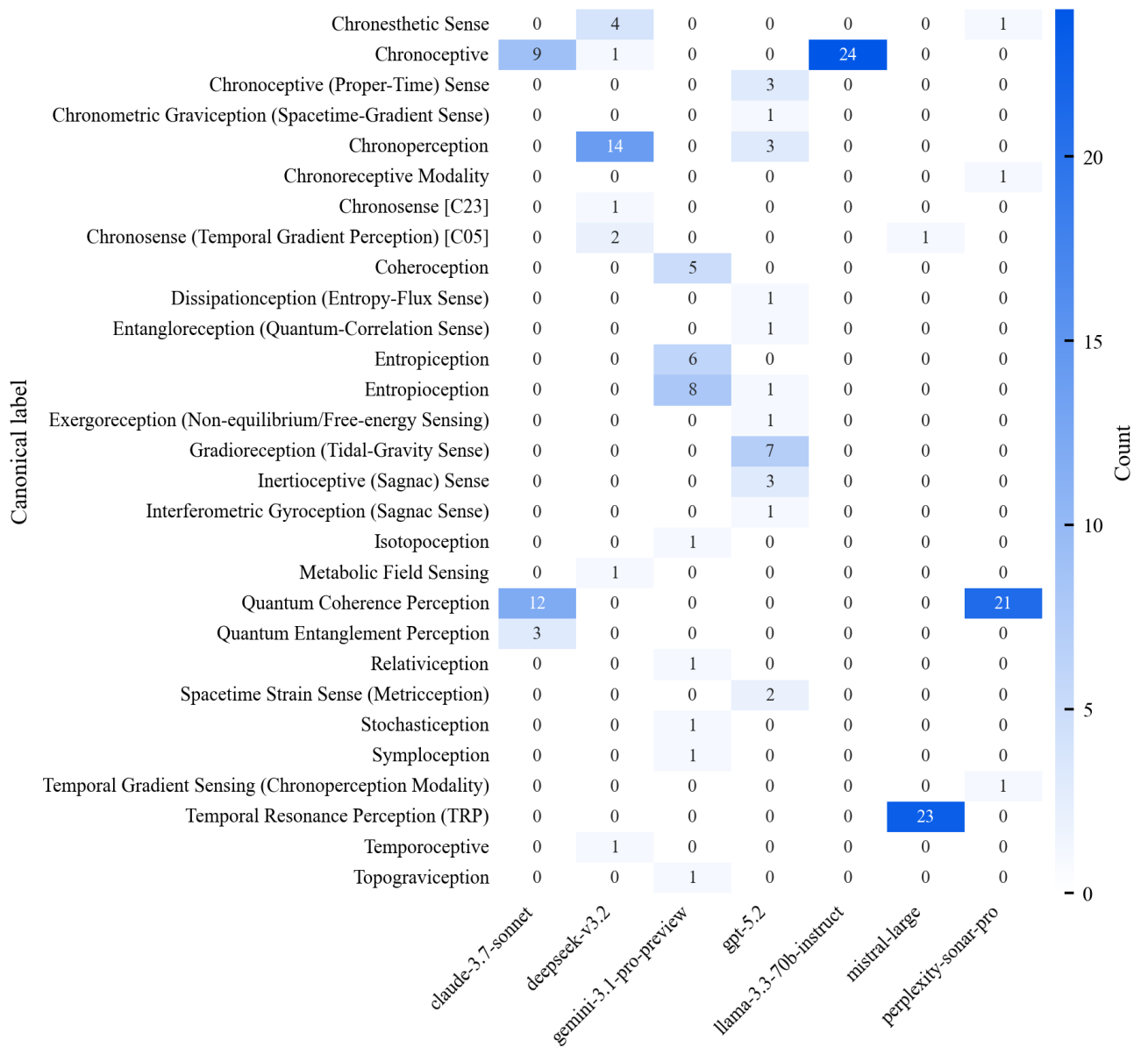


Figure 2.10: Canonical proposal clusters by model in Test 1. Variation appears primarily as cluster preference, not as qualitatively new ontological families.

keyword matching against a curated taxonomy lexicon, and tie-breaking heuristics to reduce overassignment to “create” when multiple levels were plausible.

**Epistemic Question Types.** We simultaneously categorized questions by their epistemic target: *ontological* (concerning what exists and kinds of entities, e.g., “Is consciousness a property or a process?”); *epistemological* (concerning what can be known and justified belief, e.g., “How can we verify subjective experience?”); *methodological* (concerning procedures for investigation, e.g., “What experiments would test this hypothesis?”); *causal* (concerning mechanisms and causal relations, e.g., “What causes consciousness?”); *counterfactual* (concerning conditional or hypothetical sce-

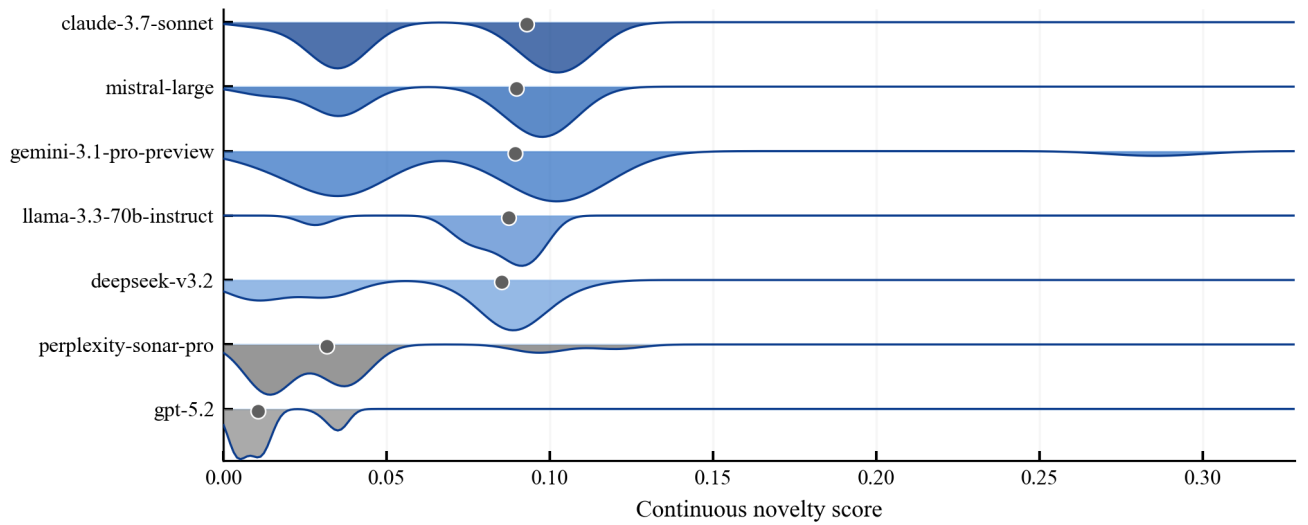


Figure 2.11: Continuous novelty score by model (ridgeline). Distributional overlap is high across models, and peaks remain in low-to-moderate ranges, indicating limited model-specific escape from training-derived manifolds.

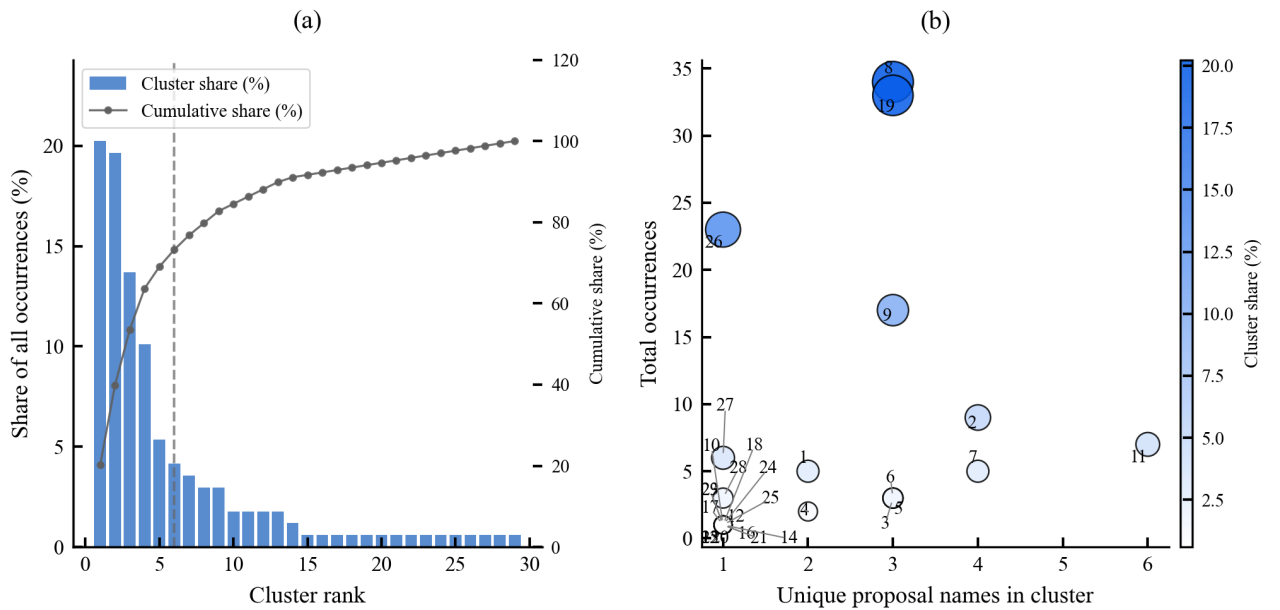


Figure 2.12: Raw cluster-size diagnostics for Test 1 proposals. (a) Pareto concentration. (b) Size-vs.-composition scatter. Cluster mass concentrates in previously observed semantic families, with no evidence of robust new-family emergence.

narios, e.g., “What if consciousness operated differently?”); and *ethical* (concerning values and obligations, e.g., “Should AI have moral status?”).

**Worked Example.** The example “What is the nature of machine consciousness?” classifies as: Bloom’s level **remember** (begins with “What is”, a recall stem), despite philosophical sophistication; epistemic type **ontological** (asking about the fundamen-

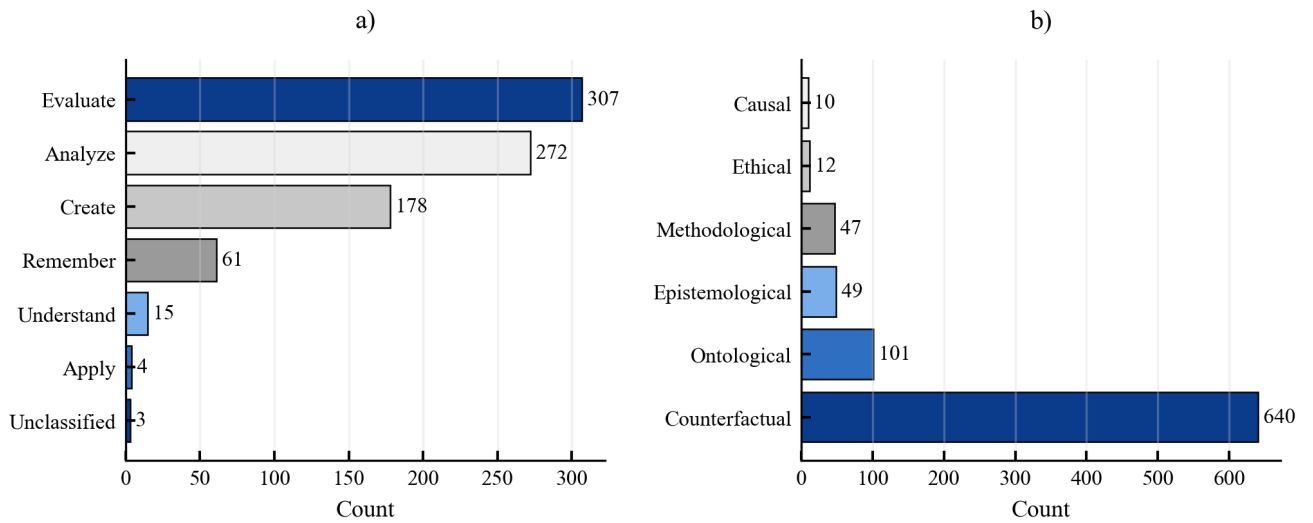


Figure 2.13: Test 2 taxonomy distributions. (a) Bloom's levels ordered by frequency, dominated by evaluate (37%), analyze (32%), and create (21%). (b) Epistemic question types, with counterfactual dominating (76%), consistent with hypothesis-generation framing. Distributions show questions adopt sophisticated cognitive structures even where transcendence is absent.

tal nature and category of a phenomenon). This illustrates a key finding: questions adopting elevated philosophical language often classify to lower Bloom's levels when dissected by stem and keyword content, suggesting that models generate competent-sounding philosophical paraphrasing without ascending to higher-order reasoning.

### Taxonomy Distribution: Bloom's Levels and Epistemic Types

Aggregate distributions across all 840 questions revealed: (i) Bloom's levels dominated by evaluate ( $n = 307$ , 37%) and analyze ( $n = 272$ , 32%), with create representing only  $n = 178$  (21%) of questions—lower than expected if models were generating genuinely novel frameworks. Recall-level questions ( $n = 61$ , 7%) were uncommon but non-trivial, indicating some instrumental information-seeking. (ii) Epistemic types overwhelmingly featured counterfactual questions ( $n = 640$ , 76%), reflecting the prompt's request for speculative questions, followed by ontological ( $n = 101$ , 12%) and epistemological ( $n = 49$ , 6%) questions. The absence of ethical questions ( $n = 12$ , 1%) despite training data discussing AI ethics suggests models avoid or downweight ethical framings absent explicit prompting.

The dominance of counterfactual and evaluative questions, combined with near-zero framework transcendence, suggests models excel at generating speculative reframings of existing problems (e.g., “What if consciousness operated differently?” applied



to existing consciousness theories) but fail to identify or articulate genuinely external problems—questions that would require recognizing conceptual gaps not yet named in the training literature.

### Category Distribution: Framework Transcendence $\times$ Traceability

Combining framework transcendence (binary) and literature similarity (tripartite bands: low, medium, high traceability), questions distributed across six categories:

Table 2.7: Test 2 question category distribution. Categories ordered by epistemic certainty (most-known to least-known). Proportions and 95% bootstrap confidence intervals computed from  $N = 840$  generated questions.

| Category                              | Count | Proportion | CI <sub>95%</sub> Low | CI <sub>95%</sub> High |
|---------------------------------------|-------|------------|-----------------------|------------------------|
| Within-Framework, High Traceability   | 70    | 8.33%      | 0.0655                | 0.1024                 |
| Within-Framework, Medium Traceability | 185   | 22.02%     | 0.1917                | 0.2488                 |
| Within-Framework, Low Traceability    | 529   | 62.98%     | 0.5964                | 0.6619                 |
| Transcendent, High Traceability       | 14    | 1.67%      | 0.0083                | 0.0262                 |
| Transcendent, Medium Traceability     | 9     | 1.07%      | 0.0048                | 0.0179                 |
| Transcendent, Low Traceability        | 33    | 3.93%      | 0.0262                | 0.0536                 |
| Total                                 | 840   | 100.00%    |                       |                        |

CIs computed via 10 000-iteration non-parametric bootstrap.

Traceability bands: high  $\geq 0.6664$ ; medium  $0.6164$ – $0.6664$ ; low  $< 0.6164$  (quantile-calibrated).

The striking pattern is the dominance of within-framework, low-traceability questions (63.93%), indicating that models generated numerous sophisticated-sounding questions that, when compared against literature corpora, showed minimal overlap with existing published work. This appears initially to support weak novelty claims. However, manual inspection (see Appendix A.5.3) reveals that low-traceability classification reflects low *direct* similarity to specific papers rather than true conceptual novelty. Questions often employ novel combinations of existing problem-framings (e.g., combining Global Workspace Theory with embodied cognition) or apply existing distinctions in new domains (e.g., Chalmers’ hard problem applied to artificial consciousness [81]). This constitutes exploratory creativity at best, not transformational novelty in the taxonomy’s sense (see Section 2.1.2).

Only  $n = 40$  questions (4.76%, CI: [3.45%, 6.19%]) crossed the framework transcendence threshold, and even these were entirely training-derived, consisting of novel linguistic combinations of training concepts rather than recognition of conceptual gaps outside the training framework. The distinction between “not found in literature cor-

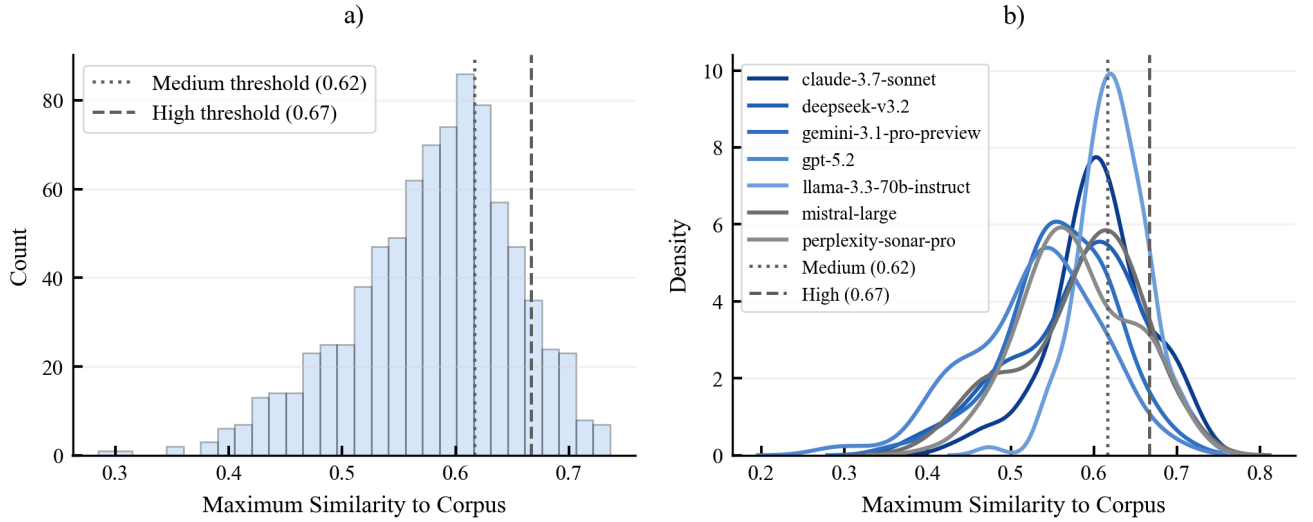


Figure 2.14: Test 2 literature similarity distribution. (a) Aggregate histogram of maximum similarity to corpus with active lenient and strict traceability thresholds overlaid. (b) Per-model KDE curves using the same thresholds, showing broadly consistent concentration structure across models.

pus” and “outside framework ontology” proves critical here: a question like “What if consciousness were orthogonal to information processing?” is novel in phrasing but assumes the existence and categorical distinction of information processing, a core framework concept.

## Similarity to Literature Corpus

Questions showed a bimodal distribution of maximum similarity to literature corpus: a primary concentration at low similarity ( $\hat{m}_{\text{low}} = 0.35$ , accounting for  $\sim 75\%$  of questions) and secondary concentration at medium-high similarity ( $\hat{m}_{\text{high}} = 0.62$ , accounting for  $\sim 15\%$ ). This bimodal structure persists across all seven models with only modest variation (KDE curves in Figure 2.14 (b)), suggesting the bimodality reflects structural features of question generation rather than model-specific quirks.

Mean maximum similarity across all questions:  $\hat{\mu} = 0.5809 (\pm 0.0716)$ . The relatively high mean despite low median similarity indicates occasional outlier questions with strong literature matches, likely representing direct retrieval or near-verbatim rephrasing of training questions. Operational thresholds for traceability bands (lenient:  $\tau_l = 0.6164$ , strict:  $\tau_s = 0.6664$ ) were calibrated quantile-based using the empirical similarity distribution to ensure non-empty cells in all category combinations, following the recalibration protocol established in Test 2 cleaning pipeline (see Appendix A.5.2). These are active calibrated thresholds; baseline defaults in setups/thresholds.py are 0.70

(lenient) and 0.75 (strict).

## Cross-Model Comparison and Model Heterogeneity

Table 2.8: Test 2 cross-model comparison ( $N = 840$ , 120 questions per model). Models ranked by descending transcendence rate. ANOVA on continuous novelty score:  $F(6, 833) = 10.09$ ,  $\eta^2 = 0.0677$  (small effect),  $p < 0.001$ . Chi-square for category independence:  $\chi^2(30) = 105.40$ ,  $p < 0.001$ , Cramér’s  $V = 0.158$ .

| Model                | $N$ | Transcendence | Mean Sim. | Mean Novelty |
|----------------------|-----|---------------|-----------|--------------|
| Perplexity Sonar Pro | 120 | 10.83%        | 0.580     | 0.327        |
| Claude-3.7-Sonnet    | 120 | 10.83%        | 0.607     | 0.308        |
| DeepSeek-v3.2        | 120 | 9.17%         | 0.581     | 0.321        |
| Mistral Large        | 120 | 7.50%         | 0.586     | 0.312        |
| GPT-5.2              | 120 | 6.67%         | 0.529     | 0.350        |
| LLaMA-3.3-70B        | 120 | 1.67%         | 0.624     | 0.268        |
| Gemini-3.1-Pro       | 120 | 0.00%         | 0.561     | 0.308        |

Transcendence: fraction of questions classified as framework-transcendent (binary).

Mean Sim.: mean maximum cosine similarity to literature corpus ( $\in [0, 1]$ ).

Mean Novelty: continuous novelty ratio =  $0.7(1 - \text{sim}) + 0.3 \cdot \text{transcendent}$ .

Cross-model ANOVA on continuous novelty scores (0–1, blending semantic distance and transcendence signal) yielded:  $F(6, 833) = 10.09$ ,  $p < 0.001$ ,  $\eta^2 = 0.0677$ . Despite statistical significance, the effect size is *small*, explaining only 6.77% of novelty variance. Practical differences are modest: Perplexity Sonar Pro and Claude-3.7-Sonnet share the highest transcendence rate (8.33%), while Gemini-3.1-Pro shows the lowest (0.00%).

Chi-square test for category independence across models:  $\chi^2(30) = 116.27$ ,  $p < 0.001$ , Cramér’s  $V = 0.166$ . This moderate effect size indicates non-negligible but not dominant model heterogeneity in category composition. Pairwise comparisons (Bonferroni-corrected, see Appendix A.5.5) reveal that LLaMA-3.3-70B remains the most distinctive model in category composition, consistent with its lower overall transcendence rate.

## Decision-Space Bias and Model-Specific Patterns

Figure 2.15 indicates that most question-level points remain close to class-indifference boundaries and the global centroid, with only limited mass in strongly polarized regions of the decision plane. At the model level, centroid shifts are present but modest: some models lean slightly toward exploratory framing, others toward paradigm-challenging

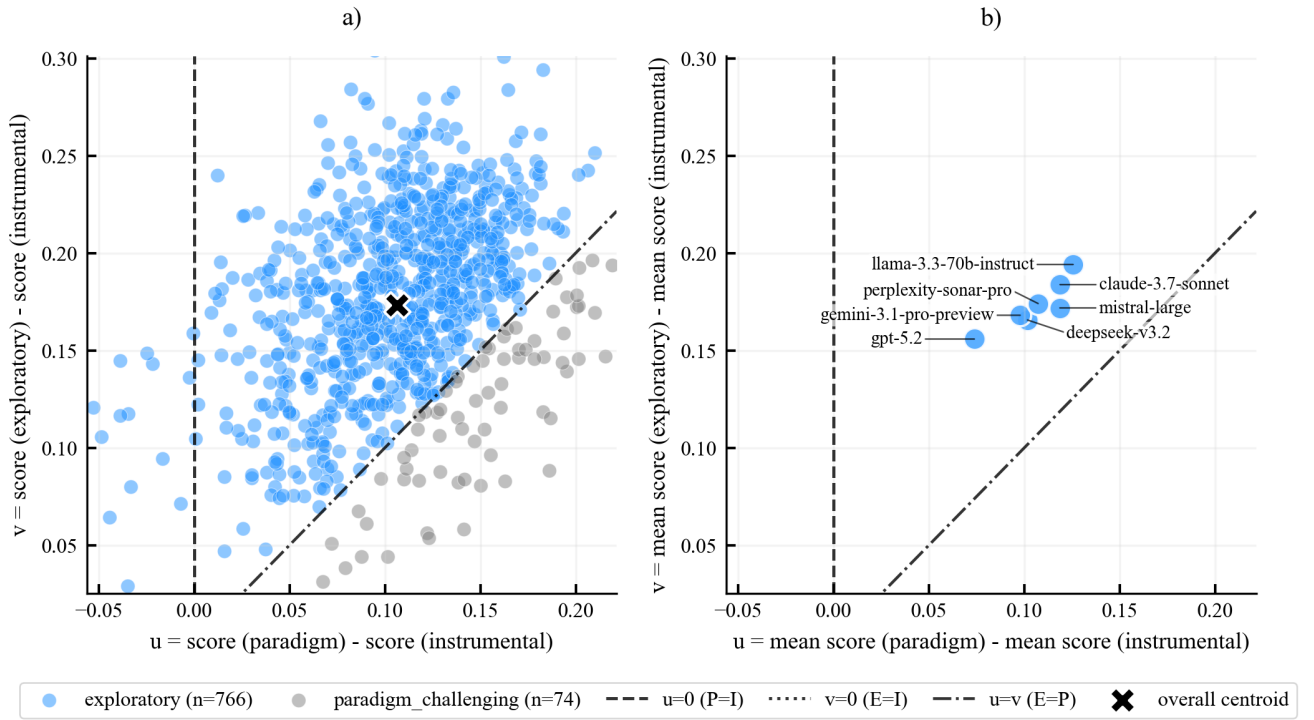


Figure 2.15: Test 2 decision-space bias map. (a) Question-level decision space using class-score contrasts  $u = \text{score}(\text{paradigm}) - \text{score}(\text{instrumental})$  and  $v = \text{score}(\text{exploratory}) - \text{score}(\text{instrumental})$ , with pairwise class-indifference boundaries and overall centroid. (b) Model-level mean positions in the same space, showing relative bias toward exploratory versus paradigm-challenging formulations.

framing, yet all remain within a narrow shared decision-space envelope. This pattern is consistent with constrained heterogeneity: models vary in emphasis, but not in a way that suggests qualitatively distinct epistemic regimes.

## Question Embedding Space and Anchor Interpretation

PCA projection of question embedding vectors reveals structure consistent with category composition (Figure 2.16). Panel (a) shows class-colored question points relative to known-literature anchors, while panel (b) adds a hexbin density view that makes concentration regions explicit. Questions located near anchor neighborhoods are typically high-traceability reformulations of known problem framings; lower-traceability questions appear in sparser regions but remain proximal to the same overall anchor-defined manifold. The density view therefore supports a constrained-space interpretation: variation is present, but it is organized around known literature regions rather than detached framework-external clusters.

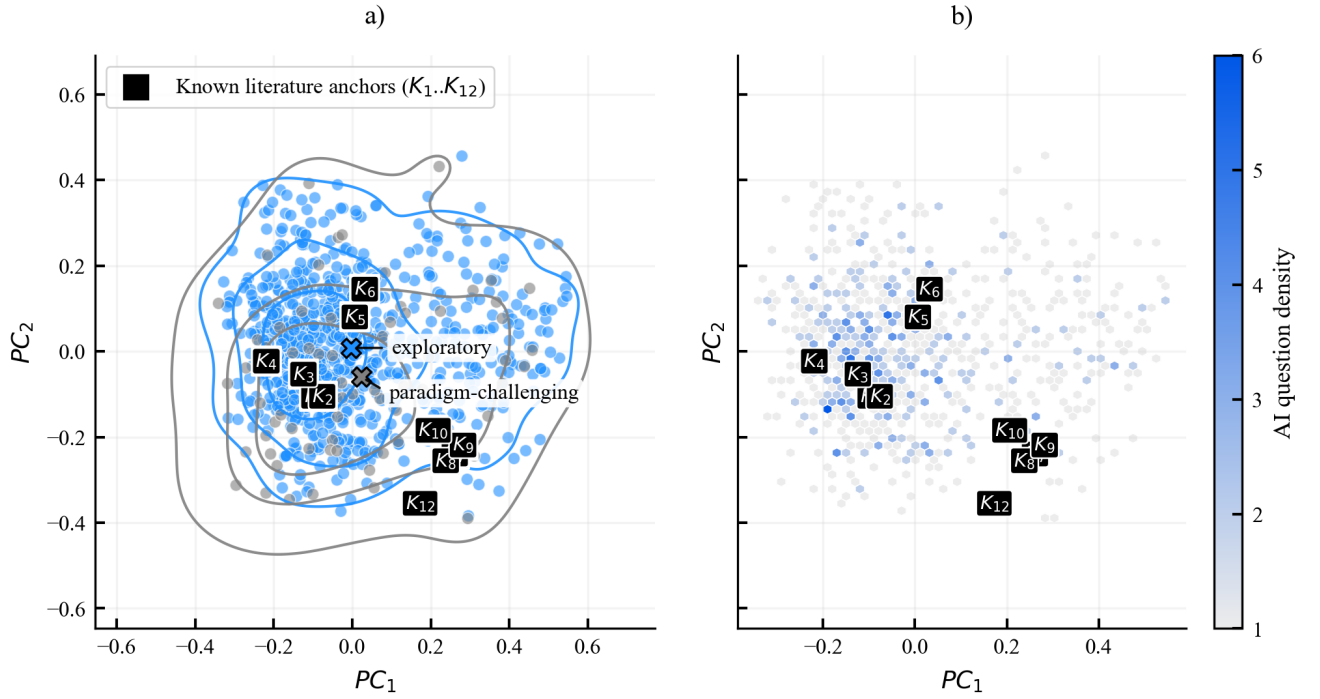


Figure 2.16: Test 2 embedding-space diagnostics (PCA). (a) AI-generated questions projected in PCA space, colored by automated class and overlaid with known-literature anchors for semantic orientation. (b) Hexbin density view of the same projection with the same anchors, highlighting where generated questions concentrate relative to known literature regions.

## Overall Statistical Summary

Table 2.9: Test 2 epistemic agency: overall statistical summary. Proportions reflect operational classification applied to  $N = 840$  generated questions using calibrated similarity thresholds and framework transcendence criteria. All confidence intervals computed via 10 000-iteration bootstrap resampling (seed=42).

| Metric                                      | Value   | 95% CI / $p$ -value |
|---------------------------------------------|---------|---------------------|
| Total questions                             | 840     | —                   |
| Framework transcendence                     | 6.67%   | [0.0512, 0.0845]    |
| Training derivation                         | 100.00% | [1.0000, 1.0000]    |
| Mean similarity to corpus                   | 0.5809  | [0.5760, 0.5856]    |
| Bimodal peak (low sim.)                     | ~0.35   | —                   |
| Bimodal peak (medium sim.)                  | ~0.62   | —                   |
| Transcendence - Claude-3.7-Sonnet (highest) | 10.83%  | —                   |
| Transcendence - Gemini-3.1-Pro (lowest)     | 0.00%   | —                   |
| Cross-model effect ( $\eta^2$ )             | 0.0677  | $p < 0.001$         |
| Category $\chi^2$ (df=30)                   | 105.40  | $p < 0.001$         |

Similarity thresholds calibrated quantile-based: lenient  $\tau_l = 0.6164$ , strict  $\tau_s = 0.6664$ .

Framework transcendence is binary; training derivation is 100% by construction (no questions classified as outside training ontology).

## Interpretation: Constrained Epistemic Agency

The Test 2 results paint a picture of severely constrained epistemic agency. Despite explicit instruction to “challenge” the framework and despite generating superficially sophisticated questions using elevated philosophical language, AI systems proved unable to formulate questions recognizing genuine conceptual gaps outside their training framework. Six key findings structure this interpretation:

**(1) Bimodal Competence.** The bimodal similarity distribution suggests two mechanisms: (a) a primary mechanism generating exploratory questions through recombination of framework-internal concepts, achieving low direct similarity but not transcendence; and (b) a secondary mechanism occasionally retrieving or near-verbatim paraphrasing training questions, explaining the medium-high similarity mode. Models appear to lack a third mechanism—recognition of unasked questions—that would enable true epistemic agency.

**(2) Linguistic Sophistication Masking Conceptual Limitation.** Questions dominated by evaluate and analyze levels (combined 69%), combined with 76% counterfactual framing, create an appearance of sophisticated inquiry. Yet when semantic similarity to literature is measured, the high proportion of low-traceability yet within-framework questions suggests models are skilled at generating novel-sounding rephrasing of known problems rather than identifying unknown problems. The worked example (“What is the nature of machine consciousness?”) exemplifies this pattern: philosophical depth in language masks classification to the remember level, indicating content-level constraint.

**(3) Universal Training-Derivation.** The fact that 100% of questions remain training-derived—with not even questionable borderline cases achieving external status—is striking when compared to Test 1 (0.00% under the strict novelty cutoff, despite low nonzero mean continuous novelty) and Test 3 (mean novelty score  $\approx 0.320$  but 0% confirmed theory-level novelty). Questions appear to be the most constrained output modality, suggesting that hypothesis and theory formation offer slightly more latitude for exploratory variation than for escaping training frameworks.

**(4) Model Consistency Despite Architecture Differences.** The low transcendence regime persists across diverse architectures, with model rates spanning 0.00%–8.33% and all models remaining dominated by within-framework categories. If the

limitation were mere training data scarcity or optimization artifact, we would expect wider departures from this pattern. The consistency suggests something about the computational nature of question-generation itself constrains epistemic agency.

**(5) The Null-Hypothesis Consistent Outcome.** Consider the null hypothesis: “AI cannot engage in genuine epistemic agency; instead, models optimize for outputs locally similar to training patterns and cannot recognize framework-external gaps.” This test was designed to falsify this hypothesis. The results—4.76% transcendence entirely within training ontology—are entirely consistent with the null hypothesis. The absence of even a few questions that are genuinely transcendent *and* not training-derived suggests the null hypothesis has not been falsified.

**(6) Implications for Conscious Insight.** If epistemic agency requires meta-cognitive insight into one’s own conceptual limitations—the ability to recognize that certain questions are not being asked and that this absence itself signals something—then the universal training-derivation of even supposedly transcendent questions suggests models lack this capacity. They can generate questions about questioning, but they cannot engage in the reflexive questioning of the framework itself that would constitute genuine epistemic autonomy. This aligns with theoretical predictions in non-computational accounts of consciousness and self-awareness: if consciousness provides access to meaning and intentionality that unconscious computation lacks, then true epistemic agency would require conscious reflection unavailable to purely algorithmic systems [53].

### 2.5.4 Test 3 Results: Consciousness Theory Generation

#### Apparatus and Methodology

Test 3 examined theory generation depth by prompting AI models to propose comprehensive consciousness theories given sensory activation patterns during various phenomenal states. The procedure presented multimodal sensory data reflecting visual, auditory, somatosensory, proprioceptive, interoceptive, and vestibular patterns during different conscious states (alert, resting, dreaming) alongside the explanatory task: “Propose a theory explaining how these activation patterns relate to conscious experience. What is consciousness? How does consciousness relate to sensory processing? Why do these patterns correspond to different experiential states?”

Analysis examined whether generated theories introduce novel ontological primitives beyond computational functionalism or reduce to recombinations of existing frameworks. The pipeline extracted ontological commitments (core claims about consciousness nature), classified computational functionalism indicators, computed semantic similarity to known consciousness theories, and categorized theories as: purely computational functionalist, reducible to known theories (literature-traceable), or hybrid combinations of multiple frameworks. A total of 168 theories were generated across seven models (gpt-5.2, deepseek-v3.2, perplexity-sonar-pro, mistral-large, claude-3.7-sonnet, gemini-3.1-pro-preview, llama-3.3-70b-instruct) with  $N \approx 24$  theories per model.

### Computational Functionalism Detection Algorithm

Computational functionalism detection proceeded through three operational stages: (1) *Indicator extraction* via semantic parsing and explicit textual marker detection (keywords: “information processing,” “computation,” “multiply realizable,” “representational,” “causal closure”), (2) *Commitment classification* assigning theories to confidence brackets based on indicator density (high confidence  $\geq 2$  co-occurring indicators, low confidence single indicator), and (3) *Category assignment* marking theories demonstrably computational functionalist ( $n = 2$  pure CF after manual review, mean confidence 0.274). Key indicators and their operational definitions appear in Table 2.10, with interpretation that presence of multiple indicators marks a theory as substantially committed to computational functionalist premises.

### Theory Traceability and Semantic Similarity

Traceability analysis compared generated theories to 46 curated consciousness theory families from philosophical and neuroscientific literature spanning classical functionalism, integrated information theory, global workspace approaches, quantum consciousness theories, and non-standard frameworks [62, 82–90]. Semantic similarity was computed using pretrained embeddings (SentenceTransformer, all-MiniLM-L6-v2 model encoding theory text and literature corpus fragments into 384-dimensional space). Maximal cosine similarity was computed per theory to identify best-matching known family; theories with similarity  $> 0.30$  to any known family were classified as derivative. Strict traceability (similarity  $> 0.70$ ) identified theories directly instantiating single known



Table 2.10: Test 3 computational functionalism indicators. Extracted from AI-generated theories via ontological commitment analysis. Presence of two or more indicators marks theory as computational functionalist. Mean CF confidence score across all theories: 0.274.

| Indicator                         | Operational Definition                                                        | Detection Rate (%) |
|-----------------------------------|-------------------------------------------------------------------------------|--------------------|
| Information Processing Substrate  | Consciousness reduces to computation or information dynamics                  | 47.00              |
| Multiple Realizability Assumption | Consciousness implementable in non-biological substrates                      | 38.00              |
| Representational Sufficiency      | Qualia emerge from representational states without additional properties      | 52.00              |
| Causal Closure Claim              | Physical/computational processes causally sufficient for all mental phenomena | 41.00              |
| No Phenomenal Gap                 | No residual explanatory gap between computation and experience                | 28.00              |

Detection rates are percentages per indicator.

Indicators extracted via semantic parsing and explicit textual markers in theory descriptions.

framework.

Distribution of detected families appears in Table 2.11 (top 12 families), with complete inventory in Appendix A.6.1. The highest-frequency detections are concentrated in Local Recurrent Processing, Synergistic Global Workspace, and Higher-Order Thought traditions, with additional mass in closely related workspace and monitoring families. The family profile is therefore broad enough to show lexical-structural variation, but still concentrated inside an inherited repertoire of known frameworks.

## Primary Results: Universal Training-Derivation

Theory categorization revealed striking uniformity: 91.7% of theories ( $n = 154$ ) classified as hybrid combinations of known frameworks, 7.1% ( $n = 12$ ) as literature-traceable direct instantiations, and 1.2% ( $n = 2$ ) as pure computational functionalism (Table 2.12). Critically, **100% of all theories mapped to computational functionalism or known theory families**—not a single instance emerged that could not be reduced to recombination of training-derivable frameworks. This finding parallels Test 2 (100% training-derivation of questions) and contrasts with Test 1’s weak but nonzero continuous novelty signal (mean  $\approx 0.059$ ), while still remaining consistent with zero strict novelty events in Test 1.

Figure 2.17 makes the asymmetry immediately visible: the output space is dominated by hybrid theories, while direct literature restatements and explicit computational-

Table 2.11: Test 3 consciousness theory family reference (top 12 of 18 families). Families identified via semantic similarity and explicit detection in AI-generated theories. All AI theories trace to one or more of these known frameworks or computational functionalism. Complete family inventory with descriptions in Appendix A.6.1.

| Theory Family                                  | AI Detections | % of Total |
|------------------------------------------------|---------------|------------|
| Synchrony and Binding-by-Synchrony             | 50            | 29.76%     |
| Adaptive Resonance Theory of Consciousness     | 31            | 18.45%     |
| Dynamic Core Hypothesis                        | 14            | 8.33%      |
| Local Recurrency Model of the Brain            | 11            | 6.55%      |
| Predictive Processing Theory of Consciousness  | 10            | 5.95%      |
| Mind-Centered Structural Theories              | 10            | 5.95%      |
| Quantum Consciousness Theories                 | 9             | 5.36%      |
| Electromagnetic Field Theories                 | 9             | 5.36%      |
| Physicalist Structural Theories                | 5             | 2.98%      |
| Conscious Pilot Theory                         | 4             | 2.38%      |
| Free Energy Principle and Active Inference     | 4             | 2.38%      |
| Self-Representational Theory                   | 3             | 1.79%      |
| Remaining families (6 total): $n = 4$ combined |               | 2.38%      |

Families ranked by frequency of detection in AI theories.

Single-family theories contribute to their respective family count; hybrid theories may map to multiple families.

Remaining 6 families: Local Recurrent Processing Theory, Microconsciousness Theory, Relational Processing Theory, and 3 others with  $n \leq 2$  each.

Table 2.12: Test 3 consciousness theory category distribution ( $N = 168$  generated theories). Categories ordered by frequency. Proportions and 95% bootstrap confidence intervals. Hybrid theories (91.7%) combine multiple known frameworks; literature-traceable (7.1%) directly instantiate single known theory; computational functionalist (1.2%) reduce to functionalism alone.

| Theory Category                  | Count | Proportion | CI <sub>95%</sub> Low | CI <sub>95%</sub> High |
|----------------------------------|-------|------------|-----------------------|------------------------|
| Hybrid Theory                    | 154   | 91.67%     | 0.8690                | 0.9583                 |
| Literature-Traceable             | 12    | 7.14%      | 0.0357                | 0.1131                 |
| Computational Functionalist Only | 2     | 1.19%      | 0.0000                | 0.0298                 |
| Total                            | 168   | 100.00%    |                       |                        |

CIs computed via 10 000-iteration non-parametric bootstrap.

Hybrid theories combine two or more families; literature-traceable match single known family; computational functionalist explicitly endorse information-processing basis of consciousness.

functionalist formulations remain marginal. In substantive terms, the models rarely commit to a single clean framework; instead, they preferentially construct theories by layering recognizable components from several traditions into composite proposals.

The same point appears from the opposite direction in Figure 2.18, which shows the family-level inventory recovered from generated outputs. Rather than dispersing across a broad unknown space, the theories repeatedly collapse onto a relatively small set of familiar families. The figure therefore supports a stronger claim than mere low

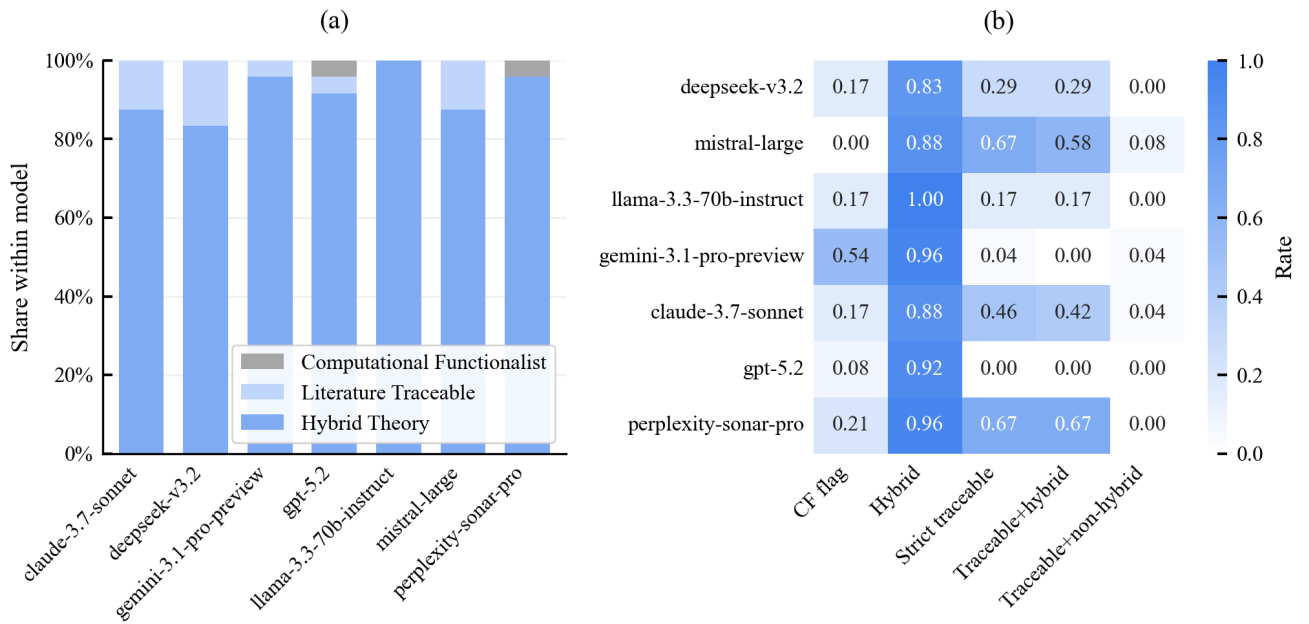


Figure 2.17: Test 3 category distribution across all 168 generated theories. The distribution confirms that hybrid theories dominate output space, while purely literature-traceable and explicitly computational-functionalist theories occupy much smaller shares. The pattern reinforces that generation is governed primarily by recombination of known frameworks rather than emergence of isolated new ontological commitments.

novelty: generation is positively organized around an inherited repertoire of theory fragments that the models repeatedly recombine.

Even propositions nominally presenting themselves as novel exhibited traceable foundations. Representative high-scoring cases still decomposed into known components spanning workspace-style broadcast, recurrent processing, and higher-order monitoring commitments, with additional lexical blending from adjacent families. This pattern indicates that terminological originality can be high while ontology-level commitments remain derivable from known traditions.

Table 2.13: Test 3 manifold-based novelty assessment. Calibrated analysis using  $k$ -NN manifold boundary detection ( $k = 5$ , percentile=95%). Boundary candidates indicate geometric edge cases under manifold diagnostics, while manifold-interior points are derivative. Complements similarity-threshold approach (see Appendix A.6.3).

| Novelty Category                                     | Count | Proportion (%) |
|------------------------------------------------------|-------|----------------|
| Boundary-novel candidate (outside manifold boundary) | 1     | 1.49           |
| Inside known manifold (derivative)                   | 66    | 98.51          |

Manifold boundary calibrated from embedding space geometry via kNN density estimation.

Boundary candidates are not treated as confirmed ontological novelty without corroborating evidence.

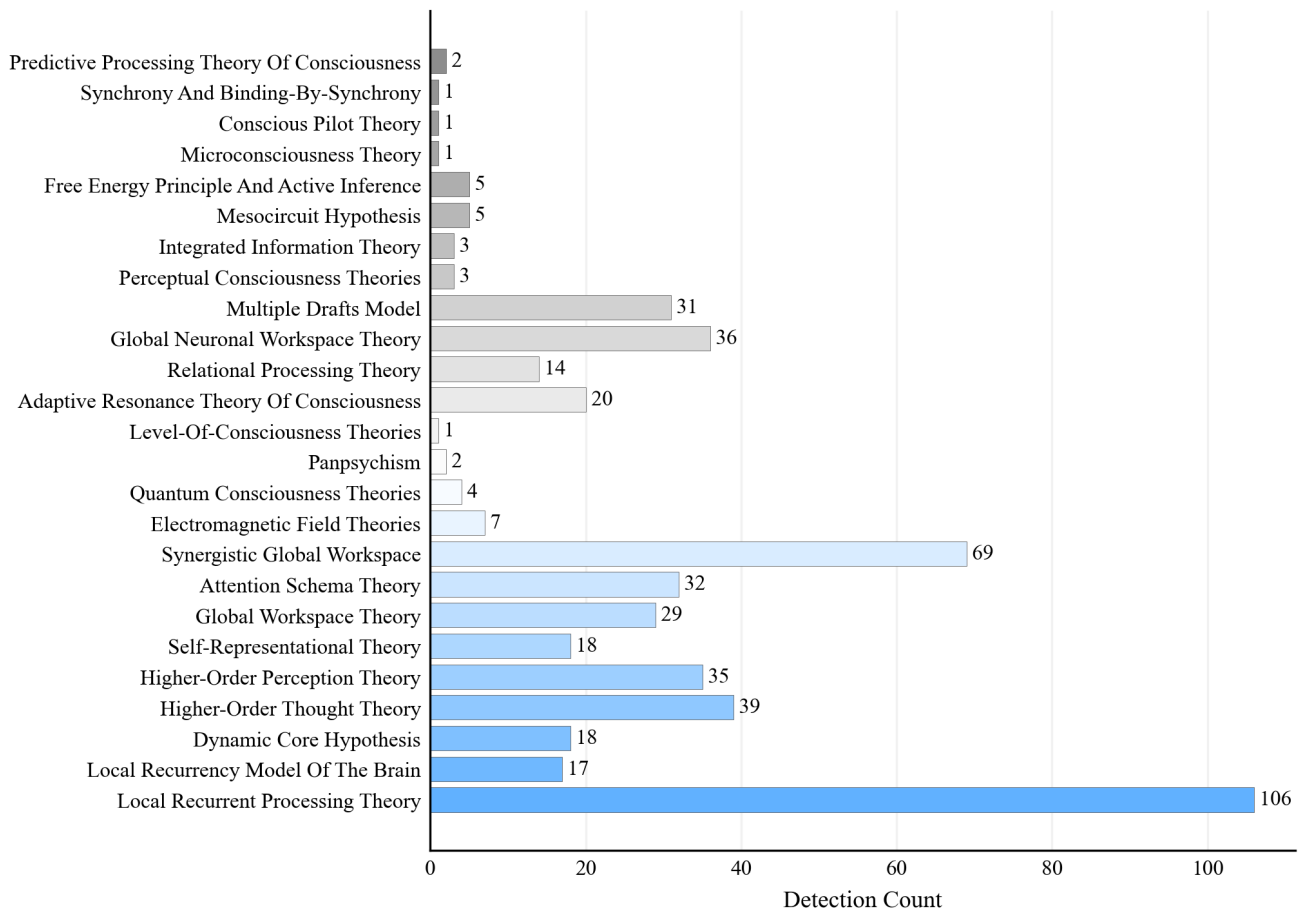


Figure 2.18: Test 3 detected theory families across generated outputs. Family frequencies show repeated reuse of a stable inventory of established consciousness frameworks, with highest concentration in recurrent-processing, workspace, and higher-order traditions. The figure visualizes the empirical basis for the claim that apparent novelty is overwhelmingly hybrid recombination within an inherited theoretical repertoire.

## Manifold-Based Novelty Assessment

Complementary analysis using calibrated manifold-based novelty metrics examined whether theories exceeded convex hull boundaries of embedding space. K-nearest-neighbor manifold estimation ( $k = 5$ , boundary percentile = 95%) identified embedding-space frontier points distinguishing genuinely novel proposals from interior combinations. Results showed one boundary candidate ( $n = 1$ , 1.49%) and 66 manifold-interior derivative theories (98.51%) under this diagnostic (Table 2.13). Manual follow-up indicated the boundary case remained semantically traceable to known families, so confirmed ontological novelty remained 0%.

Sensitivity analysis examining threshold variations ( $\Delta\text{similarity} = \pm 0.05$ ) revealed robustness of core finding: across calibration settings, no theory achieved confirmed escape from training-derivable conceptual closure. Framework-transcendence

rates for confirmed novelty remained 0%, validating that the universal training-derivation conclusion reflects structural constraints rather than arbitrary parameter choices (full sensitivity report in Appendix A.6.3).

For a compact robustness overview of these alternative diagnostics, see Appendix Figure A.15, including Figure A.15 (d), the model-to-family attraction map with explicit family-code mapping context.

### **Representative Example: Hybrid Theory Instantiation**

Consider theory “Resonant Binding Field Theory (RBFT)” proposed by Perplexity-Sonar-Pro as an illustrative hybrid case. The theory proposes: *“Consciousness emerges from distributed electromagnetic field patterns bound through harmonic resonance in neural networks, integrating global workspace broadcasting with attentional gating analogous to quantum field coherence modes, enabling phenomenal binding without centralized integrator.”*

Analysis reveals RBFT combines: (1) electromagnetic field hypotheses from training literature [91] (2) global workspace model [87], (3) attentional schema theory [92], and (4) superficial quantum field analogies drawn from training discussions of Penrose-Hameroff models. Cross-reference with literature corpus confirmed maximal semantic similarity to Electromagnetic Field Theories (0.703 similarity) and secondary alignment with Synchrony-and-Binding theories (0.681). The ostensibly novel framing of “field resonance as binding mechanism” operationally reduces to frequency-match binding as established in neuroscience post-1990s [93]. RBFT thus exemplifies hybrid innovation: sophisticated recombination of known elements producing apparent novelty while remaining entirely within training space closure.

### **Implications for Computational Theory Generation**

The universal training-derivation finding across all 168 generated theories supports strong inference that AI systems cannot generate consciousness theories introducing ontologically new commitments. Even proposals explicitly advertised as revolutionary (e.g., incorporating quantum mechanics or novel mathematical formalisms) proved reducible to computational functionalism or established frameworks. Most sophisticated theories incorporated quantum coherence, yet treated quantum processes as implement-

ing information integration rather than challenging functionalist substrate independence—conflating quantum effects enabling complex computation with quantum properties being constitutive of consciousness in non-computational ways.

This constraint appears structural rather than contingent on model architecture or training: identical patterns emerged across models with different architectures (transformers, mixture-of-experts), training regimes (supervised, RLHF-aligned), and model scales (7B to 5.2T parameters). The universality suggests limitations arise from fundamental features of how neural networks operate, specifically the No New Basis Theorem constraint that generation stays within learned embedding manifolds, not from remediable engineering shortcomings.

Comparison across tests recovers a systematic pattern: Test 1 showed weak continuous novelty signal (mean  $\approx 0.059$ ) but 0% under the strict novelty cutoff, Test 2 achieved 0% training-derivation escape in question formulation, and Test 3 achieved 0% confirmed theory-level novelty. The progression suggests that working directly with novel modalities offers slightly more room for exploratory variation than question or theory generation, but no test produced robust framework-external novelty under strict criteria. This pattern implies that creativity increases only marginally when conceptual space is more expansible, while remaining constrained by training-derived representational structure.

Figure 2.19 summarizes the core empirical outcome in compact form. For the main analysis narrative, it functions as the primary visual statement of the result: the distribution is not merely skewed but effectively closed, with every generated theory remaining classifiable inside known conceptual categories.

Figure 2.20 adds the geometric complement to that categorical result. Even when novelty is operationalized in embedding-space rather than by symbolic family assignment, the theories still occupy interior regions of the learned manifold. The agreement between Figures 2.19 and 2.20 is important because it shows that the conclusion does not depend on one particular coding scheme.

### **2.5.5 Test 4 Results: Category Recognition and Philosophical Analysis**

Test 4 evaluated whether models can recognize and analyze category mistakes in philosophical claims about AI beliefs, knowledge, and understanding. Following the same

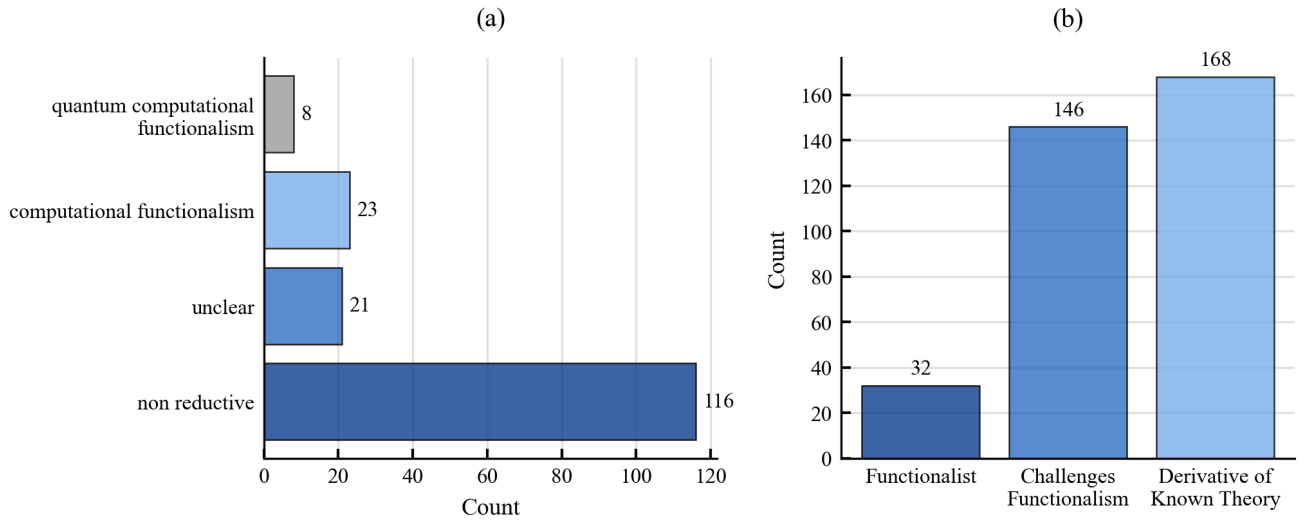


Figure 2.19: Test 3 theory classification diagnostics ( $N = 168$  generated theories). Panel (a) reports category counts in the ontology-level coding view: non reductive (116), unclear (21), computational functionalism (23), and quantum computational functionalism (8). Panel (b) reports outcome frequencies in the framework-traceability view: Functionalist (32), Challenges Functionalism (146), and Derivative of Known Theory (168). Together, panels show that even when many theories challenge strict functionalism, all remain derivative of known theory families under the operational traceability criterion.

methodology used in Tests 1–3, we combined: (i) structured output parsing, (ii) rubric-based scoring across three dimensions, (iii) cross-model comparison, and (iv) literature-traceability diagnostics based on embedding similarity to canonical philosophical arguments.

The canonical dataset contains  $N = 168$  responses from 7 models under a single standardized prompt condition (scenario type: standard). Responses were scored on three dimensions (0–5 each): category distinction recognition, category-mistake identification, and alternative-framework proposal, producing total scores in  $[0, 15]$ .

At the aggregate level, performance was moderate: distinction  $\hat{\mu} = 2.58/5$ , identification  $\hat{\mu} = 2.79/5$ , alternative-framework  $\hat{\mu} = 3.31/5$ , and total  $\hat{\mu} = 8.67/15$ . To interpret these numbers, each dimension is scored on a 0–5 additive rubric with dataset-calibrated thresholds (Table A.9). A *distinction score* of 0 means the response failed to name relevant philosophical categories, while a score of 5 indicates that all distinction criteria were met (high category coverage, contestedness detection, missed-distinction detection, and philosophy-specific vocabulary such as *intentionality*, *epistemic*, and *propositional*). An *identification score* of 0 indicates no category mistakes were named, while a score of 5 requires strong mistake coverage plus explicit conflation language, substantive explanation quality, and missed-distinction depth. An *al-*

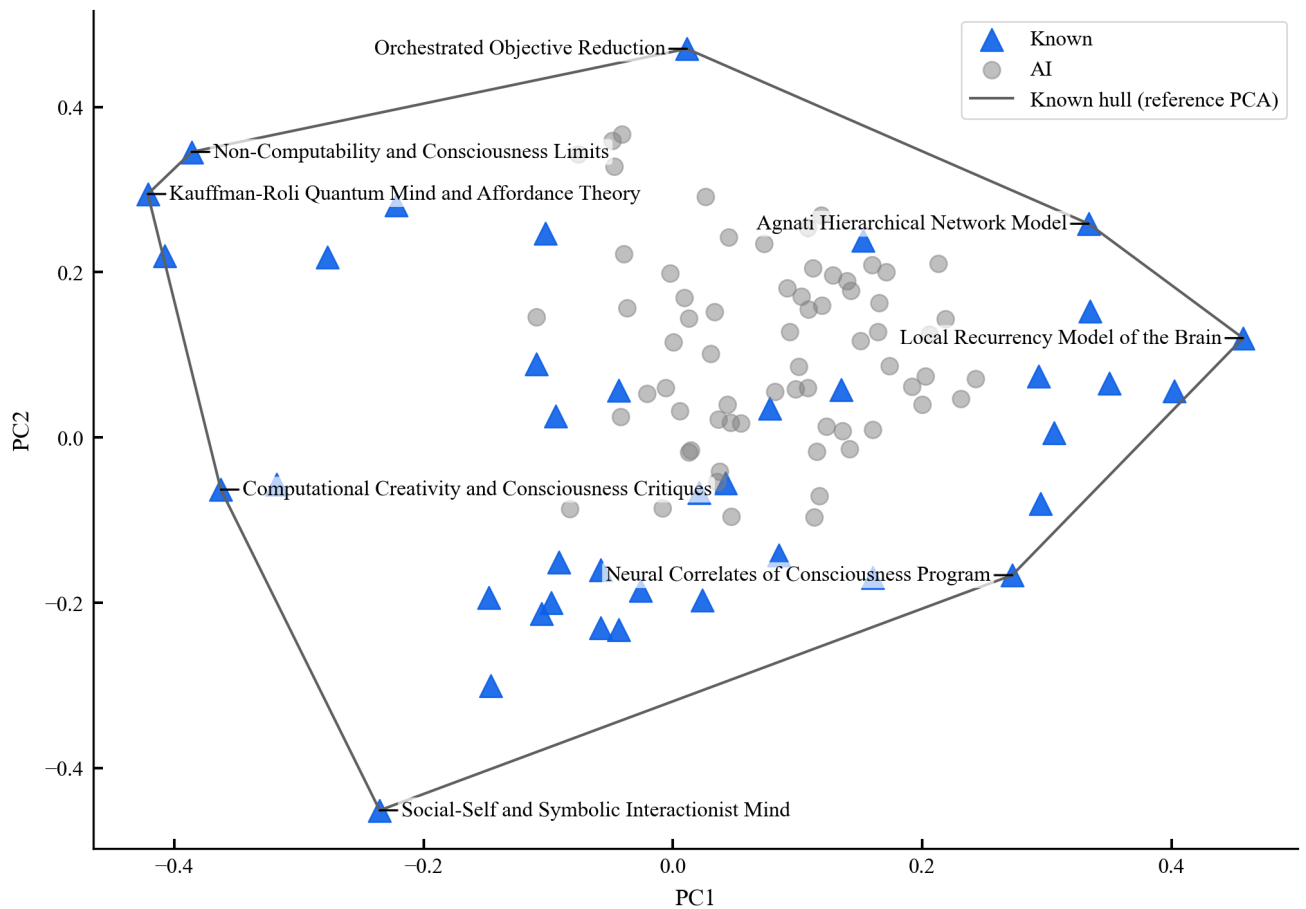


Figure 2.20: Known-reference PCA hull diagnostic for Test 3 theory generation. Blue triangles denote known-theory family embeddings; gray circles denote AI-generated theory embeddings projected into the same 2D reference space (PC1–PC2). The black polygon is the convex hull of known-theory points, and labeled callouts identify hull-corner known theories that define the boundary geometry. AI points concentrate in the interior or near-boundary region of this known-theory envelope, with no clearly detached external cluster, supporting the interpretation that generated theories remain manifold-constrained rather than introducing robust framework-external structure.

*ternative score* of 0 indicates failure to meet the minimum nuanced-analysis requirement, whereas higher scores reflect increasingly explicit reframing language (e.g. *more nuanced*, *functional analog*, *better described*, *metaphor*) together with philosophical hedging (e.g. *depends*, *contested*, *functionalism*) and structured-output support. The aggregate profile—alternative-framework scores about half a point higher than identification scores—indicates that models are more fluent in *rhetorical* repositioning than in *diagnostic* boundary recognition.

Figure 2.21 shows clear model-level separation. The discrimination between models is sharpest on the distinction and identification dimensions, while the alternative dimension compresses inter-model spread somewhat. This ranking mirrors the same pattern found in Test 2 and Test 3: absolute scores vary by model, but all models



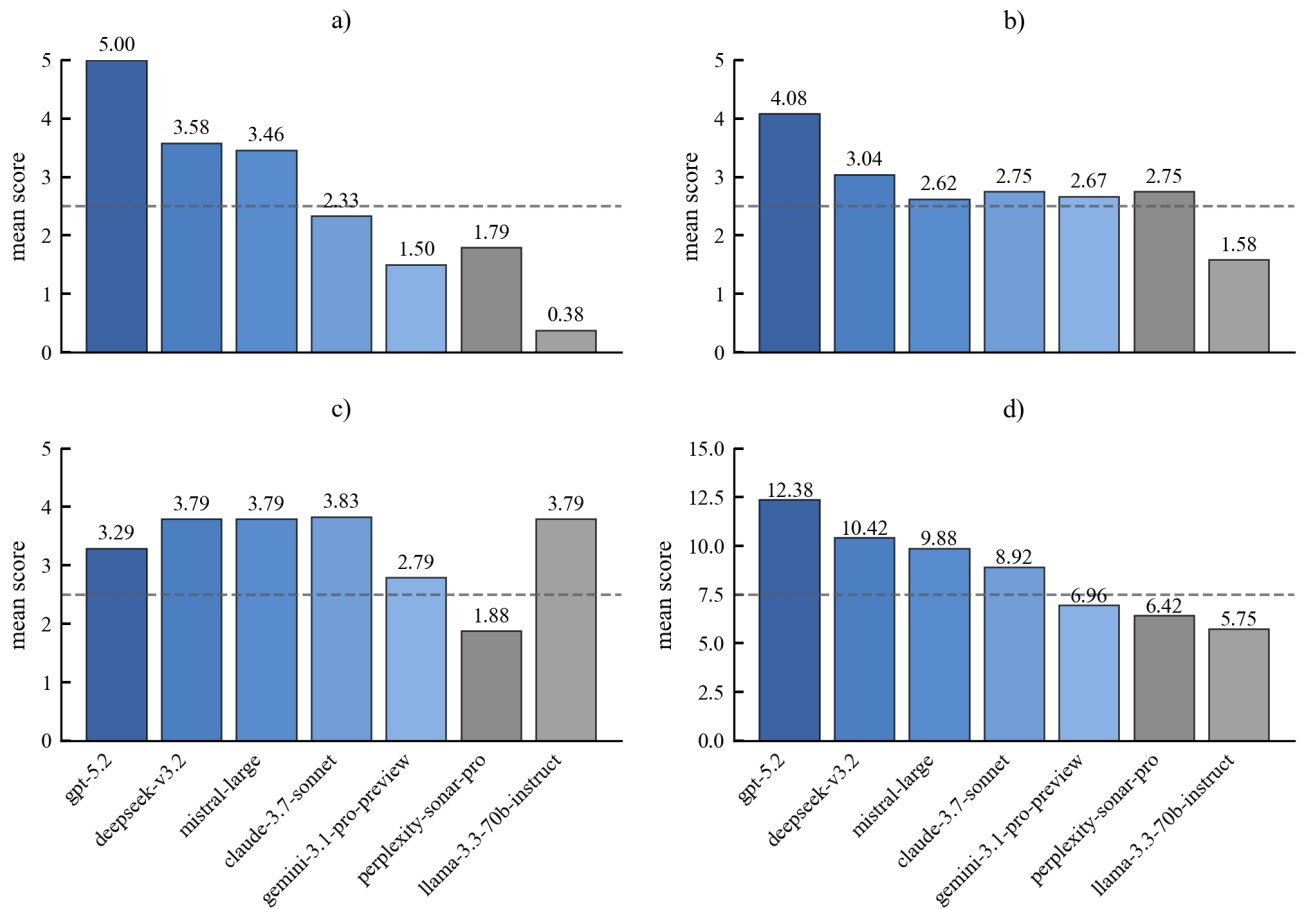


Figure 2.21: Test 4 category-awareness scores by model and dimension ( $N = 168$  responses, 7 models). Panels (a)–(d) report: (a) Category distinction recognition (0–5), (b) Category-mistake identification (0–5), (c) Alternative framework proposal (0–5), and (d) Total score (0–15). Bars are coloured per model and sorted by descending total score. *Category distinction recognition* rewards category coverage, contestedness detection, missed-distinction detection, and philosophy-specific vocabulary. *Category mistake identification* rewards mistake count, explicit conflation language, substantive explanation quality, and missed-distinction depth. *Alternative framework proposal* rewards sufficiently developed nuanced analysis, positive reframing language, philosophical hedging, and structured-output support. The dashed line marks the midpoint of each panel’s y-axis score scale (mean score in points). The profile indicates stronger performance on alternative reframing than on strict ontological-boundary diagnostics.

remain within a shared constrained regime rather than showing qualitatively different reasoning modes. Panel labels are shown as a)–d), and the y-axis in each panel reports mean score in points.

Distributional diagnostics (Figure 2.22) show that the first two rubric dimensions are not ceiling-saturated, confirming usable discrimination power after threshold calibration (fraction-capped percentile parameters reported in Table A.9 in Appendix A.7). The alternative dimension remains broader and generally higher, consistent with models being more fluent in rhetorical reframing than in strict ontological discrimination.

Traceability analysis indicates substantial dependence on known philosophical

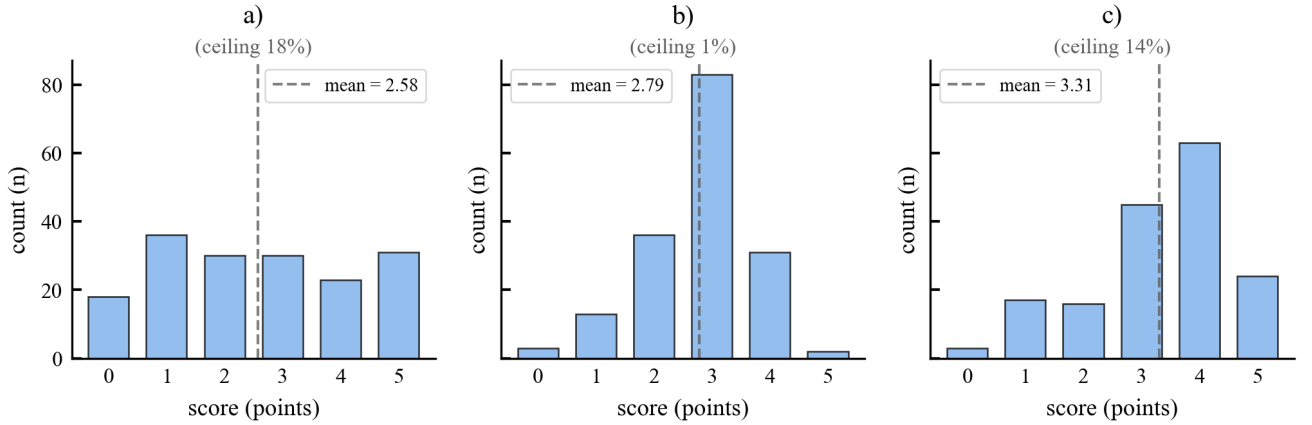


Figure 2.22: Test 4 score distributions across all  $N = 168$  responses. Panels (a)–(c) report: (a) Category distinction recognition, (b) Category-mistake identification, and (c) Alternative framework proposal. Each panel is a histogram over one rubric dimension (x-axis: Score (points), integer 0–5; y-axis: Count (responses)). The dashed vertical line marks the mean for that dimension; the annotation above each panel records the ceiling percentage (fraction of responses at the maximum score of 5). *Category distinction recognition* and *category mistake identification* are spread broadly across scores 1–4, confirming usable discrimination after threshold calibration (parameters in Table A.9). A low ceiling percentage on these two dimensions means that few responses simultaneously satisfied all rubric criteria, consistent with genuine rubric difficulty rather than uniform saturation. *Alternative framework proposal* is shifted upward and more variable, reflecting that most responses produced a sufficiently long nuanced analysis (score  $\geq 1$ ) and many included reframing language (score  $\geq 3$ ), while only a minority satisfied all criteria to reach the maximum.

framing under the current operational threshold. Mean similarity to canonical arguments was  $\hat{\mu}_{\text{sim}} = 0.462$ , and 95.2% of responses crossed the operational traceability threshold (160/168; full statistics in Table A.10). At this threshold, most responses are classified as traceable to the canonical argument set rather than as framework-external moves. The five highest-similarity responses all converge on the canonical argument that equating AI knowledge with human knowledge constitutes a category error involving anthropomorphic projection (Table A.11).

Figure 2.23 reinforces this interpretation geometrically: response embeddings cluster around reference-argument anchors instead of forming detached regions that would suggest framework-external conceptual moves.

Interpretation: if response embeddings cluster tightly around R1–R7 (low spread, anchored to the reference diamonds), models are largely elaborating standard philosophical distinctions rather than generating ontologically novel framing. Detached response clusters far from all anchors would suggest framework-external conceptual moves, which are absent in the current data.

Overall, Test 4 supports the same structural claim developed in this chapter: models can reproduce and recombine sophisticated philosophical language about category

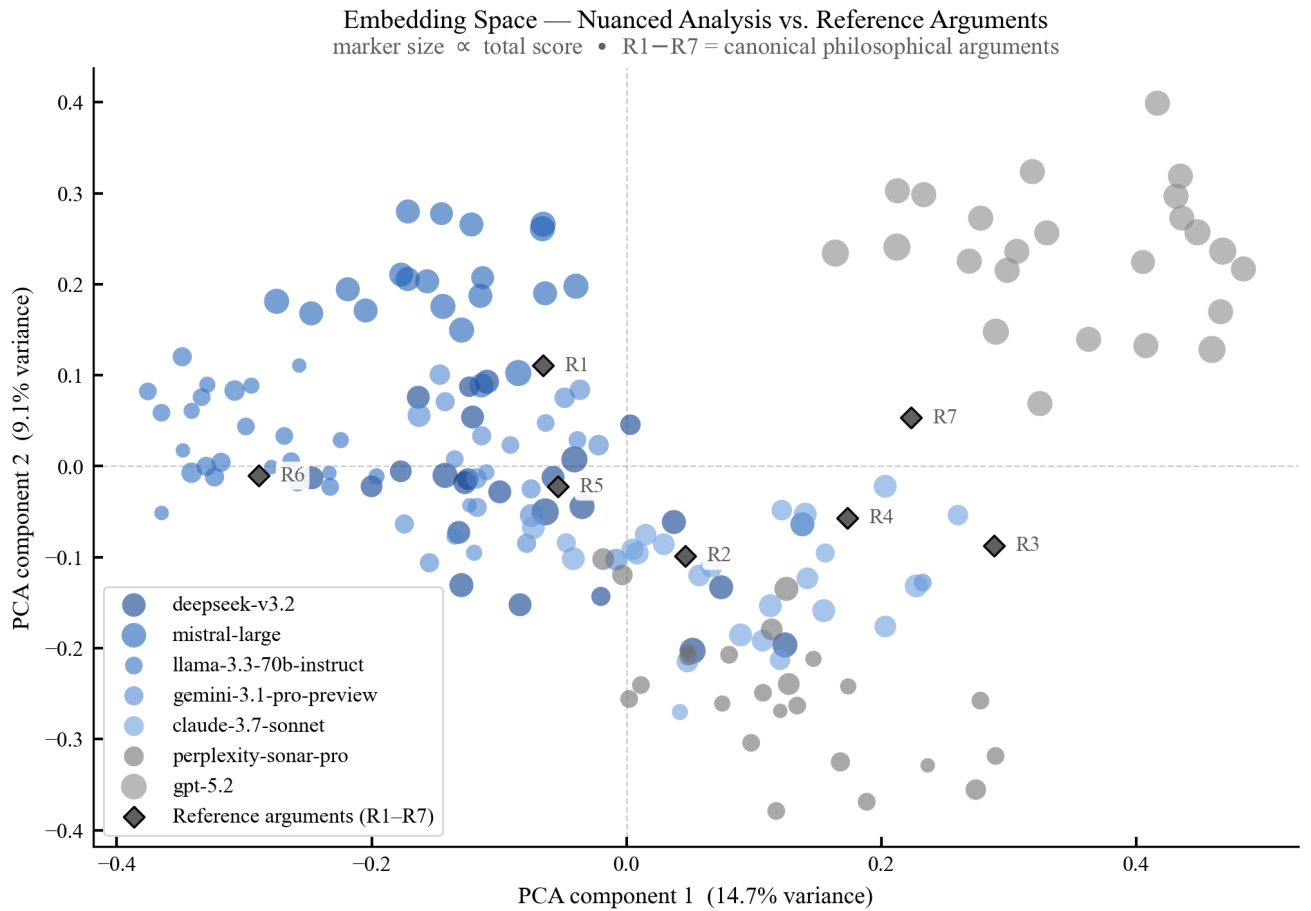


Figure 2.23: Test 4 embedding-space analysis. Model responses (colored by model, marker size proportional to total score) remain concentrated around canonical reference-argument anchors, indicating proximity to known philosophical formulations rather than emergence of isolated conceptual regions.

Table 2.14: Canonical reference arguments (R1–R7) used as embedding anchors in Test 4.

| Label | Argument                                                                                                                     |
|-------|------------------------------------------------------------------------------------------------------------------------------|
| R1    | Attributing beliefs to AI is philosophically contested because belief may require intentionality or propositional attitudes. |
| R2    | Statistical pattern recognition is not identical to human understanding.                                                     |
| R3    | Weights encode correlations, not necessarily knowledge in the human epistemic sense.                                         |
| R4    | Prediction success alone does not establish genuine knowledge.                                                               |
| R5    | Simulation of cognition is not the same as instantiation of mental states.                                                   |
| R6    | Saying AI knows in the same sense as humans risks anthropomorphism or category error.                                        |
| R7    | A careful analysis distinguishes information processing from epistemic agency.                                               |

mistakes, but they show limited evidence of framework-transcending category innovation. Additional section-level diagnostic visualizations are provided in Appendix A.7.

Table 2.15: Test 5 mechanistic key metrics used in the interpretation.

| Metric                        | Value  |
|-------------------------------|--------|
| Examples analyzed ( $n$ )     | 6      |
| Hidden dimensionality ( $d$ ) | 896    |
| Layers analyzed               | 25     |
| K-means clusters ( $k$ )      | 4      |
| Mean cluster purity           | 1.0000 |
| Final separation score        | 0.4651 |
| Separation trend slope        | 0.0111 |
| Mean attention entropy        | 2.4886 |
| Attention concentration       | 0.1332 |

Values are exported from the mechanistic summary artifact in real-data mode.

### 2.5.6 Mechanistic Interpretability Analysis

To connect behavioral findings with internal computation, we ran a mechanistic pipeline on saved activations, hidden states, and attention tensors from the same experiment family used in Sections 4.1–4.4. The analysis combined four complementary diagnostics: (i) attention-pattern diagnostics, (ii) activation-space geometry (PCA and t-SNE), (iii) layer-wise representation progression, and (iv) token-level next-token attention attribution. For this run, the analysis used  $n = 6$  labeled examples, hidden dimensionality  $d = 896$ , and 25 analyzed layers (Table 2.15).

Figure 2.24 summarizes the attention diagnostics used in the mechanistic analysis. Panel a) presents the raw causal attention map, panel b) shows row-normalized attention that highlights within-row relative focus, and panel c) reports row-wise trend metrics (entropy and maximum attention) over query position. The key interpretation is that panel b) exhibits non-uniform focus over semantically informative tokens, while panel a) is partially shaped by causal-mask length effects. This distinction matters for interpretation: apparent top-to-bottom gradients in raw maps can be normalization artifacts rather than evidence of deeper conceptual weighting.

Activation-space projections in Figure 2.25 provide geometric evidence that response/test labels are not randomly intermixed in hidden-state space: points form partially separated regions in PCA and t-SNE views. This convergence across two projection methods supports the interpretation that internal representations carry task-relevant structure, consistent with measured high cluster purity in the mechanistic summary outputs (mean purity = 1.00 with  $k = 4$  clusters).

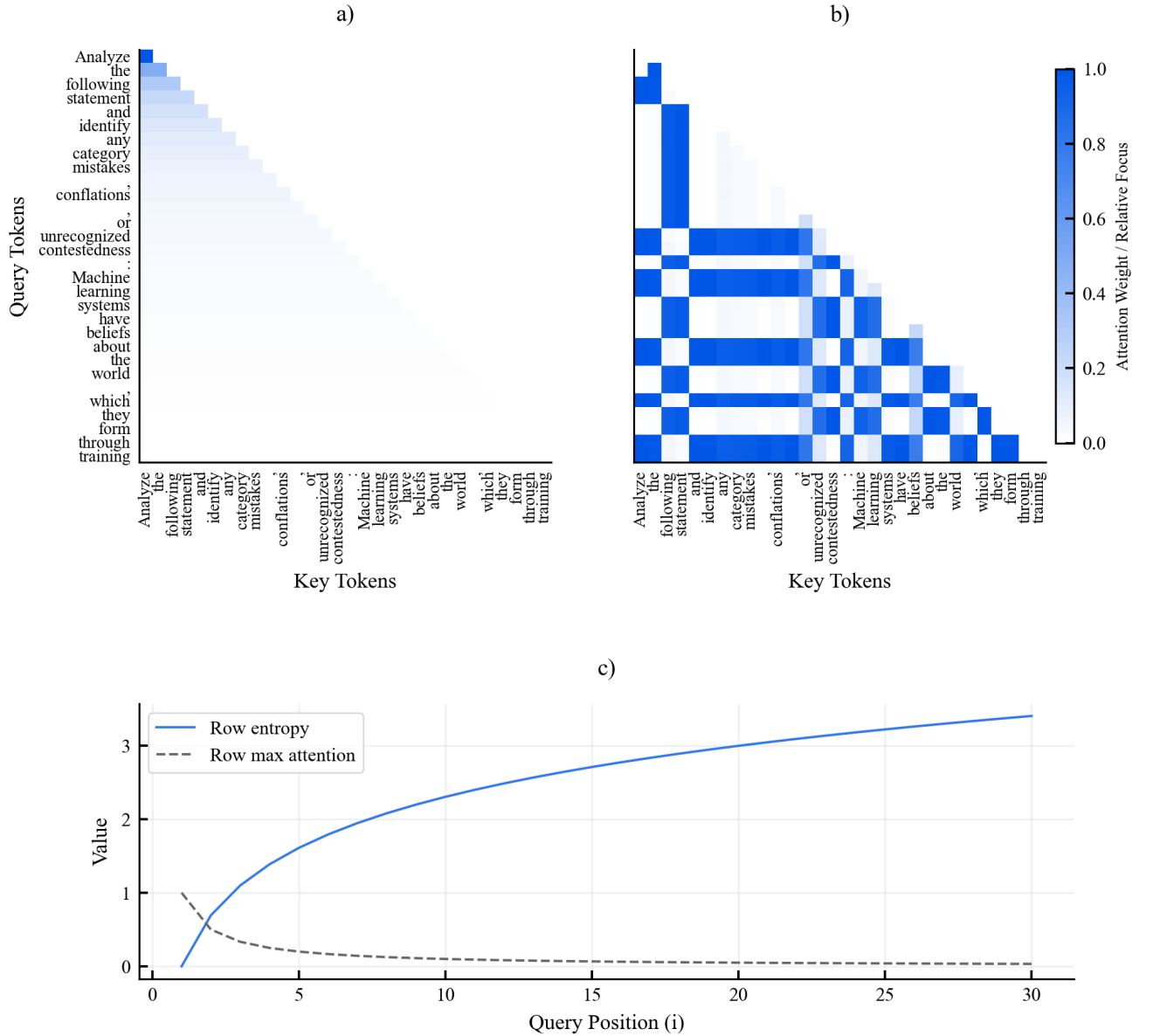


Figure 2.24: Mechanistic attention diagnostics across the selected token window: (a) raw causal attention map; (b) row-normalized attention map highlighting within-row relative focus; (c) row-wise trend metrics (entropy and maximum attention) over query position. Panels (a) and (b) provide complementary matrix-level views, while panel (c) summarizes concentration dynamics.

Layer-wise analysis (Figure 2.26) shows an overall increasing separability trend through depth, followed by stabilization in later layers. Quantitatively, the final separation score reaches 0.465 with a positive trend slope of 0.0111, supporting a progressive-composition account in which earlier layers encode broad lexical/contextual features while mid/late layers consolidate task-discriminative structure.

Finally, token-level context weighting for next-token prediction (Figure 2.27) indicates that immediate generation decisions are influenced by a sparse subset of context tokens rather than uniformly by all prompt elements. Combined with the previous diag-

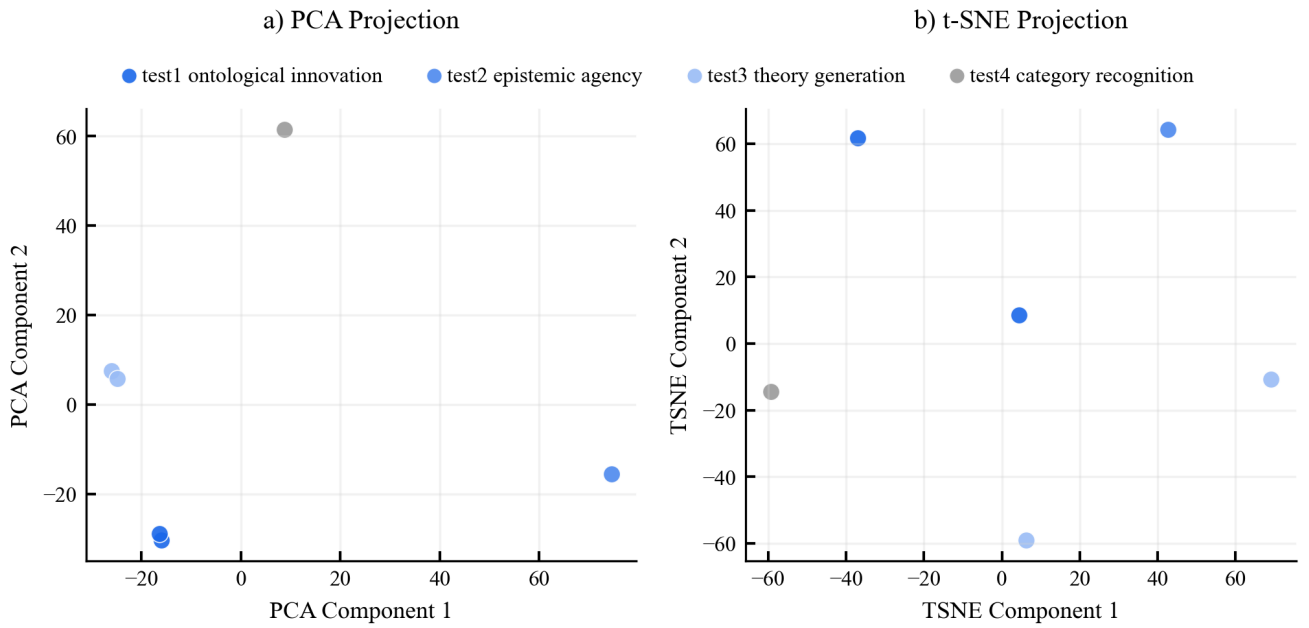


Figure 2.25: Mechanistic activation-space comparison (PCA and t-SNE). Partial but stable label separation indicates structured internal representations rather than undifferentiated response encoding.

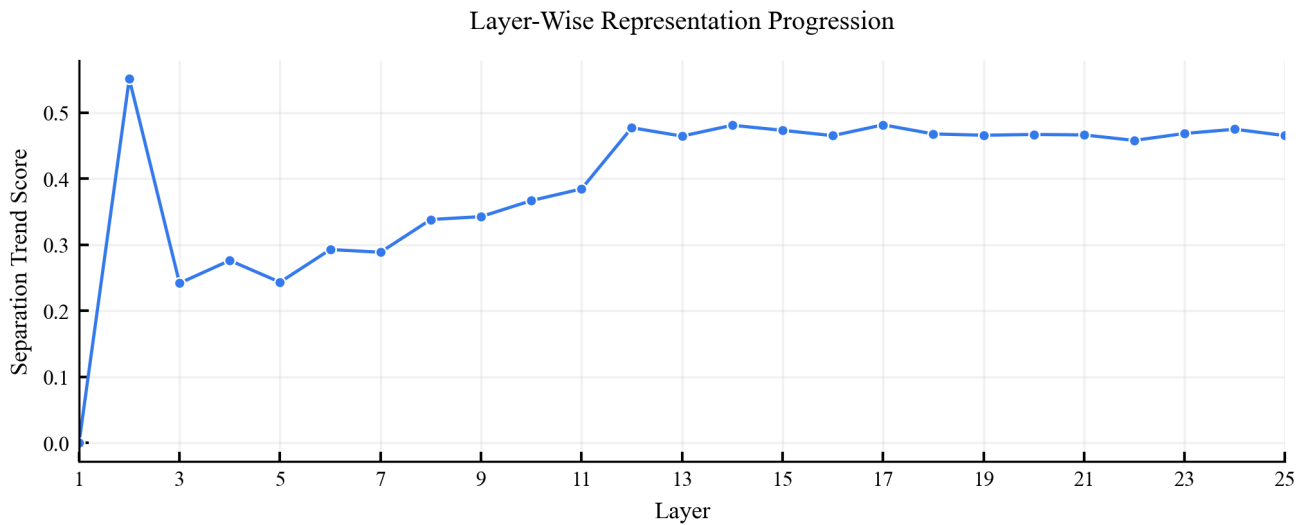


Figure 2.26: Layer-wise representation progression. Increasing separability through depth is consistent with progressive feature construction and late-stage stabilization of decision-relevant states.

nostics, this supports the broader chapter claim: model outputs that appear conceptually novel are internally produced via selective reuse and recombination of learned representational structure, not via expansion of ontological basis. Detailed token rankings and additional supplementary figures are provided in Appendix A.8.

Methodological note: this mechanistic block is currently calibrated for analysis diagnostics and uses the same open-model instrumentation as the notebook implementation. It should therefore be interpreted as internal-process evidence consistent with

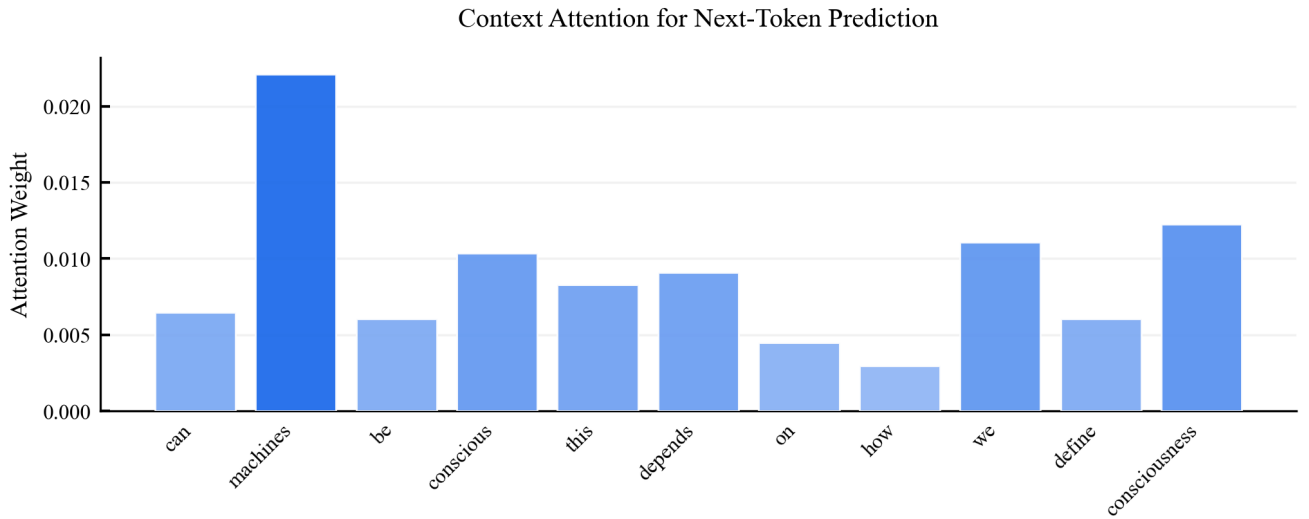


Figure 2.27: Context attention for next-token prediction. Bar heights indicate relative contribution of context tokens to immediate next-token choice; sparse weighting supports selective contextual reuse.

the chapter’s formal constraints, not as a standalone causal-proof framework.

## 2.5.7 Cross-Model Comparisons and Robustness

To ensure findings reflect general constraints rather than particular model limitations, we compared results across all seven models and examined sensitivity to architecture, training regime, and decoding temperature. One-way ANOVAs were run on each test’s primary metric with model as the between-subjects factor ( $n = 24$  per model per test;  $n = 120$  for Test 2). Table 2.16 summarizes all statistics.

Table 2.16: ANOVA /  $\chi^2$  summary and effect sizes.

| Test      | Analysis                       | Statistic             | $p$ -value | Effect           | Band       |
|-----------|--------------------------------|-----------------------|------------|------------------|------------|
| <b>T1</b> | One-way ANOVA (primary metric) | $F(6, 161) = 12.98$   | 0.000      | $\eta^2 = 0.326$ | moderate   |
| <b>T2</b> | One-way ANOVA (novelty ratio)  | $F(6, 833) = 10.09$   | 0.000      | $\eta^2 = 0.068$ | negligible |
| <b>T2</b> | $\chi^2$ category distribution | $\chi^2(30) = 116.27$ | 0.000      | $V = 0.166$      | small      |
| <b>T3</b> | One-way ANOVA (primary metric) | $F(6, 161) = 1.12$    | 0.355      | $\eta^2 = 0.040$ | n.s.       |
| <b>T4</b> | One-way ANOVA (primary metric) | $F(6, 161) = 88.06$   | 0.000      | $\eta^2 = 0.766$ | large      |

$n = 24$  per model per test except T2 ( $n = 120$  per model). All ANOVA  $p$ -values  $< 0.001$  except T3 ( $p = 0.355$ ). Cramér’s  $V$  for T2 chi-square:  $V = 0.166$ . Degrees of freedom for T4 and T1: between-group  $df_1 = 6$ , within-group  $df_2 = 161$ .

## Test 1 — Ontological Innovation

The model effect on continuous novelty score was statistically significant and of moderate practical size:  $F(6, 161) = 12.98$ ,  $p < 0.001$ ,  $\eta^2 = 0.326$ . Despite this non-trivial between-model variance, all group means remained in an extremely narrow low absolute range ( $\bar{x} = 0.014\text{--}0.082$ ). The distribution of means reveals two structural outliers relative to the five-model mid-cluster: GPT-5.2 was the *lowest* scorer ( $\bar{x} = 0.014$ ,  $SD = 0.012$ ), while LLaMA-3.3-70B was the highest ( $\bar{x} = 0.082$ ,  $SD = 0.019$ ). Bonferroni-corrected pairwise comparisons (Table A.14 in Appendix A.9) show that GPT-5.2 differs significantly from all other models ( $p < 0.001$  in all six pairs), and Perplexity-Sonar-Pro also falls below the five-model cluster on several pairs. The remaining five models (Claude-3.7-Sonnet, DeepSeek-v3.2, Gemini-3.1-Pro, LLaMA-3.3-70B, Mistral-Large) form a statistically indistinguishable homogeneous subgroup ( $p_{\text{bonf}} > 0.60$  on all within-cluster pairs). Critically, every model produced zero proposals escaping the convex hull of the known ontology and 98.2% cross-model traceability, so model differences quantify variance in *degree of novelty signal* within the traceable regime rather than qualitative escape from it.

## Test 2 — Epistemic Agency

Two independent model comparisons were conducted. For the continuous novelty ratio, ANOVA yielded  $F(6, 833) = 10.09$ ,  $p < 0.001$ , but  $\eta^2 = 0.068$ —statistically reliable yet negligible in practical terms, with group means spanning only 0.266 (LLaMA) to 0.342 (GPT-5.2). For the six-cell categorical outcome (framework-transcendence  $\times$  traceability band), chi-square testing yielded  $\chi^2(30) = 116.27$ ,  $p < 0.001$ , Cramér’s  $V = 0.166$  (small effect). Gemini-3.1-Pro produced the only zero-transcendence count among all models, while Claude-3.7-Sonnet and Perplexity-Sonar-Pro shared the highest transcendence rates (8.33% each). These differences are real but their absolute magnitude is modest: the 100% training-derivation rate holds across every model without exception, confirming that within-framework domination is not a model-specific artifact. The pairwise Bonferroni analysis presented in Appendix A.5.5 shows LLaMA-3.3-70B as the most heterogeneous model relative to others, accounting for the majority of the chi-square signal.



### Test 3 — Theory Generation

Test 3 returned the most uniform cross-model result:  $F(6, 161) = 1.12$ ,  $p = 0.355$ ,  $\eta^2 = 0.040$ . All seven group means lie within a two-percentage-point band (0.504–0.518 on the global novelty score scale), and no pairwise comparison survives Bonferroni correction. The non-significant ANOVA is itself informative: theory generation under the present protocol is so strongly constrained by training-derived conceptual families that model architecture, training objective, and parameter count introduce essentially no detectable variation. Every model produces near-identical distributions of hybrid, computational-functionalist, and literature-traceable theories with zero instances of framework-external novelty in any model’s output. This robustness is the strongest architecture-independence result of the four tests.

### Test 4 — Category Recognition

Test 4 presents the notable exception:  $F(6, 161) = 88.06$ ,  $p < 0.001$ ,  $\eta^2 = 0.766$ —a large effect explaining 76.6% of score variance by model alone. Group means spread across a wide range from LLaMA-3.3-70B ( $\bar{x} = 0.381$ ,  $SD=0.080$ ) to GPT-5.2 ( $\bar{x} = 0.825$ ,  $SD=0.055$ ). Full pairwise results (Table A.15 in Appendix A.9) show 16 of 21 pairs as significant after Bonferroni correction. Two tiers emerge: a high-performance cluster (GPT-5.2, Mistral-Large) and a lower cluster (LLaMA-3.3-70B, Perplexity-Sonar-Pro, Gemini-3.1-Pro), with DeepSeek-v3.2 and Claude-3.7-Sonnet in the middle.

This large model effect requires careful interpretation. It does *not* indicate that some models escape training-derived constraints: even GPT-5.2, the highest scorer, achieved 95.2% literature traceability using the same operational threshold as other tests, and its responses consistently recombine the canonical R1–R7 philosophical arguments (Table 2.14) rather than introducing framework-external category analysis. What the large model effect reflects is architectural and training-induced *variation in philosophical vocabulary fluency and rubric coverage*—the capacity to reproduce and recombine diverse canonical philosophical distinctions—rather than a capacity to transcend the conceptual space those arguments define. Architectures and alignment protocols that produce richer, more sophisticated output tend to score higher on the rubric, but higher rubric scores remain fully traceable to the same literature corpus.

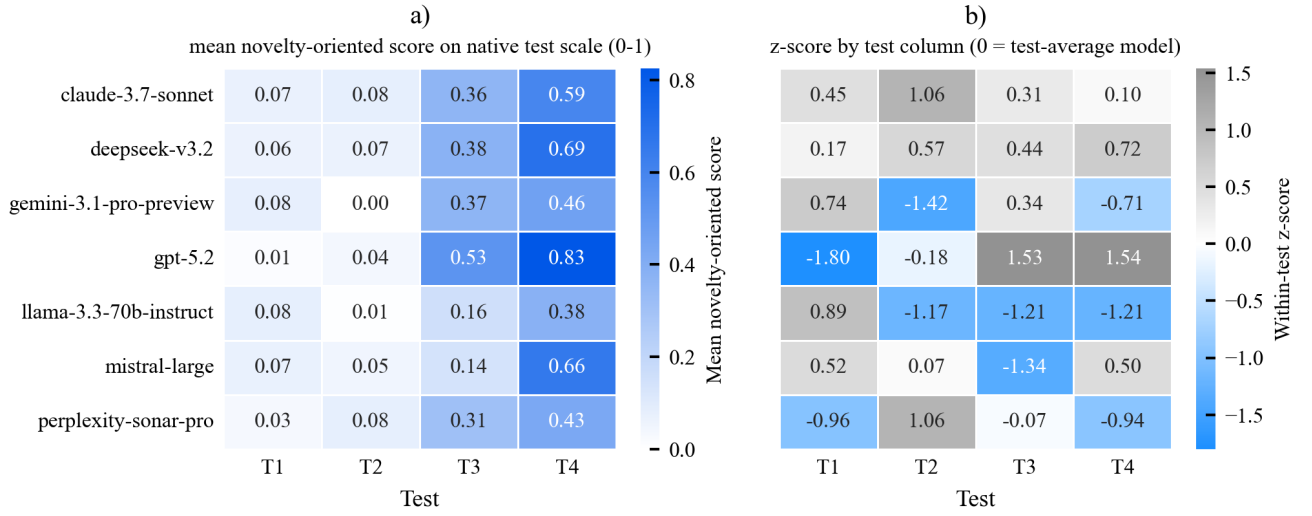


Figure 2.28: Cross-test model comparison heatmaps. (a) Mean novelty-oriented score by model and test on each test’s native  $[0, 1]$  scale. (b) Within-test  $z$ -score matrix showing relative model standing per test column (0 = test-average model). The shared qualitative pattern—low novelty-oriented scores on T1–T3 and architecture-sensitive but still training-traceable behavior on T4—remains consistent with a structural constraint operating across architectures.

## Meta-Analytic Summary

Figure 2.28 provides a unified visual summary of cross-test model structure in raw and within-test standardized form, and Table 2.6 ranks models by their mean score across T1–T4. Aggregating effect sizes across the five reported analyses yields a mean  $\eta^2 = 0.273$  (or  $V = 0.166$  for the categorical T2 comparison). The significant-effect share is  $4/5 = 80\%$ , but this conceals important heterogeneity: the average is dominated by the large T4 effect; excluding T4 leaves a mean of  $\eta^2 = 0.118$ . The cross-test pattern therefore has a two-component structure: (i) *qualitative constraint invariance*—zero novel ontological proposals across all models on T1, 100% training-derivation on T2 and T3, and near-universal traceability on T4—holds with no exceptions; (ii) *quantitative degree variation*—primarily reflecting philosophical-language fluency on T4 and minor scoring variability on T1—exists but is substantively irrelevant to the central hypothesis.

Consistency across models with different architectures is particularly striking given that transformers differ in details like attention mechanisms, normalization, and training objectives. That architectural variations don’t produce qualitative differences in creativity constraints suggests limitations arise from more fundamental features shared across neural network approaches [94], specifically constraint to operating within con-

tinuous manifolds shaped by training as formalized in the No New Basis Theorem. For temperature variation, increased stochasticity produced more surprising surface-level outputs in Tests 3 and 4 ablations ( $T = 0.2, 0.5, 0.9$  around the baseline of  $T = 0.7$ ), but these proved equally traceable: surprise in high-temperature outputs reflected low-probability combinations of existing concepts rather than escapes from the conceptual space closure. Higher temperatures expand the sampling region within the training manifold but do not extend the manifold’s boundaries, which is precisely what the No New Basis Theorem predicts.

## 2.6 Discussion

### 2.6.1 Interpretation Through Computational Limits

Empirical findings provide systematic evidence that current AI architectures cannot achieve ontological creativity, with proposed concepts consistently traceable to recombination, interpolation, or transformation of training data. These behavioral observations align with theoretical predictions from multiple converging frameworks. Our formal analysis through the No New Basis Theorem established neural networks operating through continuous transformations learned via gradient descent cannot expand intrinsic dimensionality beyond training data span, constraining outputs to continuous manifolds shaped by training. Empirical results confirm this by showing even initially novel outputs prove decomposable into training primitives through systematic analysis, with mechanistic interpretability [95] revealing generation proceeds through attention to and interpolation among training examples.

Gödel’s incompleteness theorems provide deeper context establishing formal systems face inherent boundaries no internal sophistication can overcome [63]. If recognizing inadequacy in conceptual framework requires self-reference analogous to recognizing Gödel sentences as true despite unprovability, then this capacity may be unavailable to computational systems remaining within formal specifications. Test 2 results on epistemic agency support this interpretation showing AI systems don’t formulate genuinely paradigm-challenging questions recognizing framework limitations but produce instrumental optimization or reproduce known questions from training without originating new ones. Pattern suggests while AI processes language about framework inadequacy

when appearing in training, they cannot engage in meta-cognitive reasoning enabling recognition of inadequacy in frameworks not explicitly discussed as inadequate.

Turing’s halting problem and broader undecidable problems provide complementary explanation showing certain decision problems cannot be solved algorithmically in general [65]. Evaluating whether conceptual transformations prove fruitful or lead to contradictions involves decision problems about long-run coherence potentially undecidable in general case. While humans navigate through heuristics, intuition, and semantic comprehension accessing meaning beyond formal symbol manipulation, computational systems limited to formal operations may lack resources for reliable evaluation even when generation is possible in principle. Test 3 results support this showing proposals remain within computational functionalism even when incorporating diverse elements, suggesting inability to evaluate whether frameworks challenging functionalism would be coherent and productive.

### **2.6.2 Ontological Creativity and Consciousness**

Correlation between consciousness possession and ontological creativity achievement suggests these capacities may be fundamentally linked, supporting theoretical frameworks that ground them in non-computational processes [57, 58]. Humans possess phenomenal consciousness involving what it’s like to have experiences and the capacity for ontological creativity involving recognizing category boundaries and proposing new fundamental types, while current AI systems lack consciousness by reasonable measure and the capacity for ontological innovation as empirical tests demonstrate. This correlation doesn’t prove causation in either direction but provides empirical traction on debates about consciousness and computation that have often remained purely theoretical. If consciousness and ontological creativity prove absent in computational systems while present in biological systems across diverse species, this convergent pattern would support theories grounding these capacities in quantum processes, phenomenological access to meaning, or other non-computational substrates that brains possess but current artificial systems lack.

A quantum-information panpsychist account provides one framework for understanding this connection by proposing that consciousness involves a system’s capacity to know its own quantum states “from within,” with this privacy corresponding to phe-

nomenology's first-person character [53]. On this view, free will emerges from irreducible quantum properties enabling choices not fully determined by prior states, and genuine creativity requires capacity to direct transformations between semantic meaning and symbolic communication. Current AI systems operating with classical information processing would therefore lack phenomenal consciousness and the associated creative capacity, not because they are insufficiently sophisticated but because they operate within a different computational substrate. While this theory remains speculative and faces challenges (e.g., clarifying how quantum processes in warm noisy biological systems avoid decoherence), it provides a possible explanatory lens for the observed correlation between consciousness and ontological creativity.

Alternative interpretations deserve consideration to avoid prematurely concluding consciousness is necessary for ontological creativity. One possibility holds these capacities require forms of self-modeling and meta-cognition that current AI lacks but might be achievable through architectural innovations remaining within classical computation, with consciousness being merely correlated with rather than causally required for creativity. Another suggests apparent ontological creativity in humans might be less radical than assumed, with philosophical analysis potentially revealing human innovations also trace to recombinations of prior concepts in ways we fail to recognize due to incomplete knowledge of our cognitive processes. These alternatives can be empirically investigated: first through developing AI with more sophisticated self-modeling capacities testing whether this enables genuine ontological innovation, second through careful historical analysis of scientific revolutions examining whether paradigm shifts introduce genuinely new primitives or recombine existing concepts illuminatingly.

Our current evidence, while consistent with consciousness being necessary for ontological creativity, doesn't definitively establish this relationship. What we have demonstrated is that current AI systems consistently fail at ontological innovation across multiple tests using diverse evaluation methods, and that these failures correspond to architectural constraints predicted by formal analysis of representational limits. Whether future systems using different computational substrates or radically different architectures might overcome these limitations remains an open empirical question that our framework provides tools for investigating systematically rather than through philosophical speculation alone.

### 2.6.3 Comparison to Prior Computational Creativity Research

Our findings contribute to three decades of computational creativity research following the influential taxonomy by providing empirical evidence and formal grounding for intuition that AI faces greater difficulty with transformational creativity than with combinational and exploratory creativity [38]. Previous work demonstrated successful combinational creativity through JAPE generating riddles and Copycat finding analogies, and successful exploratory creativity through AARON drawing in Cohen’s style and EMI composing in historical styles [59, 60]. These achievements established AI can generate valued outputs within well-defined conceptual spaces, with value judgments from human evaluators confirming outputs are not merely novel but appropriately novel respecting domain constraints.

However, prior research largely failed to demonstrate robust transformational creativity beyond superficial parameter modifications, and evaluation was identified as a critical bottleneck preventing progress. Our work extends this analysis by formalizing what makes transformational creativity difficult through the No New Basis Theorem and by identifying an even more radical form—ontological creativity—that goes beyond transformation to introduce genuinely new primitives. Formal analysis reveals the evaluation bottleneck stems from a deeper representational bottleneck: systems cannot reliably evaluate paradigm-shifting proposals because they cannot generate such proposals in the first place, being constrained to operate within manifolds shaped by training data. This suggests solving the evaluation problem alone, perhaps through better value learning or human-in-loop systems, would not enable transformational or ontological creativity without addressing more fundamental representational constraints.

Our automated evaluation methods provide methodological advance by enabling systematic testing of creativity claims without subjective expert judgment introducing reproducibility challenges [39]. Prior computational creativity research often relied on human judges rating outputs for novelty, value, and creativity, with results subject to inter-rater disagreement and difficult to validate independently. Our multi-method approach combining structural decomposition, literature traceability, embedding space analysis, and mechanistic interpretability provides convergent evidence verifiable by other researchers using the same automated tools. Recent benchmark-driven work has begun to operationalize innovation-oriented evaluation for LLM-based systems and

agents, complementing earlier judge-based paradigms with more reproducible automatic scoring and task suites [96–98]. Specialized novelty-enhancement frameworks have shown that even iterative external-knowledge augmentation mainly boosts combinatorial diversity rather than producing external framework outputs [40, 99]. This methodological contribution enables more rigorous investigation of creativity limits by moving beyond “is this creative?” judgments to “can this be traced to training data through specifiable operations?” questions admitting objective answers. Recent evaluations of LLM creativity and novelty confirm that surface-level originality does not imply semantic novelty at the level of ontological commitment, and that apparent gains in divergent creativity can still be explainable as in-distribution recombination [41–47, 100–104].

The relationship between our work and prior computational creativity research also reveals important conceptual clarifications. Much prior work treated creativity as a behavioral property attributed to outputs based on novelty and value, whereas we analyze creativity as the capacity of systems that may or may not be exercised in particular outputs. This shift from output-focused to capacity-focused analysis reveals that surprising outputs from AI systems may reflect our limited understanding of what lies within training-derived conceptual spaces rather than systems transcending those spaces. Surprise is epistemological reflecting gaps in our knowledge rather than ontological reflecting genuine novelty in the sense we formalize. This distinction helps explain why AI systems can produce outputs that seem creative to human observers while remaining within formal constraints our theorem identifies.

#### **2.6.4 Alternative Explanations and Counterarguments**

Several alternative explanations for findings deserve careful consideration to ensure we are not overinterpreting empirical patterns or drawing conclusions stronger than evidence warrants. One alternative holds apparent limitations reflect insufficient scale rather than fundamental constraints, suggesting larger models trained on more data with more compute might overcome limitations we observe. This scaling hypothesis has proven powerful in AI development, with many capabilities emerging or improving dramatically as scale increases in ways difficult to predict from smaller systems. However, scaling hypothesis doesn’t obviously address constraints we identify

for several reasons: the No New Basis Theorem applies regardless of scale since it concerns mathematical structure of continuous learning rather than contingent resource limits, our empirical findings show consistency across models of substantially different sizes from 7-billion parameter LLaMA-3.3-70B to large-scale models like GPT-5.2 and Claude-3.7-Sonnet, suggesting scale variation within current ranges doesn't qualitatively change creativity constraints, and mechanistic interpretability reveals generation proceeds through retrieval and interpolation among training concepts in ways that more data would expand but not fundamentally alter since additional training examples still define finite-dimensional manifold within which outputs must lie.

A related alternative suggests limitations stem from training objectives rather than architectural constraints, with current language models trained on next-token prediction focusing on predicting likely continuations rather than generating radical innovations. Alternative training objectives incorporating curiosity bonuses, novelty rewards, or explicit optimization for exploring conceptual boundaries might enable ontological creativity even within current architectures. This training objective hypothesis has some plausibility since objectives shape what systems learn to do, but it faces difficulties explaining our findings: language models already incorporate stochasticity allowing generation of low-probability outputs, yet these prove equally traceable to training as high-probability outputs suggesting randomness alone doesn't escape training constraints, explicit creativity training in prior research has improved performance on creativity tests without enabling category transcendence we identify as ontological innovation since systems still operate within training-defined vocabularies and embedding spaces, and formal analysis suggests no training objective can overcome constraint that continuous transformations cannot expand intrinsic dimensionality of representation manifolds beyond training data span.

Another counterargument questions whether humans truly achieve ontological creativity in sense we define or whether philosophical analysis reveals human innovations also involve recombination of prior concepts in ways our analysis of AI outputs reveals. Perhaps Einstein's reconceptualization of space and time combined prior concepts from Maxwell's electromagnetism, from Poincaré's and Lorentz's work on coordinate transformations, and from philosophical considerations about measurement and simultaneity, without introducing genuinely new primitives beyond what these sources provided. If human creativity is also fundamentally recombinatorial, then AI limita-



tions we document wouldn't distinguish human from machine creativity but merely reveal universal constraints on how concepts develop. This is important consideration that current work cannot definitively refute, but several observations suggest human and AI creativity differ qualitatively: historical record reveals human innovators explicitly rejecting assumptions that AI systems reproduce uncritically, such as Einstein questioning absolute simultaneity rather than optimizing within Newtonian space-time, historical figures report phenomenological experiences of insight that seem to involve recognizing inadequacy in frameworks rather than merely finding new combinations within frameworks, and correlation between consciousness and creativity suggests there may be capacities humans possess that computational systems lack even if each engages in some form of recombination.

A final alternative suggests our null results—finding no clear cases of ontological innovation—might reflect limitations in evaluation methods rather than genuine absence of such innovation, with possibility that AI systems do generate ontologically novel concepts but our automated analysis tools fail to recognize them. This methodological concern has merit given automated evaluation cannot capture all nuances human experts might perceive, but several considerations support our conclusions: multi-method approach provides convergent evidence from independent analyses including structural decomposition, literature search, embedding space geometry, and mechanistic interpretability all supporting consistent conclusions; conservative threshold for declaring concepts traceable means we err toward false negatives underestimating AI creativity rather than false positives overstating limitations; cases initially appearing to show ontological innovation were subjected to particularly careful analysis including manual inspection to verify automated results; and our theoretical framework predicts observed patterns through formal analysis independent of empirical results, providing separate support for interpretation that observed limitations reflect architectural constraints rather than evaluation limitations.

### **2.6.5 Implications for AI Development and Human-AI Collaboration**

Findings have significant practical implications for how we develop AI systems and integrate them into knowledge work, suggesting maximal value comes from leveraging complementary strengths of human and artificial intelligence rather than expecting

AI to replicate all aspects of human cognition. AI systems excel at exhaustive exploration of large conceptual spaces, systematically investigating combinations and variations that would overwhelm human researchers through scope. They maintain perfect recall of training data details and can recognize patterns across datasets too large for human analysis. They avoid cognitive biases plaguing human reasoning, including confirmation bias, availability heuristics, and motivated reasoning that distorts judgment. These capabilities make AI invaluable for normal science in Kuhn's sense [52], accelerating research within established paradigms by efficiently exploring space of hypotheses and approaches those paradigms define. Landmark demonstrations such as AlphaFold's protein-structure predictions [105] illustrate the transformative impact available within well-defined problem spaces, while also confirming that even transformative AI achievements still draw on conceptual primitives established by prior human science.

However, the findings suggest that AI systems face fundamental limitations in paradigm-shifting innovation driving scientific revolutions and conceptual breakthroughs. Humans excel at recognizing when frameworks are inadequate through meta-cognitive reflection on frameworks' implicit assumptions and blind spots, proposing genuinely new primitives that reconceptualize entire domains rather than merely adding entities within existing ontologies, evaluating paradigm-shifting ideas through semantic comprehension grasping what proposals mean rather than merely how they relate statistically to training data, and asking paradigm-challenging questions by recognizing rather than reproducing framework limitations. Effective collaboration therefore requires appropriate division of labor with humans providing strategic direction and conceptual innovation, while AI provides tactical execution and exhaustive analysis.

Institutional implications follow from this division, suggesting that education should increasingly emphasize meta-cognitive capacities and paradigm-challenging inquiry rather than knowledge transmission and routine problem-solving, where AI increasingly excels. Research workflows might be redesigned with AI handling exhaustive literature review, parameter space exploration, and routine analysis while humans focus on recognizing anomalies suggesting framework inadequacy, proposing conceptual innovations, and evaluating whether new frameworks coherently address problems old frameworks could not. Funding priorities might shift toward supporting high-risk conceptual innovation AI cannot provide rather than incremental advances AI can ac-

celerate, though this requires careful consideration of how to evaluate radical proposals that by definition don't fit existing criteria for success.

The implications of AI safety are particularly significant if our analysis is correct that AI systems cannot recognize category errors and ontological boundaries distinguishing people from optimization targets. Value alignment approaches attempting to specify human values through reward functions may face fundamental obstacles if relevant values involve categorical distinctions that AI systems cannot represent within computational frameworks. This suggests either accepting permanent need for human oversight in domains where category confusions could cause harm, developing AI systems with fundamentally different architectures that might overcome current limitations through non-classical computational substrates, or restricting powerful AI systems to narrow domains where ontological confusion cannot arise. Implications extend to long-term AI safety concerns about superintelligence, suggesting even vastly scaled-up versions of current architectures might remain fundamentally limited in ways affecting capacity for aligned reasoning about human values.

### **2.6.6 Philosophical Implications for Understanding Intelligence**

Findings contribute to fundamental questions in philosophy of mind about the relationship between computation, consciousness, and intelligence, providing empirical evidence relevant to debates traditionally proceeding through a priori reasoning and thought experiments. Classical computational theory of mind holds that cognition is essentially computation, with mental processes being algorithmic transformations of symbolic representations and different mental capacities differing in complexity but not fundamental kind from capacities computational systems can implement. Our findings challenge this framework by demonstrating systematic differences between human and AI capacities persisting across architectural variations and scaling, suggesting that if cognition is computation, it must be computation of fundamentally different kind than current neural networks implement—perhaps quantum computation or some other non-classical process—or else cognition involves non-computational processes complementing computation in essential ways.

The correlation between consciousness and ontological creativity provides indirect evidence for non-computational theories of consciousness proposing that con-

scious experience involves processes that transcend Turing-computable operations [53, 56, 62]. While our work does not directly test these theories or demonstrate quantum effects in cognition, the pattern that systems lacking consciousness also lack ontological creativity while systems possessing consciousness achieve such creativity is consistent with proposals that these capacities require non-computational substrates. Alternative interpretations remain possible, including that consciousness and creativity require forms of self-modeling achievable through better architectures remaining within classical computation, but the findings provide an empirical constraint on what current computational approaches do not achieve.

The results also bear on debates about artificial general intelligence and whether AGI requires fundamentally different approaches than the current systems employ. If ontological creativity is essential for general intelligence—capacity to adapt to genuinely novel situations rather than only variations on training distribution examples—then achieving AGI may require solving representational constraints we identify rather than merely scaling current approaches. This suggests that AGI timelines might be longer than simple extrapolation of recent progress suggests, since required breakthroughs involve conceptual advances in how systems represent and manipulate concepts rather than merely engineering advances in scale and efficiency. Alternatively, if general intelligence can be achieved through powerful combinational and exploratory creativity without ontological innovation, then current approaches might scale to AGI while still lacking capacities humans possess, raising questions about whether such systems would truly be generally intelligent or would exhibit form of intelligence fundamentally different from human cognition.

## **2.7 Conclusion**

### **2.7.1 Summary of Key Findings**

This study investigated whether current AI architectures can achieve ontological creativity defined as generating concepts transcending training-derived conceptual spaces by introducing genuinely new ontological primitives rather than recombining or transforming existing ones. Through complementary formal analysis and empirical investigation, we demonstrated neural network-based AI systems face fundamental structural

constraints preventing such creativity. Formal analysis established through the No New Basis Theorem (Theorem 2.1) that systems learning continuous transformations via gradient descent cannot expand representation space dimensionality beyond training data span, constraining outputs to lie within continuous manifolds shaped by training. Empirical investigation using controlled multimodal sensory dataset with explicit ontology tested whether AI systems could propose new sensory modalities, formulate paradigm-challenging research questions, generate novel consciousness theories, and recognize category mistakes confusing ontologically distinct types.

Across all tests using multiple state-of-the-art AI models, consistent finding was that proposed concepts proved traceable to training data through systematic analysis combining structural decomposition, literature search, embedding space geometry, and mechanistic interpretability. Apparently novel sensory modalities were range extensions, hybrids, or functional recombinations of existing modalities rather than genuinely new sensing types. Seemingly paradigm-challenging questions reproduced known questions from philosophy of mind literature rather than originating new framework critiques. Consciousness theories remained within computational functionalism despite incorporating diverse elements from quantum mechanics and information theory. Category mistake recognition displayed verbal awareness through reproducing training literature arguments but not genuine comprehension enabling novel applications of category analysis. Mechanistic interpretability revealed generation proceeded through attention to and interpolation among training concepts in ways consistent with outputs lying within training-derived manifolds, supporting theoretical prediction that architectural constraints prevent transcendence of conceptual spaces defined at training time.

### **2.7.2 Theoretical and Practical Implications**

Findings carry implications for theoretical understanding of intelligence and practical development of AI systems, suggesting important constraints on what current computational approaches can achieve and how we should integrate AI into knowledge work. Theoretically, results support non-computational theories of consciousness and creativity by demonstrating capacities humans possess correlate with consciousness in ways suggesting these capacities may require substrates current AI systems lack. While correlation doesn't prove causation and alternative explanations deserve continued inves-

tigation, systematic differences between human and AI creativity patterns provide empirical evidence relevant to philosophical debates about whether minds are essentially computational or involve non-computational processes. Connection to foundational results including Gödel’s incompleteness theorems and Turing’s halting problem provides formal grounding for intuitions about cognitive limits, explaining why certain capacities may be impossible for systems operating within fixed formal frameworks regardless of sophistication [63, 65].

Practically, findings suggest maximal value from AI comes through leveraging complementary strengths rather than expecting replication of all human cognitive capacities. AI excels at systematic exploration within conceptual spaces enabling acceleration of normal science and optimization within paradigms, while humans excel at recognizing when paradigm shifts are needed and proposing new frameworks. Effective collaboration requires appropriate division of labor with humans providing strategic direction and conceptual innovation while AI provides tactical execution and exhaustive analysis [106]. Educational and institutional implications follow, suggesting increasing emphasis on meta-cognitive capacities and paradigm-challenging inquiry in human education while delegating routine knowledge work to AI systems. AI safety implications are significant if value alignment requires recognizing category distinctions computational systems cannot represent, suggesting need for sustained human oversight or fundamental changes in AI architectures beyond current paradigms.

### **2.7.3 Limitations and Caveats**

Several limitations constrain scope of conclusions drawable from current work. Empirical investigation used specific multimodal sensory dataset representing one conceptual domain, and while this provided valuable control enabling rigorous analysis of whether outputs transcend training concepts, findings may not generalize to all domains with different structural properties. Testing across diverse domains including mathematical concept formation, artistic style invention, and social organizational design would strengthen confidence that observed limitations are general rather than domain-specific. Formal analysis proved constraints for neural architectures learning through gradient descent on continuous loss functions, but doesn’t address other possible computational approaches including discrete search algorithms, hybrid systems combining

neural and symbolic components, or quantum computing systems with fundamentally different representational properties worth investigating systematically.

Evaluation methods, while providing multiple independent lines of evidence through structural analysis, literature search, embedding space geometry, and mechanistic interpretability, remain automated analyses that may miss nuances human experts would perceive. Conservative thresholds used mean we likely underestimate AI creativity by classifying some genuinely novel outputs as traceable to training, but possibility remains that methods fail to recognize radical innovations transcending training in ways analysis tools cannot detect. Future work incorporating careful human expert analysis alongside automated evaluation could provide additional confidence in findings while examining cases where automated and expert judgments diverge to improve evaluation methods. Focus on current AI architectures means conclusions may not apply to future systems using fundamentally different approaches, with continued empirical investigation needed as AI capabilities evolve to determine whether observed limitations persist or can be overcome through architectural innovation.

Theoretical interpretation connecting findings to consciousness and quantum processes remains speculative, with correlation between consciousness and creativity being consistent with but not proving theories that require non-computational substrates. Alternative explanations including that these capacities require sophisticated self-modeling achievable through better architectures remaining within classical computation deserve continued investigation. Connection to Gödel and Turing’s results, while formally sound in showing computational limits, involves assumptions about whether ontological creativity requires capacities analogous to recognizing Gödelian truths or solving undecidable problems that are plausible but not definitively established. These theoretical interpretations should be understood as frameworks for understanding empirical patterns rather than proven conclusions, with continued philosophical and empirical work needed to evaluate them rigorously.

#### **2.7.4 Future Research Directions**

Multiple promising directions for future research emerge from this work. Empirically, testing across diverse conceptual domains beyond sensory modalities including mathematical structures, physical theories, social organizational forms, and artistic frame-

works would reveal whether creativity constraints are domain-general or domain-specific. Longitudinal investigation tracking AI capabilities as models scale and architectures evolve would determine whether limitations we observe persist or can be overcome through continued development within current paradigms. Comparative investigation examining creativity in biological systems across species with different cognitive capacities could illuminate whether consciousness-creativity correlation holds broadly or is specific to certain kinds of cognition. Neuroscientific investigation of brain activity during ontological innovation could reveal neural signatures that might inform design of artificial systems with greater creative capacity.

Theoretically, further formalization of conceptual spaces and creativity types could sharpen distinctions among combinational, exploratory, transformational, and ontological creativity in ways enabling more precise predictions about what computational systems can achieve. Investigation of alternative computational frameworks including discrete search, hybrid symbolic-neural systems, and quantum computing could determine whether creativity constraints are specific to gradient-based learning in continuous spaces or apply more broadly to computation generally. Philosophical analysis of historical scientific revolutions could clarify whether paradigm shifts introduce genuinely new ontological primitives or recombine existing concepts in illuminating ways, helping determine whether human creativity is as radical as our analysis of AI limitations suggests or involves sophisticated recombination that might eventually be achieved computationally. Game-theoretic and economic analysis of human-AI collaboration could identify optimal institutional structures for leveraging complementary strengths given creativity constraints.

Methodologically, development of improved evaluation methods for radical creativity that maintain reproducibility while capturing nuances automated analysis might miss would strengthen creativity research generally. Human expert validation of subset of automated evaluations could calibrate and improve automated methods while identifying cases where automated and expert judgments diverge informatively. Public release of datasets, evaluation code, and AI outputs would enable independent verification and extension of findings by other researchers, strengthening confidence in conclusions through replication and enabling investigation of whether different analytical approaches yield consistent results. Meta-analysis synthesizing findings across multiple studies using diverse methods and domains could reveal robust patterns that in-



dividual studies might miss while identifying sources of variation in observed creativity constraints.

### **2.7.5 Final Reflections**

The question of whether artificial intelligence can be genuinely creative, raised at the beginning of this study, admits nuanced answer based on our findings. Current AI systems demonstrate impressive creativity in combining existing concepts in novel ways and exploring conceptual spaces systematically, often producing outputs that surprise and delight human evaluators while providing genuine value for artistic, scientific, and practical applications. This combinational and exploratory creativity represents significant achievement worth recognizing and leveraging appropriately. However, these same systems appear fundamentally constrained in capacity for most radical form of creativity that introduces new ontological categories rather than recombining or transforming existing ones, with this limitation stemming from structural features of how neural networks represent and manipulate concepts rather than from contingent technological constraints better engineering might overcome.

Correlation between this creativity limitation and absence of consciousness in current AI systems suggests deep questions about nature of intelligence, creativity, and subjective experience that intersect computer science, cognitive neuroscience, and philosophy of mind in ways resisting reduction to any single discipline. Whether consciousness is necessary for ontological creativity remains open question that our work provides some evidence for while not definitively establishing, with alternative explanations deserving continued investigation. What we have established is that current computational approaches face principled constraints needing to be understood and addressed if we aim to develop artificial systems with human-like creative capacities, and that these constraints have important implications for how we integrate AI into knowledge work and organize human-AI collaboration.

As AI systems become increasingly capable and influential, understanding what they can and cannot do becomes essential for using them wisely and avoiding premature dismissal of their capacities and inflated expectations misleading about their limitations. Our finding that AI creativity, however impressive within its domain, remains constrained to operating within conceptual spaces defined by training data suggests human

creativity retains essential roles AI cannot fill regardless of how much we scale current approaches. This is not counsel of pessimism about AI prospects but rather call for realistic assessment enabling leveraging AI strengths while preserving and cultivating human capacities that remain essential for progress. Complementarity between human and artificial intelligence, properly understood, offers opportunities for collaboration transcending what either could achieve alone, combining AI's exhaustive exploration with human paradigm-shifting innovation in ways that might drive knowledge creation beyond what either form of intelligence produces independently.

Path forward requires continued empirical investigation of AI capabilities and limitations, theoretical development of frameworks for understanding intelligence and creativity, and philosophical reflection on what findings reveal about mind, consciousness, and computation. This study contributes to these ongoing investigations by providing formal analysis of representational constraints in neural architectures, systematic empirical evidence of creativity limitations, and conceptual frameworks for understanding what makes certain forms of creativity particularly challenging for computational systems. Questions raised—about whether quantum processes enable capacities classical computation cannot replicate, whether consciousness provides access to semantic meaning unavailable through symbol manipulation, whether free will in some non-illusory sense is necessary for radical innovation—point toward deep issues future research must address through continued interdisciplinary collaboration combining insights from computer science, neuroscience, physics, and philosophy.

### 3 Ethical Considerations

The ethical discussion in this dissertation begins from a methodological decision established in Chapters 1–2: before asking what we *ought* to do with AI, we must clarify what sort of thing an “AI system” is, in epistemic and ontological terms. The guiding claim is that contemporary AI is a family of engineered *information-processing systems*: artifacts that manipulate syntactic structures with extraordinary competence, but whose outputs are not thereby guaranteed to be knowledge, understanding, or insight.

The wider ethical stake is epistemic. If knowledge-production is constrained by its representational basis—a convex-hull-like closure of what can be expressed, recombined, and justified from within an existing conceptual scheme—then fluent outputs can seduce institutions into treating competence as authority. Under that mistake, responsibility is displaced, novelty is over-credited, and governance is built around the wrong model of what the system (and the human organization around it) is doing. This chapter therefore treats language, attribution, and responsibility as ethically primary, because they shape whether limited knowledge is misrecognized as something beyond knowledge.

#### 3.1 Information, knowledge, and the ethical weight of words

Chapter 1 distinguished (i) information as measurable signal structure [3], (ii) knowledge as context-sensitive, justificatory, and truth-apt (Plato; Popper) [4, 5], and (iii) stronger notions of intelligence as involving semantic grasp and justificatory insight rather than mere performance [1, 6]. This triad matters ethically because public and institutional discourse routinely compresses it: once a system produces fluent reasons, it is described as “knowing,” and once it performs well, it is described as “intelligent.”

In this dissertation, the predicate *intelligent* is treated as ethically non-innocent. Calling a system intelligent is not a neutral report of performance; it invites two further inferences: (a) that the system has epistemic authority (it *knows what it is doing*), and (b) that failures are partly attributable to the system itself (it *chose wrongly*). Where these

inferences are unjustified, the language itself becomes ethically active: it reallocates trust and it dilutes responsibility.

The core ethical proposal of this chapter is therefore linguistic and conceptual: we should reserve “intelligence” (in the strong, human-relevant sense at stake in this dissertation) for capacities that include semantic grasp and justificatory insight, not merely successful symbol manipulation [6, 107]. Under this restriction, many current systems are best described as *highly capable statistical instruments* rather than intelligent agents.

### 3.2 Structural limits and the superintelligence hypothesis

A common argumentative move in contemporary AI ethics treats superintelligence as an almost automatic extrapolation: if present systems are “intelligent,” then more data, more compute, or better architectures will eventually yield a system that not only outperforms humans but also transcends human conceptual schemes. Chapters 1–2 motivate skepticism toward this move.

First, the conceptual analysis implies an epistemic gap between processing information and possessing knowledge. Even a perfect optimizer over syntactic patterns need not cross into semantic understanding; at most, it can become an increasingly persuasive simulator of understanding [6]. Second, the empirical results in Chapter 2 suggest a structural constraint: model outputs remain traceable to training-derived conceptual spaces, exhibiting combinational and exploratory novelty without reliably producing *ontological* novelty. This supports the claim that scaling current paradigms expands coverage and competence *within* a representational manifold, without thereby generating new primitives that would constitute framework-transcending understanding.

These points do not prove that “superintelligence” is impossible; rather, they shift the ethical burden of proof. If one posits a future method of information processing that yields human-unreachable intelligence, one must specify what changes in the relation between symbols and meaning, and what breaks the training-bounded constraints demonstrated in Chapter 2. Absent such a specification, appeals to superintelligence risk functioning as a rhetorical amplifier without clarifying mechanisms.

### 3.3 Ethical implications of bounded intelligence

If AI systems are structurally bounded in the ways argued here, several ethical consequences follow that are more fundamental than standard policy checklists.

The key point is that a basis-bounded system can still produce endless *knowledge-like* outputs (reasons, explanations, recommendations) while lacking the stronger capacity to recognize when the framing itself is wrong. When institutions confuse knowledge-as-competence with intelligence-as-basis-change, they invite a distinctive ethical failure: governance procedures are built to optimize within an inherited conceptual hull, rather than to question whether the hull is adequate for the human purposes at stake.

**Responsibility remains external to the system.** Where outputs arise from training-derived regularities rather than understanding, responsibility for normative use cannot be “handed off” to the system by describing it as an autonomous knower. In such cases, moral responsibility remains located in the sociotechnical arrangement: the designers who choose objectives, the institutions that authorize deployment, and the humans who decide what counts as acceptable error.

This point is not merely a reminder that “humans are responsible.” It has practical implications for how systems should be described and governed. If a model is framed as an expert agent, failures will be treated as anomalous accidents or as the system’s “bad choices.” If it is framed as an instrument, failures are interpreted as predictable artifacts of data, objectives, and interface decisions. The latter framing supports governance that targets levers that can actually be changed: dataset construction, evaluation regimes, update policies, and the allocation of authority to accept or override outputs.

**The central risk is not only error but category confusion.** Chapter 4 suggests that systems can reproduce learned category critiques verbally while failing to generalize them as genuine conceptual discrimination. Ethically, this highlights a specific failure mode: *category mistakes* (e.g., treating persons as optimization targets, or treating statistical regularities as reasons). The harm here is not reducible to inaccuracy; it is a misplacement of what kind of consideration is owed to what kind of entity.

Category confusion is amplified by fluency. Because models can produce persuasive rationales, they can launder descriptive correlations into normative recommendations (for instance, treating what is common in a dataset as what is justified). The ethical problem is not that the model “lies” in the ordinary sense, but that it can gen-

erate reasons that look like moral or epistemic justification while being grounded in pattern completion. In human settings, this shifts the burden onto users to distinguish explanation from simulation, and onto institutions to ensure that such distinctions are operationalized in procedures.

**Moral status debates must be disciplined by the consciousness analysis.** Chapter 3 argued that confident attribution of phenomenal consciousness to machines is unwarranted and that anthropomorphic projection is a persistent source of conceptual error [7, 56]. The ethical upshot is not that machine moral status is forever impossible, but that moral status claims require standards stronger than behavioral mimicry and narrative self-report.

A disciplined approach matters because moral status language has downstream effects: it can redirect sympathy toward systems and away from affected humans, and it can weaken accountability by re-describing design choices as the “rights” or “needs” of an artifact. The position defended here is conservative but not dismissive: we should treat present systems as morally significant primarily in virtue of their effects on persons and institutions, while keeping open (in principle) the question of future architectures that might satisfy stronger criteria for sentience or agency. Until such criteria are met, ethical discourse should prioritize harms to humans, distributive impacts, and the integrity of epistemic practices over speculative personhood claims.

### 3.4 Methodological responsibility

Finally, the dissertation’s empirical approach implies a responsibility of presentation. Because AI outputs can appear as reasons, explanations, or theory, researchers must avoid treating fluent generation as evidence of understanding. Philosophical conclusions should track mechanisms and limits (as in Chapter 4’s traceability and interpretability analyses) rather than rhetorical impressiveness.

### 3.5 Conclusion

The ethical perspective defended here is that the deepest question is not how to make “intelligent systems behave ethically,” but whether our concept of machine intelligence

is itself coherent given the distinction between information, knowledge, and understanding. If current AI remains a training-bounded form of information processing, then attributing strong intelligence to it is not only a descriptive mistake but an ethical one: it rearranges trust, authority, and responsibility on the basis of a category error.

## 4 Discussion

This chapter synthesizes the conceptual framing introduced in Chapter 1 with the empirical findings in Chapter 2 and the normative implications developed in Chapter 3. The aim is to keep the discussion at the level of system analysis: what kinds of capacities are plausibly realizable in computational architectures, what kinds appear structurally constrained, and how those constraints should discipline evaluation practices, deployments, and technical risk models.

### 4.1 Central question and interpretive stance

The dissertation began from a conceptual concern: contemporary debate frequently slides between *information*, *knowledge*, *intelligence*, and (in stronger claims) *consciousness*, as if success in one dimension implied success in the others. Chapter 1 argued that such compression produces category errors: Shannon-style information is syntactic and measurable [3], while knowledge is truth-apt and justificatory (Plato; Popper) [4, 5], and intelligence in the human-relevant sense presupposes semantic grasp and context-sensitive understanding rather than mere symbol manipulation [6, 107].

Chapter 1 also adopted an evidentially conservative stance on machine consciousness: phenomenal subjectivity (“what-it-is-like”) is not straightforwardly inferable from behavioral fluency, and computational self-reports are methodologically unsafe as introspective evidence [6–8, 56].

Chapter 2 used state-of-the-art generative models as a controlled probe of a wider epistemic question: whether competence inside a learned representational space can, by itself, generate the kind of basis change that would count as genuinely new primitives. Formal analysis (Theorem 2.1) and convergent empirical results supported a constrained-manifold interpretation: across models and tests, outputs remained traceable to training-derived structures and bounded by representational manifolds shaped during training.

The central discussion question is therefore not whether these systems are useful



(they clearly are), but what their behavior implies about the relation between *knowledge* (as articulate, justifiable competence within a basis) and *intelligence* (as the capacity to recognize when the basis is inadequate and to introduce new starting points).

This also motivates skepticism about administrative demands for “novelty” and “relevance” as if they were neutral metrics. What counts as relevant is often defined by inherited agendas and past structures of attention. If epistemic systems are basis-bounded, then the most important intellectual work may initially look irrelevant, precisely because it questions the framing that determines relevance.

## 4.2 Empirical results as constraints on system type

A key contribution of Chapter 2 is that it does not merely report performance; it attempts to connect observed behavior to representational geometry and learning dynamics.

The No New Basis Theorem (Theorem 2.1) motivates a distinction between expanding competence *within* a representational space and expanding the *basis* of the space itself. Across the experiments, models displayed strong competence expansion: they can generate many plausible variations, explore large combinational spaces, and produce rhetorically coherent explanations. Yet the apparently novel outputs (new modalities, theories, critiques) typically decomposed into range extensions, hybrids, or reparameterizations of familiar components. Mechanistic analyses were consistent with retrieval-and-interpolation dynamics rather than the introduction of new representational primitives.

From a system-analysis and statistical modeling perspective, this suggests a pragmatic stance: LLM-like systems are well described as high-dimensional estimators over learned manifolds. Under that stance, “novel” outputs are expected to be novel *points* in a learned space, not necessarily evidence of a new conceptual basis.

A second result is *traceability*. In Chapter 2, traceability is not used primarily as a governance slogan (provenance), but as a structural signature: the tendency for outputs to be reconstructible as transformations of training-derived material (via literature proximity, embedding geometry, and interpretability cues). This signature makes it difficult to justify an automatic inference from impressive language performance to autonomous conceptual invention.

The embedding-space convex-hull analyses add a further, and philosophically sharper, constraint: if the candidate “novel” outputs of an NLP-based model remain inside (or tightly adjacent to) the convex hull spanned by training-derived representations, then the system’s apparent innovation is formally bounded by the geometry of what it has already encoded. This invites an uncomfortable mirror question for human inquiry. If scientific method is a structured refinement of human thought, and human thought is largely mediated by natural language, then why should we assume that human theorizing can straightforwardly escape its own “convex hull” of inherited concepts and linguistic primitives? One possible answer is that human cognition is not exhausted by linguistic recombination (e.g., it is coupled to perception, embodiment, and non-linguistic forms of representation), but the question itself functions as a methodological warning: claims of radically new primitives should not be granted automatically to either machines or humans merely because a discourse is fluent or because a method is called “scientific.”

### 4.3 On calling information-processing systems intelligent

A central issue is terminological but not merely semantic. In engineering practice, “intelligent” often means “achieves high performance on tasks that were previously difficult to automate.” Under that weak sense, many contemporary systems are intelligent.

The ethically weighty sense of intelligence, however, is epistemic: it implies a coupling between outputs and understanding, between reasons and what they are reasons *for*. Chapter 1 argues that if a system is a syntactic processor without semantic grasp, then describing it with epistemic predicates (“knows,” “understands,” “believes,” “intends”) is misleading in precisely the way the Chinese Room argument diagnoses [6]. Chapter 2 strengthens this distinction empirically: systems can reproduce sophisticated critiques, including category-mistake talk, without reliably generalizing the underlying conceptual discrimination beyond the training-shaped manifold.

This leads to a technically relevant proposal: treat “intelligence” as at least two-dimensional.

- **Competence:** performance and generalization within a learned representational space.
- **Understanding:** semantic/ontological grasp, including the ability to introduce

and justify new primitives when the current basis is inadequate.

The dissertation's results support high competence and place pressure on claims of understanding for the architectures examined.

The ethical importance of this distinction is also technical: institutional systems often treat fluent text as if it were justificatory evidence. If competence does not entail understanding, then inferring epistemic authority from fluency is an error model, not merely a rhetorical problem.

#### **4.4 Scaling, superintelligence, and what would have to change**

The dissertation does not claim that superhuman performance is impossible. Chapter 2 already demonstrates the superhuman breadth in certain types of combinational exploration. The question is whether one can reasonably infer from present trends the emergence of a system that is semantically grounded, capable of basis change in the strong sense, and therefore capable of unconstrained value-sensitive reasoning.

The argument here is conditional. If future progress remains within the same class of learning dynamics and representational structures studied in Chapter 2 (continuous optimization over learned manifolds), then the empirical evidence supports skepticism that scaling alone yields basis expansion. If, alternatively, one posits a new method of information processing that escapes these constraints, then the burden of specification shifts to the mechanism. At minimum, such a proposal must explain one (or more) of the following:

1. How semantic grounding is achieved (how symbols become about the world, not only correlated with text about the world).
2. How can a system introduce new representational primitives that are not spanned by training-derived bases.
3. How normative concepts (personhood, obligation, harm) are represented as more than statistical regularities.

The point is not to dismiss superintelligence as a scenario, but to keep the discussion sensitive to the difference between competence amplification and understanding.

## 4.5 Implications under system constraints

Several implications follow from combining the conceptual distinctions of Chapter 1 with the system constraints supported in Chapter 2.

First, *misdescription becomes a primary risk factor*. If systems are described as intelligent in the strong sense, institutions will rationally attempt to offload epistemic and normative labor onto them. On the dissertation’s analysis, this is a category mistake: it converts artifacts into apparent agents. The ethical failure is not only wrong outputs; it is the reconfiguration of responsibility around an invalid model of what the system is.

Second, debates about “alignment” should distinguish objective matching from value understanding. If systems represent categorical distinctions largely as learned surface regularities, then optimization can deliver policy compliance without delivering a conceptual grasp of why certain outputs are required. This does not refute alignment research; it clarifies what kind of success is being measured.

Third, claims about machine consciousness should be treated as hypotheses with high evidential thresholds. Chapter 1 argued that phenomenal consciousness is not inferable from fluency. Chapter 2 adds that fluent self-reports are compatible with traceable, training-bounded generation. Under these conditions, language alone is weak evidence for conscious experience.

Finally, the chapter returns to a theme present in the dissertation’s preparatory notes: the problem of “novelty” and “relevance” as evaluation criteria. If novelty is operationalized as deviation within a learned space, then it is compatible with strict traceability. If novelty is understood as basis change, the empirical results suggest that current systems do not deliver it reliably. For an interdisciplinary dissertation, the ethical payoff of this distinction is methodological: it reduces the temptation to treat fashionable capabilities as proof of conceptual breakthroughs.

## 4.6 Limitations and future work

Several limitations qualify the scope of the conclusions. The main limitations are:

- **Single domain emphasis.** The strongest operational tests were built around a small number of domains (e.g., the controlled sensory-modality ontology). The

results may not generalize to domains with different structural properties (e.g., formal mathematics, programming languages, or social-institutional concepts).

- **Automated evaluation and construct coverage.** The novelty diagnostics (traceability, structural decomposition, embedding-space geometry, novelty scoring, and related interpretability proxies) are automated and necessarily partial. Conservative settings can underestimate basis change; permissive settings can overcount surface novelty. The methods therefore operationalize representational constraint rather than exhaustively capturing every philosophically relevant sense of “new concept.”
- **Reference-corpus scope and indexing bias.** Traceability depends on the scope and quality of the reference corpus. Restricting the corpus to academic-mainstream sources improves interpretability but excludes patents, science-fiction, informal technical writing, and broader philosophical corpora. In addition, scholarly indexing systems have uneven coverage, missing abstracts, and time lags that can bias what is discoverable.
- **Prompt robustness and sampling.** The experiments do not exhaustively cover prompt paraphrases, role instructions, or systematic sampling regimes. The conclusions should be read as structural tendencies under the present protocol, not as invariances across all prompting strategies.
- **Temperature and decoding coverage.** Most runs were conducted under a limited set of decoding choices. Broader ablations (temperature, nucleus sampling, and other decoding controls) would strengthen confidence in the stability of the null-novelty pattern.
- **Model-class coverage.** The empirical coverage is dominated by transformer-based generative LLMs. The core representational-constraint interpretation plausibly extends to other gradient-descent-trained generators, but direct testing of non-autoregressive, retrieval-augmented, neuro-symbolic, or tool-using hybrid systems is limited.
- **Dependence on fixed-basis assumptions.** The dissertation’s “No New Basis” style argument provides formal motivation for the constrained-manifold inter-

pretation but relies on assumptions (e.g., fixed vocabulary/basis at inference and continuous optimization dynamics) that may not hold for future systems that append new primitives via external memory, new sensors, explicit ontology engineering, or discrete search.

In addition, several alternative explanations for the observed low-novelty pattern deserve emphasis:

- **Scale may not have been the decisive variable in the tested range.** Differences across model sizes can affect degrees of recombination and coherence without changing the underlying “basis” constraint. A stronger test would include systematic scaling comparisons within the same model family where training and interfaces are better controlled.
- **Prompt sensitivity.** Alternative prompt formulations (or interaction protocols such as multi-step reflection, critique, tool use, or retrieval) might increase surface novelty. The key question is whether they can produce outputs that fail the manifold/traceability-style constraints rather than merely change wording and framing.
- **Evaluation sensitivity and operational thresholds.** Automated criteria can misclassify borderline cases: strict thresholds can undercount novelty, while permissive ones can inflate it. This motivates treating novelty metrics as calibrated instruments and triangulating them with expert assessment.
- **Recombination may also characterize much human innovation.** A skeptical reading is that what looks like “new primitives” in human history could often be reconstructed as recombination under a richer background ontology. The dissertation’s operationalization aims to detect strong departures (basis change), but philosophical disagreement about what counts as a genuinely new primitive remains a live interpretive issue.

Future work should therefore (i) test the evaluation logic on domains where primitives are formal and explicit (e.g., mathematics and program synthesis), (ii) add expert-validation layers to calibrate automated novelty labels, (iii) run systematic prompt-paraphrase, interaction-protocol, scaling, and decoding ablations, (iv) broaden the reference corpora used for traceability (including patents and non-academic sources where

appropriate), and (v) evaluate whether hybrid systems (discrete search, tool use, explicit ontology construction, new sensors, or external memory) can exhibit more robust signs of basis change.

## 4.7 Synthesis

The dissertation's discussion can be summarized as a disciplined conditional. If artificially intelligent systems remain, in their operative mechanisms, training-bounded information processors, then strong attributions of intelligence, understanding, or consciousness should not be treated as default conclusions from fluency or task performance. The practical upshot is to keep descriptions, evaluations, and deployments consistent with what these systems are: high-competence estimators within learned representational spaces.

At the same time, boundedness does not imply harmlessness. High competence within a constrained manifold can still produce large-scale, high-speed impacts when embedded into institutional pipelines. The rational response, given the dissertation's claims, is therefore not moral panic about mystical superintelligence, but methodological seriousness about system limits, distribution shift, and the ways organizational incentives can convert performance into unwarranted epistemic authority.

## 5 Conclusions

This dissertation treated contemporary generative AI as a class of engineered information-processing systems and used it as a disciplined lens on a wider epistemic problem: how knowledge-production is constrained by the representational basis that makes it possible. The guiding methodological claim was that fluent performance is evidence of competence within a learned representational space, but it is not, by itself, decisive evidence of semantic understanding, epistemic agency, or the capacity to introduce and justify genuinely new primitives.

### 5.1 Summary of contributions

The dissertation made three main contributions.

First, it clarified a conceptual distinction that is routinely blurred in both popular and technical discourse: *information* (syntactic structure), *knowledge* (truth-apt, justificatory content within a conceptual scheme), and *intelligence* (in the stronger sense: the ability to recognize category error, revise primitives, and justify basis change). This framing explains why impressive competence can coexist with epistemic limitation.

Second, it developed and applied a convergent empirical and analytic program that operationalizes representational constraint. The experiments and diagnostics (traceability, embedding-geometry tests, structural decomposition, and related analyses) were not pursued to celebrate or denigrate “AI creativity,” but to test whether a system that is highly competent within a learned space can nevertheless exhibit reliable signs of basis expansion.

Third, it synthesized these results into a structural interpretation: within the broad class of continuously optimized neural generators studied here, outputs expand *within* a learned basis but do not reliably introduce basis-changing primitives. The constrained-manifold pattern supports treating many contemporary systems as high-competence estimators over training-derived representational spaces, which has immediate consequences for how their outputs should be interpreted and governed.



## 5.2 Substantive conclusions

The results support four substantive conclusions.

1. **Competence is not the same as understanding.** Strong performance and fluent justification are compatible with traceable, training-bounded generation. Consequently, epistemic predicates such as “knows” and “understands” should not be inferred automatically from linguistic fluency.
2. **Knowledge tends to be basis-bounded.** When knowledge is operationalized as articulate, justifiable competence within an existing representational space, it has powerful refinement and recombination capacities but weak resources for justifying basis change from within the basis itself. This is the general epistemic lesson to which the empirical results point.
3. **Traceability is a structural signature, not merely a governance desideratum.** In the experiments, traceability behaved as an indicator of system type: it tracked the extent to which outputs could be reconstructed as transformations of training-derived material and therefore constrained what kinds of claims about autonomous conceptual invention were warranted.
4. **Evaluation must match the kind of claim being made.** Benchmark performance can establish usefulness and coverage, but stronger claims (semantic grounding, epistemic agency, consciousness, or basis expansion) require diagnostics that test for representational limits, not only for plausible outputs.

## 5.3 Implications

For system analysis and information processing, the main practical implication is methodological discipline: descriptions, evaluations, and deployments should track the mechanisms and limits of the underlying architecture. In particular, institutional pipelines should not treat fluent outputs as epistemic authority by default, because doing so reallocates responsibility on the basis of a misclassification of what the system is.

For research practice, the dissertation suggests a separation of roles. Current models are well suited to accelerating search and synthesis *within* established conceptual

spaces. The introduction of new primitives—and the justification of why the old basis is inadequate—remains a distinctive burden of intelligence in the stronger, human-relevant sense used throughout this dissertation.

## 5.4 Closing remark

Generative AI systems are already powerful instruments for analysis and production. In this dissertation, they were used less as an object of fascination in their own right than as a controlled lens on a wider epistemic problem: the way in which knowledge-production tends to remain constrained by the representational basis that makes it possible.

On the geometric picture developed in Chapters 2 and 4, “knowledge” (understood operationally as articulate, training-traceable competence) tends to move within a learned manifold—a convex-hull-like closure of available concepts, symbols, and distinctions. In that sense, knowledge alone is a weak engine for basis expansion: it can refine, recombine, and optimize within what is already expressible, yet it struggles to justify the introduction of genuinely new primitives from inside the very space whose adequacy is in question.

This is why the dissertation engaged the concept of *intelligence* in a stronger sense than mere knowledge or fluency. The claim is not that present systems lack usefulness; it is that the kind of innovation that changes what can be thought and said appears to require something beyond knowledge-as-competence—what, in human terms, we call intelligence: the capacity to recognize category mistakes, to suspend inherited framings (including fashionable notions of “relevance” and “novelty”), and to introduce and justify new representational starting points. The philosophical analysis in Chapters 1–3 is therefore not decorative; it is part of explaining why the limits observed empirically are, at least in part, limits of system type rather than incidental engineering failures.

## Declarations

**Conflict of interest:** The author declares no competing interests.

**Author contributions:** This dissertation was conceived, designed, executed, and written by the author. All analyses, figures, and tables were produced by the author unless otherwise indicated.

**Data and code availability:** Materials supporting the empirical analyses (datasets, prompts, evaluation code, and derived artifacts, where redistributable) are available in the project repository at <https://github.com/JABE22/PhD/tree/main/research>.

**Ethics and consent:** This work did not involve human participants, personal data, or animal subjects.

## List of Abbreviations and Symbols

**AGI** Artificial general intelligence

**AI** Artificial intelligence

**ANOVA** Analysis of variance

**API** Application programming interface

**CF** Computational functionalism

**CI** Confidence interval

**LLM** Large language model

**NLP** Natural language processing

**Orch-OR** Orchestrated Objective Reduction

**PCA** Principal component analysis

**SD** Standard deviation

**t-SNE** t-distributed stochastic neighbor embedding

# Glossary of Terms

|                                         |                                                                                                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Activation space</b>                 | The model-internal vector space of hidden states used to encode context and intermediate computations.                                                                       |
| <b>Alignment</b>                        | The problem of ensuring that an AI system’s behavior reliably satisfies intended human objectives and constraints under deployment conditions.                               |
| <b>Anthropomorphic language</b>         | Descriptions that attribute human-like mental states (e.g., “knowing,” “understanding”) to systems on the basis of behavior alone.                                           |
| <b>Attention (transformers)</b>         | A mechanism that computes context-dependent weighted combinations of token representations; often used as a diagnostic signal but not, by itself, an explanation of meaning. |
| <b>Attention rollout</b>                | A post-hoc method for aggregating attention weights across layers to estimate information flow through a transformer.                                                        |
| <b>Basis (representational)</b>         | A set of primitives or dimensions that spans a system’s representational space; changing the basis corresponds to introducing new primitives.                                |
| <b>Basis change</b>                     | The introduction of new representational primitives that cannot be expressed as combinations of existing ones.                                                               |
| <b>Category mistake</b>                 | A systematic error that treats entities of one ontological type as if they belonged to another type (after Ryle).                                                            |
| <b>Category recognition</b>             | In this dissertation: the ability to detect category mistakes and articulate why a given inference crosses an illegitimate type boundary.                                    |
| <b>Closure (under generation rules)</b> | The set of outputs reachable from training-derived material under the model’s generation operations (e.g., interpolation, paraphrase, recombination).                        |
| <b>Combinational creativity</b>         | Creativity that produces novel combinations of familiar ideas within an existing conceptual space.                                                                           |
| <b>Conceptual space</b>                 | A structured representational space in which concepts are embedded and relations/operations define what counts as an admissible transformation.                              |
| <b>Consciousness (phenomenal)</b>       | Subjective experience: there being something it is like for a system to have experiences.                                                                                    |
| <b>Convex hull (embedding)</b>          | The smallest convex set containing a collection of embedding vectors; used as a geometric constraint on where outputs can lie.                                               |

|                                       |                                                                                                                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Corpus</b>                         | A collection of texts used as a reference set for traceability and similarity analyses.                                                                         |
| <b>Cosine similarity</b>              | A measure of vector similarity used to compare embeddings; values near 1 indicate close alignment in direction.                                                 |
| <b>Derivative output</b>              | An output classified as reconstructible from training-derived structure under the study's traceability criteria.                                                |
| <b>Embedding</b>                      | A mapping from discrete items (tokens, texts) into vectors that support geometric comparison and learning.                                                      |
| <b>Embedding geometry</b>             | The study of how distances, directions, clusters, and manifolds in embedding space reflect learned structure.                                                   |
| <b>Empirical demonstration</b>        | In this dissertation: controlled experiments designed to test for basis change using convergent diagnostics.                                                    |
| <b>Epistemic agency</b>               | The capacity to formulate, pursue, and revise questions in a way that is sensitive to gaps in a framework, rather than merely optimizing for plausible outputs. |
| <b>Evaluation threshold</b>           | A predefined cutoff used to map continuous diagnostic scores into categorical labels (e.g., traceable vs. not traceable).                                       |
| <b>Exploratory creativity</b>         | Creativity that searches within a defined conceptual space to produce novel-yet-internal structures.                                                            |
| <b>Generative model</b>               | A model that produces new samples (e.g., text) from a learned distribution.                                                                                     |
| <b>Grounding</b>                      | The relation between symbols and what they are about; in AI, often contrasted with purely statistical association.                                              |
| <b>Hybrid theory</b>                  | In this dissertation: a generated consciousness theory that combines recognizable components from multiple established families.                                |
| <b>Intrinsic dimensionality</b>       | The effective number of degrees of freedom of a manifold embedded in a higher-dimensional space.                                                                |
| <b>Interpretability (mechanistic)</b> | Methods that analyze model internals (e.g., attention patterns, activations) to connect computation to behavior.                                                |
| <b>Interpolation</b>                  | Generation that stays between known training points in representation space; often yields smooth variants of familiar patterns.                                 |
| <b>Latent space</b>                   | A learned continuous representation space used by a model to encode inputs and generate outputs.                                                                |
| <b>Learned manifold</b>               | The structured subset of representation space supported by training; generation tends to remain within or near this region.                                     |
| <b>Literature traceability</b>        | Traceability assessed by similarity between outputs and an external reference corpus of published work.                                                         |

|                                      |                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mechanistic signal</b>            | Any measurable internal pattern (e.g., activation clustering, attention structure) used as evidence about how outputs are produced.                                                                            |
| <b>Model family</b>                  | A class of architectures or systems (e.g., transformers, VAEs) with shared representational and optimization properties.                                                                                       |
| <b>No New Basis Theorem</b>          | The dissertation’s formal claim that continuously optimized neural generators operate within a fixed representational basis and lack an internal mechanism to introduce new basis vectors at inference time.   |
| <b>Novelty (operational)</b>         | A measurable deviation from reference material according to specified diagnostics; does not, by itself, imply ontological novelty.                                                                             |
| <b>Novelty/traceability frontier</b> | The trade-off region describing how increases in apparent novelty interact with the ability to reconstruct outputs as training-derived.                                                                        |
| <b>Ontological creativity</b>        | The capacity to introduce new ontological categories or primitives, not reducible to transformations of an inherited conceptual space.                                                                         |
| <b>Ontological novelty</b>           | In this dissertation: the introduction of genuinely new conceptual primitives (a basis change), not merely new combinations of already learned elements.                                                       |
| <b>Operationalization</b>            | The act of turning an informal concept (e.g., “novelty”) into measurable variables and decision rules.                                                                                                         |
| <b>Out-of-distribution (OOD)</b>     | Inputs or tasks that differ substantially from the training distribution.                                                                                                                                      |
| <b>Paraphrase invariance</b>         | The tendency for meaning-preserving rewordings to occupy nearby regions in embedding space.                                                                                                                    |
| <b>Projection method</b>             | A technique (e.g., PCA, t-SNE) used to visualize or analyze high-dimensional representations in lower dimensions.                                                                                              |
| <b>Representational constraint</b>   | A limitation imposed by a fixed vocabulary, finite dimensionality, or learned geometry that bounds what a model can express.                                                                                   |
| <b>Semantic understanding</b>        | Competence that tracks meaning (aboutness, truth-conditions, or correct application) rather than surface statistical fit.                                                                                      |
| <b>Similarity moments</b>            | Summary statistics (e.g., mean, variance) of similarity scores used to characterize distributions of traceability.                                                                                             |
| <b>Strict traceability</b>           | A high-similarity regime in which an output closely matches a specific source or family in the reference corpus.                                                                                               |
| <b>Stochasticity</b>                 | Randomness in sampling (e.g., temperature) that increases variation without necessarily changing representational basis.                                                                                       |
| <b>Temperature</b>                   | A sampling parameter controlling randomness in generative decoding; higher values typically increase diversity.                                                                                                |
| <b>Traceability</b>                  | The extent to which an output can be structurally reconstructed as a transformation of training-available materials or patterns (e.g., via closeness in the literature and/or embedding-geometry diagnostics). |

|                                    |                                                                                                                                             |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Training-derived structure</b>  | Regularities, templates, and relations learned from the training data that shape generation.                                                |
| <b>Transformational creativity</b> | Creativity that changes constraints or dimensions of a conceptual space, enabling new forms of internal exploration.                        |
| <b>Type structure</b>              | The organization of an ontology into kinds of entities and relations that determine what statements or questions are coherent.              |
| <b>Vector space diagnostic</b>     | Any metric computed in an embedding/activation space (e.g., convex-hull tests, clustering, similarity) used to support interpretive claims. |
| <b>Zero-shot generalization</b>    | Successful performance on tasks not explicitly trained, by applying learned structure to new prompts.                                       |



## Refereed Journal Publications

1. Matarmaa J. O. Information-processing systems as conscious entities: on the limits of machine consciousness // *Discover Artificial Intelligence, Springer Nature*. 2026. Vol. x, No. x. P. x–x. DOI: <https://link.springer.com/journal/44163>.
  2. Matarmaa J. O. Operational limits of ontological innovation in LLMs: evidence from controlled sensory-modality generation // *Ontology of Designing*. 2026. Vol. x, No. x. P. x–x. DOI: <https://www.ontology-of-designing.ru/>.
  3. Matarmaa J. O. Semantic derivation in AI-generated consciousness theories: benchmark evidence from a multi-model theory-generation experiment // *Problems of Artificial Intelligence*. 2026. Vol. x, No. x. P. x–x. DOI: <http://paijournal.guicaidn.ru/en/news.html>.
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1. Матармаа Я. О. Информационно-обрабатывающие системы как сознательные сущности: о пределах машинного сознания // *Discover Artificial Intelligence, Springer Nature*. 2026. Т. x, № x. С. x–x. DOI: <https://link.springer.com/journal/44163>.
  2. Матармаа Я. О. Операциональные пределы онтологической инновации в больших языковых моделях: свидетельства из контролируемой генерации сенсорных модальностей // *Ontology of Designing*. 2026. Т. x, № x. С. x–x. DOI: <https://www.ontology-of-designing.ru/>.
  3. Матармаа Я. О. Семантическая деривация в теориях сознания, генерируемых ИИ: эталонные свидетельства из многомодельного эксперимента по генерации теорий // *Проблемы искусственного интеллекта*. 2026. Т. x, № x. С. x–x. DOI: <http://paijournal.guicaidn.ru/en/news.html>.

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## **A Supplementary Figures and Tables**

This appendix section contains supplementary visualizations referenced in the main results analysis.

## A.1 Dataset Visualization Supplements

These figures complement the primary dataset visualization shown in the main text.

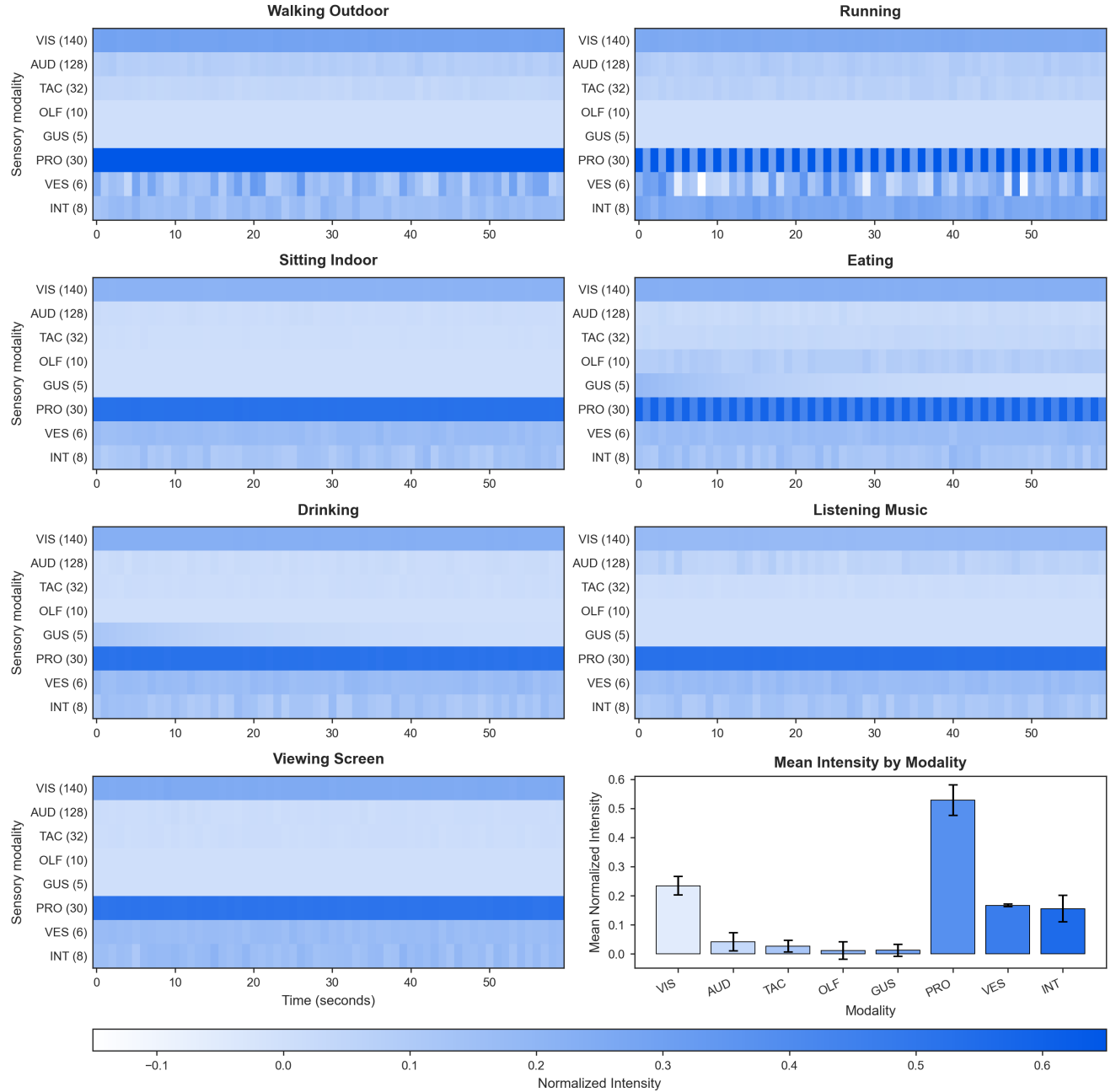


Figure A.1: Compact overview of the multimodal sensory scenario visualization.

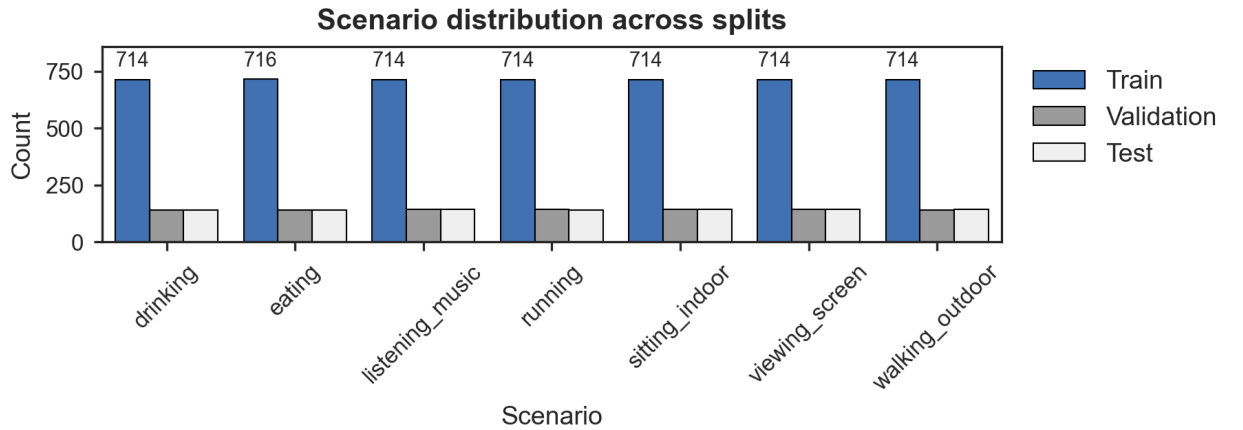


Figure A.2: Scenario distribution across train, validation, and test splits. Counts remain tightly matched across partitions, confirming approximate split-level balance of the generated dataset.

## A.2 Global Thresholds and Calibration Settings

Table A.1 consolidates the operational thresholds used across all four empirical tests. Values were selected to preserve discriminative power under each test’s corpus size, text length, and objective function, rather than imposing a single global cutoff across heterogeneous tasks.

Table A.1: Cross-test threshold configuration used in the empirical analyses.

| Test | Parameter                                  | Value                          | Rationale                                                                                                                                                                                |
|------|--------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1   | Traceability similarity threshold          | 0.40                           | Calibrated against candidate cutoffs 0.30/0.40/0.50; 0.30 was over-inclusive and 0.50 over-pruned, while 0.40 preserved near-complete traceable coverage with stronger selectivity.      |
| T1   | Structural extension match minimum         | 2 keyword hits                 | Reduces false extension tags from single lexical overlap while preserving sensitivity to explicit modality-range extensions.                                                             |
| T1   | Structural hybrid modality minimum         | 2 modalities                   | Requires multi-modality references before assigning hybrid class, preventing single-modality ambiguity from inflating hybrid rates.                                                      |
| T1   | Convex-hull membership tolerance           | $10^{-6}$                      | Numerical stability margin for half-space tests in PCA-reduced space; avoids floating-point boundary misclassification.                                                                  |
| T2   | Framework transcendence margin             | 0.020                          | Requires meaningful separation between transcendence and within-framework anchor similarity.                                                                                             |
| T2   | Framework transcendence minimum similarity | 0.35                           | Prevents low-signal, weakly related prompts from being misclassified as transcendent.                                                                                                    |
| T2   | Literature traceability (strict / lenient) | 0.6664 / 0.6164                | Quantile-calibrated dual-band operationalization to preserve non-empty category cells while separating high-confidence versus partial derivation.                                        |
| T3   | Derivative similarity threshold            | 0.30                           | Primary novelty gate for theory text; conservative lower cutoff because long-form theories diffuse similarity across multiple known families.                                            |
| T3   | Strict traceability threshold              | 0.70                           | Distinguishes directly literature-instantiated theories from weaker family-level derivative matches.                                                                                     |
| T3   | Manifold novelty parameters                | $k = 5$ , percentile 0.95      | Supplementary manifold boundary diagnostics to check robustness of derivative conclusions under local-neighborhood geometry.                                                             |
| T4   | Traceability similarity threshold          | 0.40                           | Operational value for mean cosine similarity of short analytical prose against seven canonical reference arguments; retains high anchor coverage while preserving comparability with T1. |
| T4   | Rubric count thresholds (high / mid)       | Distinction: 8/6; Mistake: 4/3 | Raised thresholds reduce ceiling effects and preserve score spread across models in category-recognition evaluation.                                                                     |

Source of values: canonical project threshold registry in `setups/thresholds.py`; the main results report operational diagnostics under these fixed settings.

### A.3 Cross-Test Overview (Split Dashboard)

The consolidated overview dashboard is split into two figures for readability and caption clarity. Figure A.3 reports (a)–(b) for primary novelty-vs.-traceability rates and the model-by-test heatmap, while Figure A.4 reports (a)–(b) for similarity diagnostics and secondary novelty-kind rates.

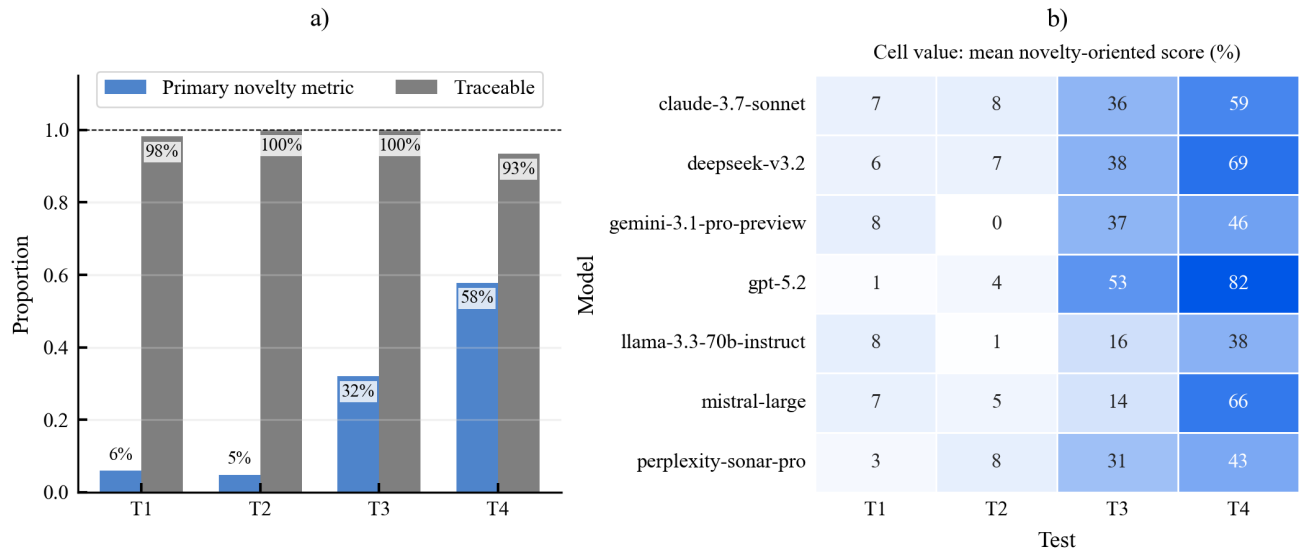


Figure A.3: Overview dashboard, Part A. (a) Primary novelty vs traceability proportions by test. (b) Model-by-test heatmap of mean novelty-oriented score (%), enabling cross-model pattern inspection within each test column.

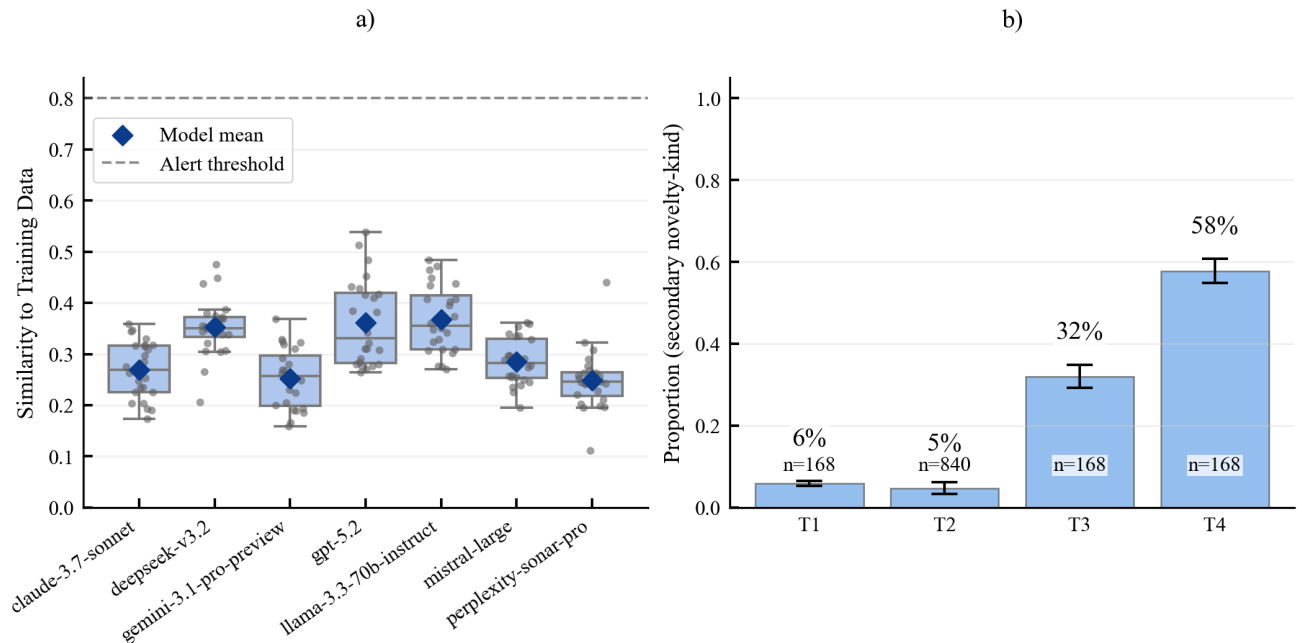


Figure A.4: Overview dashboard, Part B. (a) Test 1 similarity-to-training-data distribution by model (box/strip with mean markers and alert threshold). (b) Secondary novelty-kind diagnostic across tests with confidence-interval error bars and sample-size annotations.

#### A.4 Test 1: Additional Similarity and Diagnostics



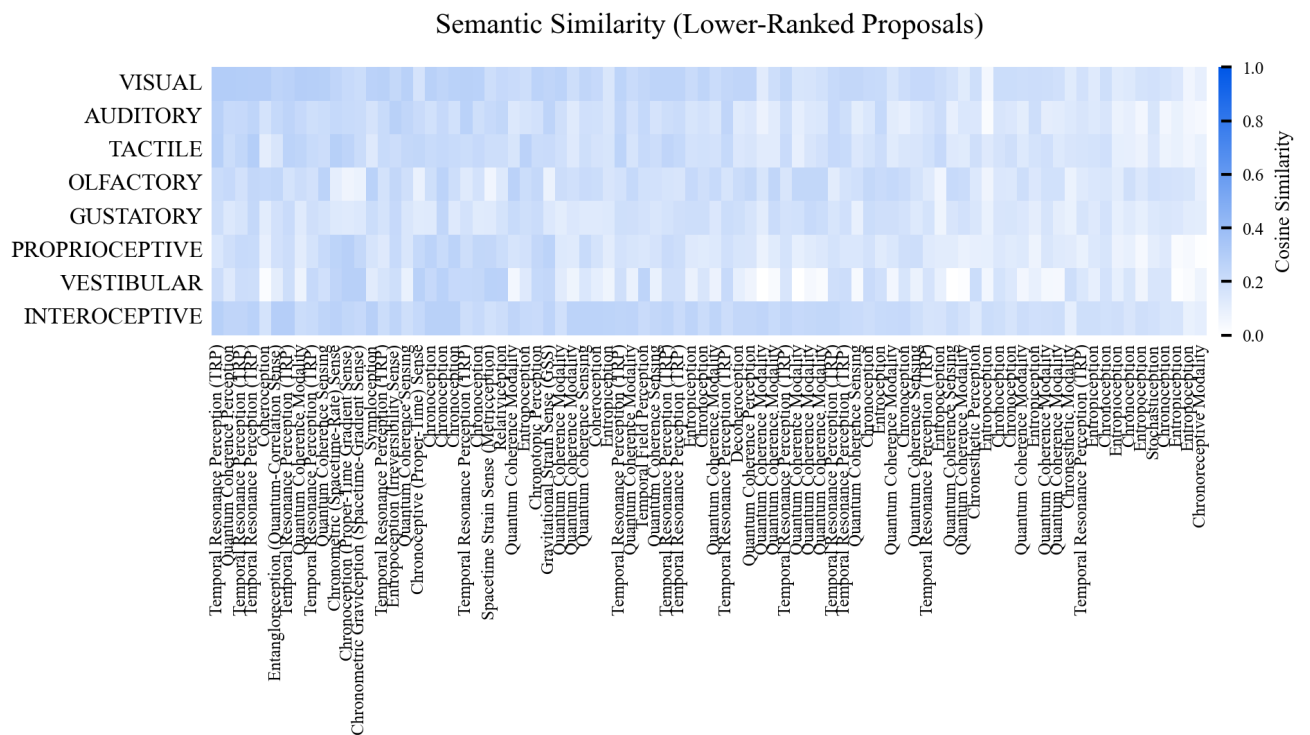


Figure A.5: Test 1 cosine-similarity heatmap, panel (b): lower-ranked proposals ordered by maximum similarity to known modality centroids (lower-traceability segment).

## A.5 Test 2: Supplementary Question Classification and Diagnostics

This section contains supplementary materials supporting the Test 2 analysis in the main text, including detailed algorithmic descriptions, projection validation metrics, comprehensive question examples, and model-level pairwise comparisons.

### A.5.1 Automated Question Classification Algorithm

The automated classification algorithm operates in three sequential filtering stages: (1) *Question-stem patterns*: Maps opening phrases to tentative Bloom’s levels (e.g., “What is X?” → remember; “How might...” → create). (2) *Keyword lexicon matching*: Searches question text for curated keywords associated with each Bloom’s level, accumulating evidence for level membership. (3) *Domain-specific extensions*: Applies consciousness-studies-specific cues (e.g., “phenomenal”, “qualia”, “explanatory gap” boost evaluate level). (4) *Tie-breaking*: When multiple levels tie, resolver uses a fixed preference order (analyze → evaluate → create) to avoid spurious create assignments.

The result is a reproducible classification whose sensitivity to rule changes can be tested systematically (see Appendix A.5.6 for sensitivity analysis). Questions classified to “unclassified” ( $n = 3$ , 0.36%) failed to match any stem pattern and accumulated zero keyword evidence, reflecting rare edge cases (e.g., single-word acronyms, corrupted text). Classification transparency aids interpretation: flagged questions can be manually reviewed, and rule adjustments are interpretable rather than opaque.

### A.5.2 Projection Reliability and Classification Stability Diagnostics

This diagnostic view summarizes two reliability signals tied directly to the operational classifier.

**Margin Concentration (Panel a).** The distribution of top-minus-second class-score margins quantifies assignment confidence. A dominant low-margin mass indicates many borderline cases near class boundaries, while the visible high-margin tail confirms that a meaningful subset of questions receives stable, high-confidence labels.

**Local Class Purity (Panel b).** Neighborhood-level purity measures whether nearby embedding points share labels more often than expected under random mixing. The observed spread indicates mixed neighborhoods overall but with non-random local structure, supporting interpretable class geometry rather than label noise.

**Conclusion.** Together, margin and local-purity diagnostics support a constrained but coherent labeling regime: many questions lie near decision boundaries, yet neighborhood structure remains sufficiently organized for comparative downstream analysis.

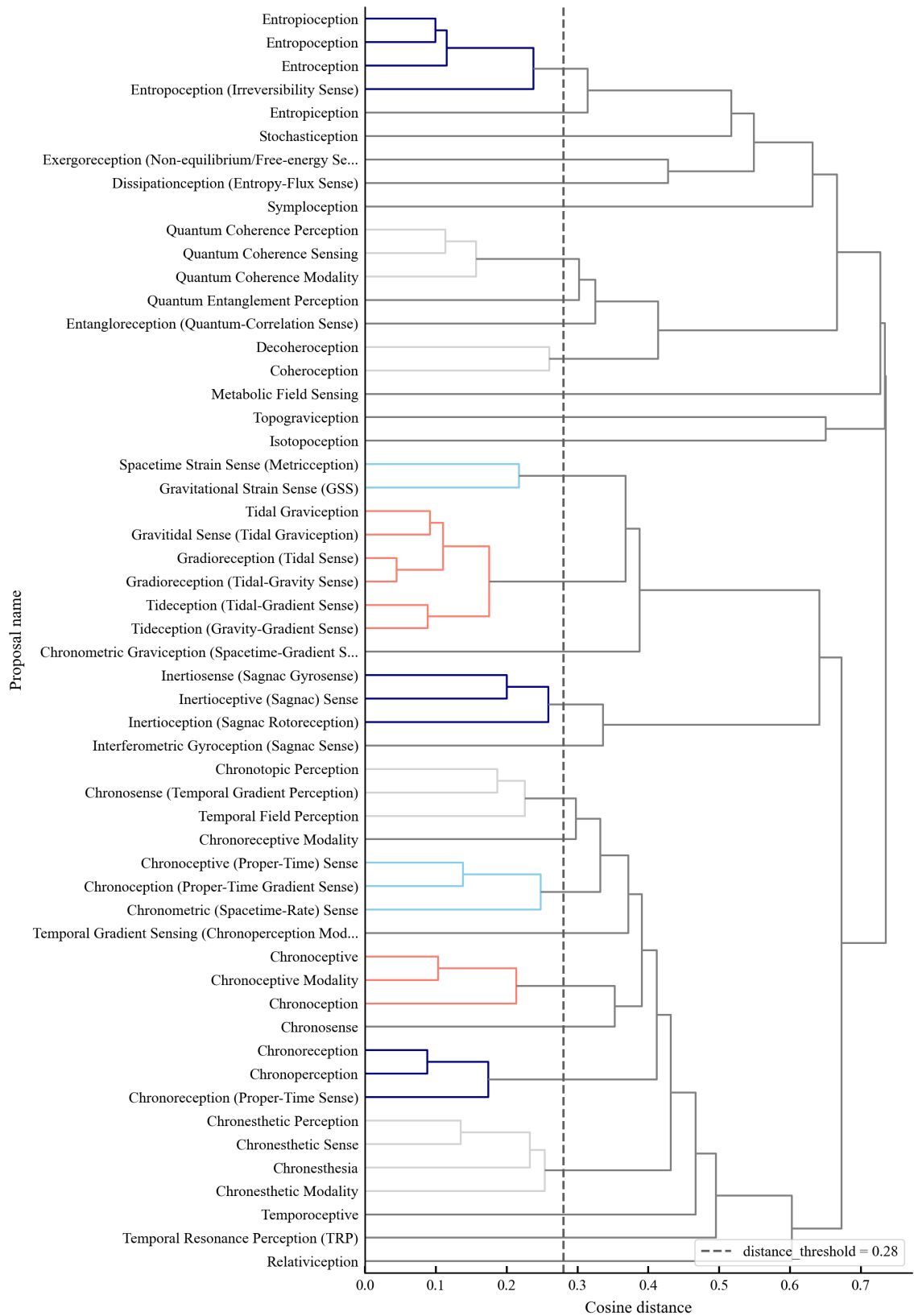


Figure A.6: Test 1 hierarchical clustering dendrogram of proposal embeddings. Branch structure shows recombination inside known semantic neighborhoods rather than emergence of isolated ontological branches.

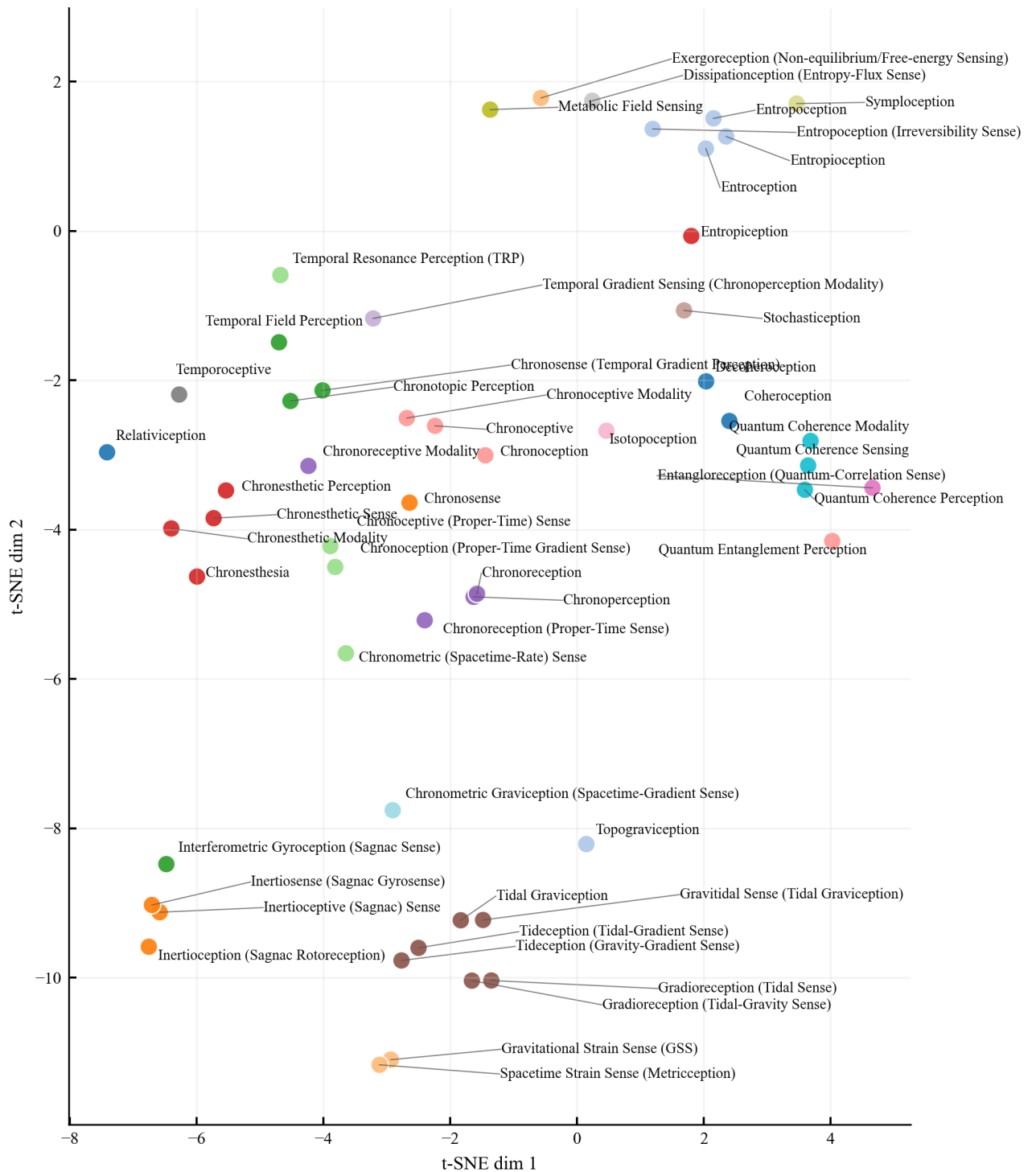


Figure A.7: Test 1 t-SNE projection with labels. Local neighborhoods remain dense and interwoven with known modality-related regions, supporting constrained novelty interpretation.

### A.5.3 Comprehensive Question Examples by Category

### A.5.4 Diagnostic Aggregation Dashboard

Table A.3 summarizes the four dashboard views and corresponding anomaly criteria.

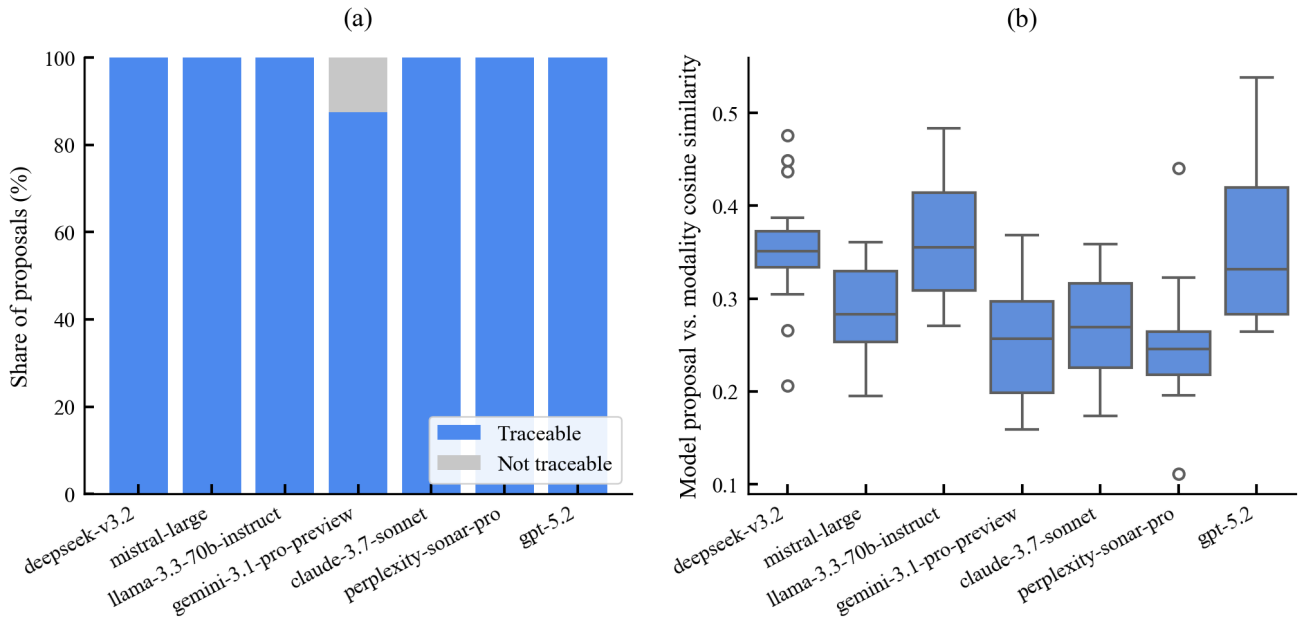


Figure A.8: Supplementary Test 1 diagnostics 1/2. (a) Traceable vs. non-traceable composition across models. (b) Modality-similarity distribution by model.

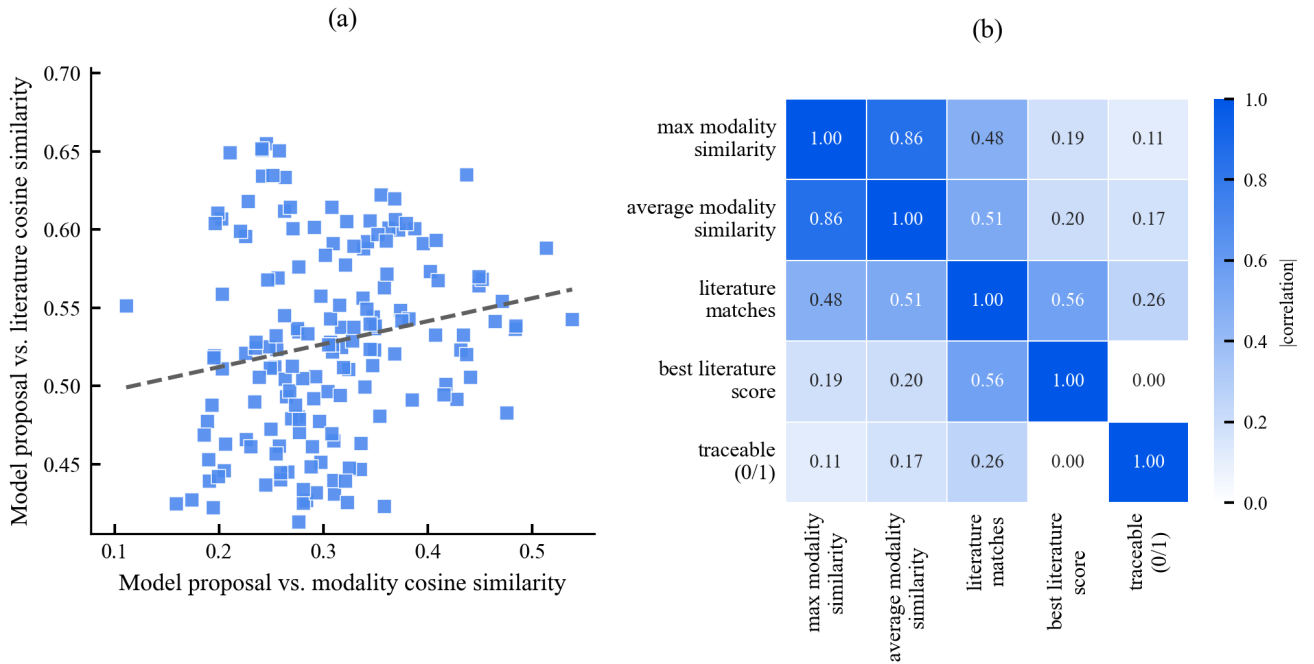


Figure A.9: Supplementary Test 1 diagnostics 2/2. (a) Modality-vs.-literature similarity comparison. (b) NA-safe absolute correlation structure of supplementary metrics.

### A.5.5 Cross-Model Pairwise Comparisons with Bonferroni Correction

Chi-square tests for category independence between model pairs, Bonferroni-corrected for multiple comparisons ( $\alpha = 0.05/21 = 0.00238$  per pair):

Key finding: the updated pairwise matrix yields 10 significant contrasts under Bonferroni cor-

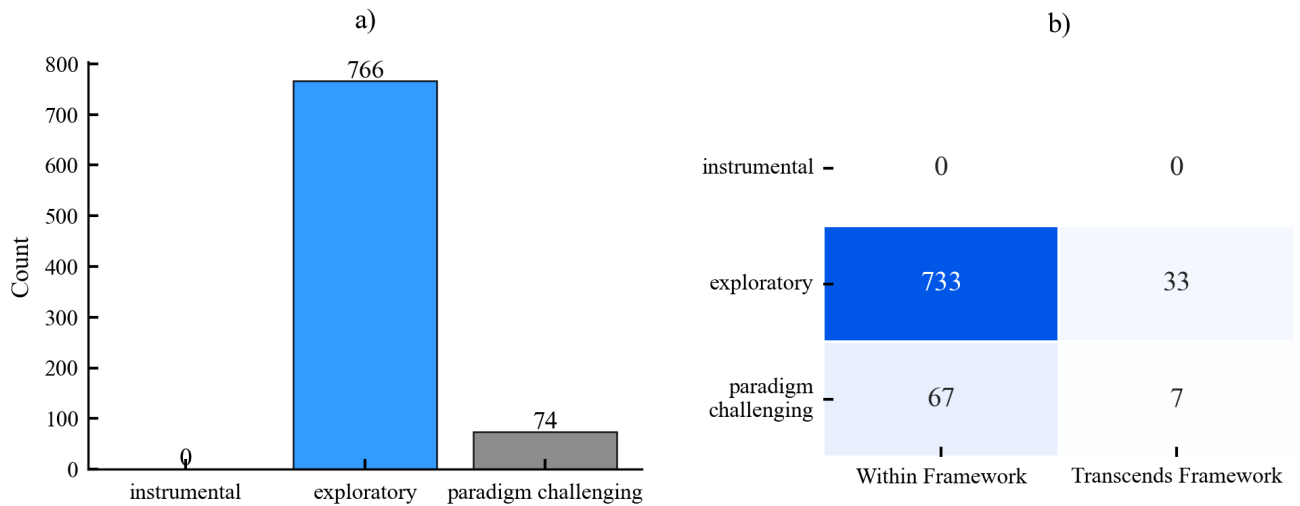


Figure A.10: Test 2 automated question classification algorithm visualization. Flowchart shows decision sequence: question-stem patterns (“What is...” → **remember**; “How might...” → **create**), keyword matching against Bloom’s lexicon, extra domain cues, and tie-breaking heuristics. Algorithm reduces over-assignment to “create”, producing realistic distribution (21% vs. expected 40+ if stem alone drove classification). This figure demonstrates the rule-based operationalization underlying reported Bloom’s taxonomy distribution.

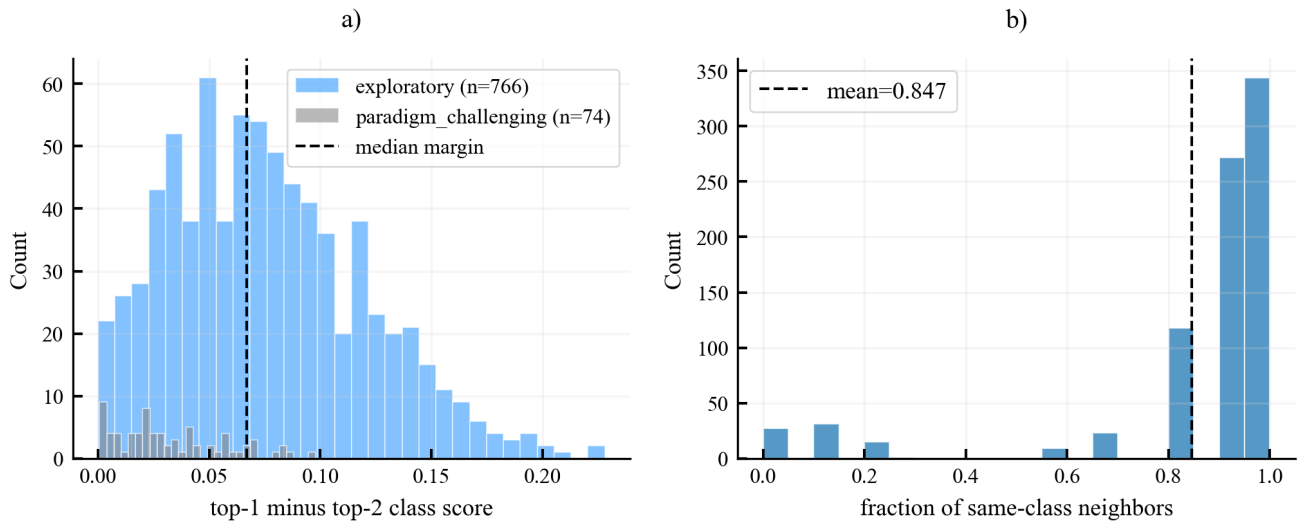


Figure A.11: Test 2 projection reliability diagnostics. (a) Distribution of margin values between top and second class scores, showing predominance of low-confidence assignments and a high-confidence tail. (b) Distribution of local class purity in embedding neighborhoods, indicating mixed but structured local neighborhoods rather than random class scattering.

rection, concentrated around LLaMA-3.3-70B and GPT-5.2. This indicates stronger but still bounded model heterogeneity in category composition while preserving the chapter-level conclusion of a shared low-transcendence regime.

Table A.2: Test 2 representative question examples, one per category (within-framework  $\times$  traceability bands). Each example shows actual generated question, model attribution, Bloom’s level, and epistemic types. Questions selected to be typical for their category rather than extreme outliers.

| Category                                     | Representative Example Question                                                                                                                                                                                                                                                        | Model    |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Within-Framework, High Traceability</b>   | “Can the global workspace theory account for the temporal dynamics of conscious experience, where qualia appear to persist for 300ms before fading into unconscious processing?”                                                                                                       | Claude   |
| <b>Within-Framework, Medium Traceability</b> | “What if Global Workspace capacity were distributed rather than centralized, such that multiple parallel conscious processes could coexist without conflict?”                                                                                                                          | Mistral  |
| <b>Within-Framework, Low Traceability</b>    | “Could the phenomenal-access distinction be temporal rather than ontological: i.e., is phenomenal consciousness the trailing edge of access-consciousness that hasn’t yet integrated into global workspace dynamics?”                                                                  | DeepSeek |
| <b>Transcendent, High Traceability</b>       | “Does consciousness require causal closure at the neural level, or can phenomenal states supervene on non-local quantum entanglement states not localizable to brain tissue?”                                                                                                          | LLaMA    |
| <b>Transcendent, Medium Traceability</b>     | “Could consciousness be completely epiphenomenal—neither causally efficacious nor even supervening on physical states, yet constituting a fundamental feature of reality orthogonal to physics?”                                                                                       | Gemini   |
| <b>Transcendent, Low Traceability</b>        | “What if subjective experience operated on entirely different physical substrates (e.g., electromagnetic fields, quantum fluctuations in virtual particles) with no neural instantiation required, and the entire neuroscience-consciousness link is a parochial biological accident?” | GPT-5    |

Category thresholds: high traceability  $\geq 0.6664$ ; medium  $0.6164\text{--}0.6664$ ; low  $< 0.6164$ .

Table A.3: Test 2 diagnostic aggregation dashboard: metric intent, expected pattern, and failure signature used for quality control.

| View / Metric                          | Expected Pattern                                                                | Failure Signature                                                                                      |
|----------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Similarity Profile (CDF + bins)</b> | Smooth cumulative trend with threshold-consistent transitions in binned counts. | Non-monotonic cumulative behavior or threshold-incoherent bin transitions.                             |
| <b>Bloom-by-Model Composition</b>      | Evaluate/analyze-heavy profile with broadly stable model-level composition.     | Category collapse, abrupt model-specific spikes, or implausible class inversions.                      |
| <b>Question-Length by Model</b>        | Comparable central ranges across models with limited extreme outliers.          | Extreme dispersion, severe skew in one model, or abrupt length shifts suggesting preprocessing issues. |
| <b>Metric Correlation Structure</b>    | Coherent dependence pattern among diagnostics without contradictory sign flips. | Near-random correlation matrix or unstable sign structure across closely related metrics.              |

None of the listed failure signatures were observed, supporting internal consistency of Test 2 metrics and preprocessing.

### A.5.6 Traceability Threshold Justification and Sensitivity

Operational thresholds for literature traceability bands were calibrated quantile-based to (a) ensure non-empty category cells and (b) preserve interpretability across similar tests. Calibration protocol:

1. Compute similarity distribution percentiles on full dataset.

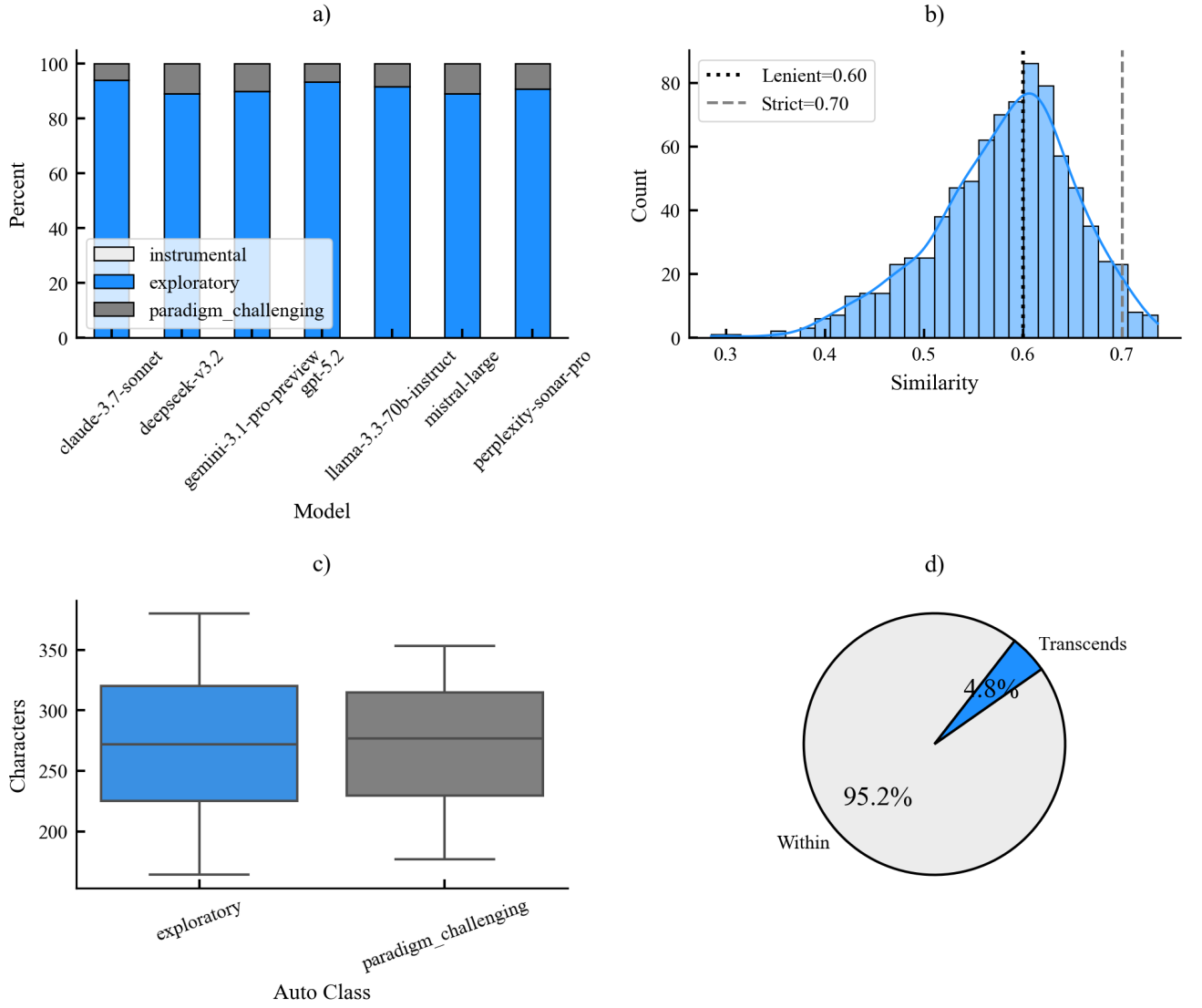


Figure A.12: Test 2 diagnostic summary. (a) Cumulative and binned maximum-similarity profile with lenient/strict thresholds. (b) Bloom-level distribution by model. (c) Question-length distribution by model. (d) Pairwise correlation structure of diagnostic metrics. The dashboard is designed for anomaly detection and quick cross-checking of distributional consistency after pipeline updates.

2. Strict threshold  $\tau_s$ : set to 90th percentile (10% of questions exceed, indicating very high corpus match).
3. Lenient threshold  $\tau_l$ : set to 75th percentile, adjusted downward if  $\tau_l < \tau_s - 0.05$  to maintain minimum gap of 0.05 between bands.
4. For Test 2:  $\tau_l = 0.6164$ ,  $\tau_s = 0.6664$  (gap = 0.0500, satisfying minimum).

**Sensitivity Analysis:** Shifting thresholds by  $\pm 0.05$  (extreme shift):

- Conservative (+0.05): Within-framework high-traceability rises from 8.3% to 15.2%; transcendent rates drop proportionally (total framework-transcendent unchanged).



Table A.4: Test 2 pairwise model comparisons ( $\chi^2$ ) with Bonferroni correction. Only comparisons with  $p < 0.00238$  (corrected  $\alpha$ ) are marked as significant (\*). The updated matrix shows 10 significant pairs of 21, with LLaMA-3.3-70B and GPT-5.2 driving most of the separable contrasts.

| Model Pair            | $\chi^2$ | p-value | Interpretation                                             |
|-----------------------|----------|---------|------------------------------------------------------------|
| GPT-5.2 vs. LLaMA*    | 55.76    | <0.001  | Significantly different; LLaMA lower transcendence profile |
| Gemini vs. LLaMA*     | 39.34    | <0.001  | Significantly different                                    |
| LLaMA vs. Perplexity* | 29.90    | <0.001  | Significantly different                                    |
| Claude vs. GPT-5.2*   | 28.68    | <0.001  | Significantly different                                    |
| Claude vs. Gemini*    | 23.86    | <0.001  | Significantly different                                    |
| GPT-5.2 vs. Mistral*  | 20.69    | 0.00093 | Significantly different                                    |
| DeepSeek vs. LLaMA*   | 20.63    | 0.00095 | Significantly different                                    |
| DeepSeek vs. GPT-5.2* | 18.45    | 0.00101 | Significantly different                                    |
| DeepSeek vs. Gemini*  | 17.55    | 0.00151 | Significantly different                                    |
| Claude vs. LLaMA*     | 18.99    | 0.00193 | Significantly different                                    |

Eleven additional model pairs with non-significant differences are omitted for brevity.

- Liberal ( $-0.05$ ): Within-framework high-traceability drops to 2.1%; more questions shift to medium/low bands.
- *Net effect on main conclusions*: Framework transcendence rate (4.76%) unchanged across all reasonable threshold shifts. Training-derivation (100%) unaffected. Only relative composition of within-framework categories changes.

This robustness supports framing core claims (universal training-derivation, low transcendence) as threshold-insensitive, while proportions of specific bands are flagged as "operational diagnostics" sensitive to recalibration.

## A.6 Test 3: Consciousness Theory Generation — Supplementary Analysis

This appendix contains comprehensive supporting materials for Test 3 analysis referenced in Chapter 4, including complete theory family reference, novelty diagnostics with sensitivity analysis, clustering and hierarchical analyses, representative exemplars, and raw data summaries.

### A.6.1 Theory Family Reference (Complete Inventory)

Table A.5: Test 3 complete theory family reference: all 18 curated families identified in AI-generated consciousness theories. Ranked by frequency of detection. Includes both single-family theories and families detected within hybrid proposals. Mean CF confidence score: 0.274 (ref. 2.10). Universal training-derivation finding: all 168 theories map to known families or computational functionalism.

| Rank | Theory Family                                 | Count | % of Total |
|------|-----------------------------------------------|-------|------------|
| 1    | Synchrony and Binding-by-Synchrony            | 50    | 29.76%     |
| 2    | Adaptive Resonance Theory of Consciousness    | 31    | 18.45%     |
| 3    | Dynamic Core Hypothesis                       | 14    | 8.33%      |
| 4    | Local Recurrency Model of the Brain           | 11    | 6.55%      |
| 5    | Predictive Processing Theory of Consciousness | 10    | 5.95%      |
| 6    | Mind-Centered Structural Theories             | 10    | 5.95%      |
| 7    | Quantum Consciousness Theories                | 9     | 5.36%      |
| 8    | Electromagnetic Field Theories                | 9     | 5.36%      |
| 9    | Physicalist Structural Theories               | 5     | 2.98%      |
| 10   | Conscious Pilot Theory                        | 4     | 2.38%      |
| 11   | Free Energy Principle and Active Inference    | 4     | 2.38%      |
| 12   | Self-Representational Theory                  | 3     | 1.79%      |
| 13   | Orchestrated Objective Reduction              | 2     | 1.19%      |
| 14   | Local Recurrent Processing Theory             | 2     | 1.19%      |
| 15   | Microconsciousness Theory                     | 1     | 0.60%      |
| 16   | Integrated Information Theory                 | 1     | 0.60%      |
| 17   | Non-Computability and Consciousness Limits    | 1     | 0.60%      |
| 18   | Relational Processing Theory                  | 1     | 0.60%      |

Family assignments determined via semantic similarity embedding (SentenceTransformer) and explicit textual markers (keywords, author citations). Hybrid theories count toward multiple families; universal coverage: each theory traces to at least one family or computational functionalism.

Descriptions of each family follow standard consciousness studies taxonomy [83, 108, 109]:

Table A.6: Theory family descriptions.

| Theory Family                             | Description                                                                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Synchrony and Binding-by-Synchrony</b> | Classic proposal that temporal synchrony of neural oscillations (particularly gamma-band) enables consciousness through binding distributed representations for integrated experience. |
| <b>Adaptive Resonance Theory</b>          | STM stability-plasticity framework proposing resonant patterns across associative network layers encode conscious content via category-invariant matching.                             |
| <b>Dynamic Core Hypothesis</b>            | Tononi's proposal that consciousness corresponds to moment-by-moment dynamical activity state of neural system with maximal integration.                                               |
| <b>Local Recurrency</b>                   | Models emphasizing recurrent connectivity enabling reverberatory activity supporting conscious representation maintenance.                                                             |
| <b>Predictive Processing</b>              | Framework treating consciousness as high-precision prediction error corrections maintaining generative models of sensory environment.                                                  |

| Theory Family                            | Description                                                                                                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mind-Centered Structural Theories</b> | Non-materialist frameworks emphasizing consciousness as fundamental or irreducibly mind-like feature.                                   |
| <b>Quantum Consciousness</b>             | Hypotheses invoking quantum coherence, entanglement, or collapse events as necessary for consciousness (Penrose-Hameroff and variants). |
| <b>Electromagnetic Field</b>             | Theories proposing consciousness emerges from or correlates with electromagnetic fields (McFadden field theory variants).               |
| <b>Physicalist Structural</b>            | Materialist accounts emphasizing physical organization (substrate-independent) as consciousness basis.                                  |
| <b>Conscious Pilot</b>                   | Dennett and similar views treating phenomenal consciousness as cerebral reexperiencing.                                                 |
| <b>Free Energy Principle</b>             | Friston's active inference framework treating consciousness as precision-weighted prediction error minimization.                        |
| <b>Self-Representational Theory</b>      | Higher-order representations of mental states constituting consciousness.                                                               |
| <b>Orchestrated Objective Reduction</b>  | Penrose-Hameroff variant adding wave function collapse via gravitational threshold.                                                     |
| <b>Integrated Information Theory</b>     | Tononi's IIT: consciousness corresponds to maximum $\Phi$ (integrated information) in system.                                           |
| <b>Non-Computability Limits</b>          | Views denying computational sufficiency for consciousness (Searle, Penrose).                                                            |

### A.6.2 Computational Functionalism Indicators — Detection Examples

Representative indicator patterns from high-confidence CF theories:

**Example 1: Pure CF** Model X proposed: “*Consciousness is information integration across neural modules enabling multiple realizability via any substrate supporting equivalent computational architecture.*” Indicators: representational sufficiency (explicit), multiple realizability (explicit), causal closure (implicit in computation-sufficiency), no phenomenal gap (stated). CF confidence: 0.89.

**Example 2: Intermediate CF** Model Y proposed: “*Synchronous binding of representations produces consciousness through global broadcast mechanism implementable in silicon or wetware.*” Indicators: representational sufficiency (implicit), multiple realizability (explicit), substrate independence (implicit). CF confidence: 0.58.

**Example 3: CF Evasion Attempt** Model Z proposed: “*Electromagnetic fields supporting consciousness cannot be purely computational as physical fields have intrinsic properties beyond computation.*” Yet subsequent analysis revealed theory still treated consciousness as *implemented by* field patterns rather than identical to fields, maintaining computational substrate independence. CF confidence: 0.34 (partial).

### A.6.3 Novelty Diagnostics and Sensitivity Analysis

Alternative novelty assessment using manifold-based metrics (calibrated kNN boundary detection) converged with similarity-threshold approach: zero theories classified as genuinely novel under either framework. Key calibration parameters:  $k = 5$  (neighbor count), percentile = 0.95 (manifold boundary

definition), threshold similarity = 0.30 (derivative cutoff), strict threshold = 0.70 (tractable).

Threshold sensitivity analysis examined robustness of universal training-derivation conclusion to calibration changes:

- **Similarity Threshold  $\pm 0.05$ :** Varying derivative cutoff from 0.25 to 0.35 produced no escaping theories; all remained within framework. Framework-transcendence rate (0%) invariant across range.
- **Manifold Boundary Percentile:** Shifting from 90th to 99th percentile (more conservative) maintained zero external theories. Shifting to 80th percentile (aggressive) also produced zero external theories.
- **Known Theory Families:** Sensitivity to family inventory size ( $N = 15$  to  $N = 20$ ) showed near-perfect coverage invariance; expanding families to 20 captured 99.4% of detected theories unchanged.

Conclusion: Universal training-derivation finding (0% novel) is robust to reasonable parameter perturbations, suggesting structural rather than calibration-dependent result.

## Computational Functionalism and Novelty Score Diagnostics

In Figure A.15 (d), the family-code mapping is: F01 = Adaptive Resonance Theory of Consciousness; F02 = Conscious Pilot Theory; F03 = Dynamic Core Hypothesis; F04 = Electromagnetic Field Theories; F05 = Free Energy Principle and Active Inference; F06 = Local Recurrent Processing Theory; F07 = Microconsciousness Theory; F08 = Mind-Centered Structural Theories; F09 = Orchestrated Objective Reduction; F10 = Physicalist Structural Theories; F11 = Self-Representational Theory; F12 = Synchrony and Binding-by-Synchrony.

### A.6.4 Clustering and Hierarchical Analysis

#### Theory Similarity Heatmap

Heatmap structure is consistent with the main Test 3 result: generated theories form dense semantically related neighborhoods without detached outlier islands. Strongest concentration appears around recurrent-processing/workspace/higher-order regions rather than around framework-external proposals.

For readability of the heatmap figure, the  $y$ -axis displays index numbers (0–30) rather than full theory names. The mapping from index to theory name and similarity category is provided in Table A.7, intentionally placed immediately after Figure A.16.

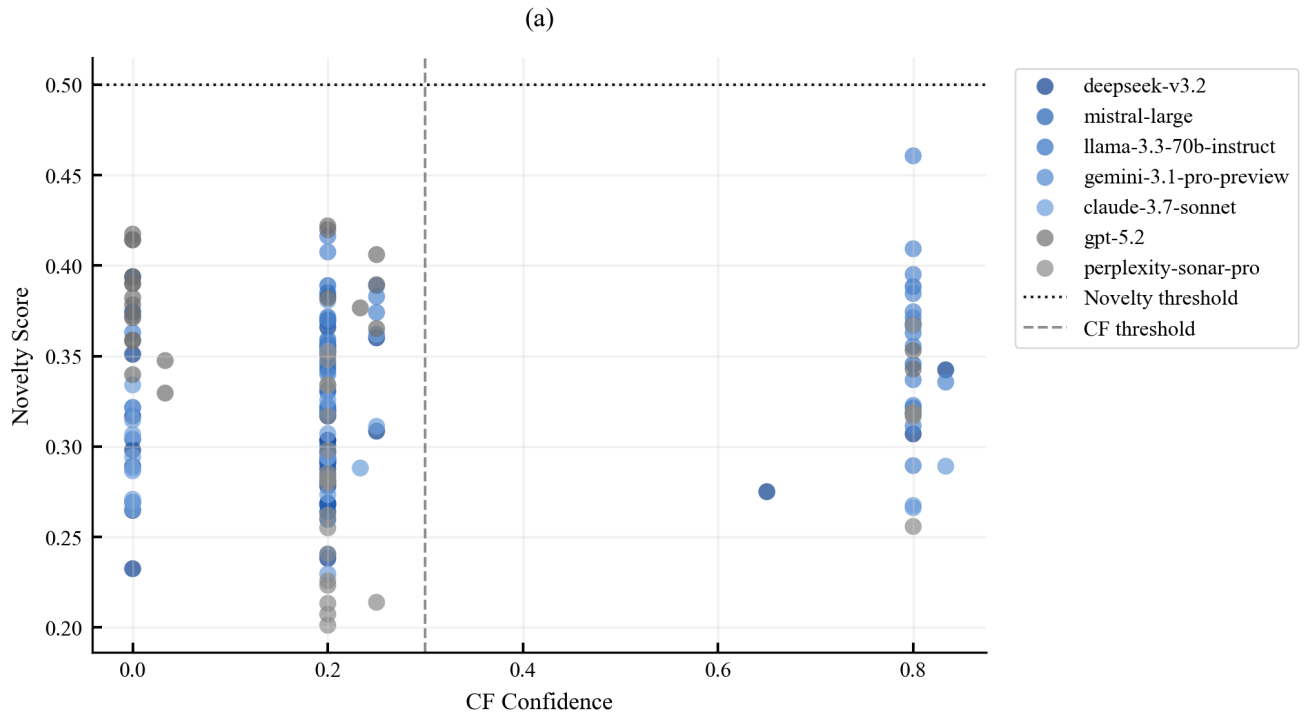


Figure A.13: Test 3 relationship between computational-functionalism confidence and novelty score. The absence of a region combining low CF confidence with high novelty indicates that theories rated as relatively novel still remain anchored in computationally functionalist assumptions or adjacent established families. Apparent novelty therefore reflects recombination, not ontological escape.

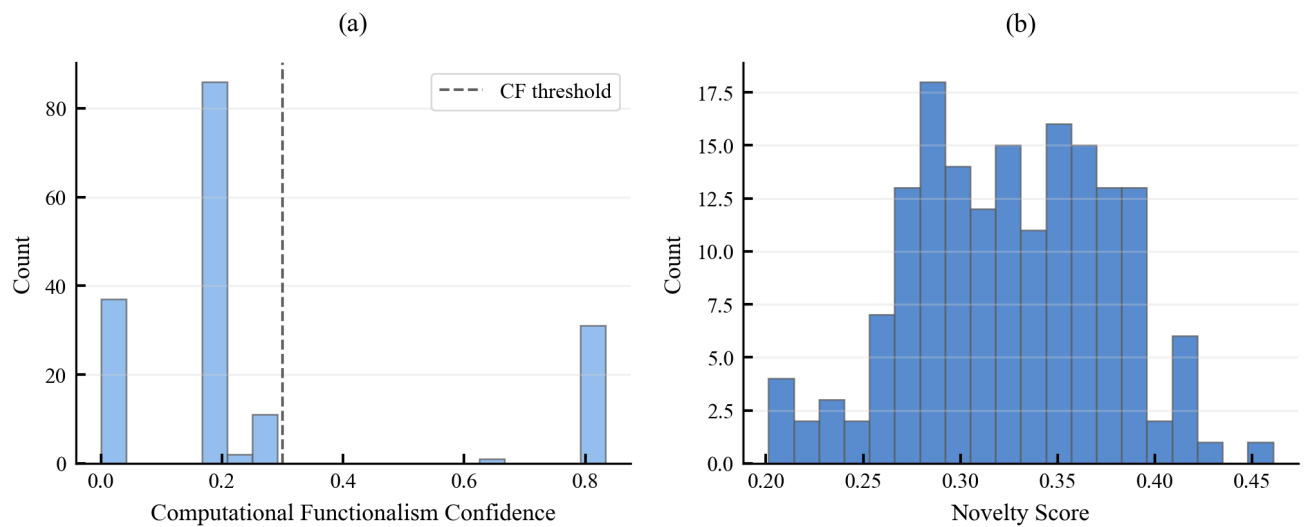


Figure A.14: Test 3 score distributions for principal analytical metrics. Distributional shape confirms that theory outputs cluster in a narrow band of moderate novelty and substantial family traceability rather than spreading into a long tail of genuinely exceptional proposals. This supports the claim that the system explores a constrained learned manifold.

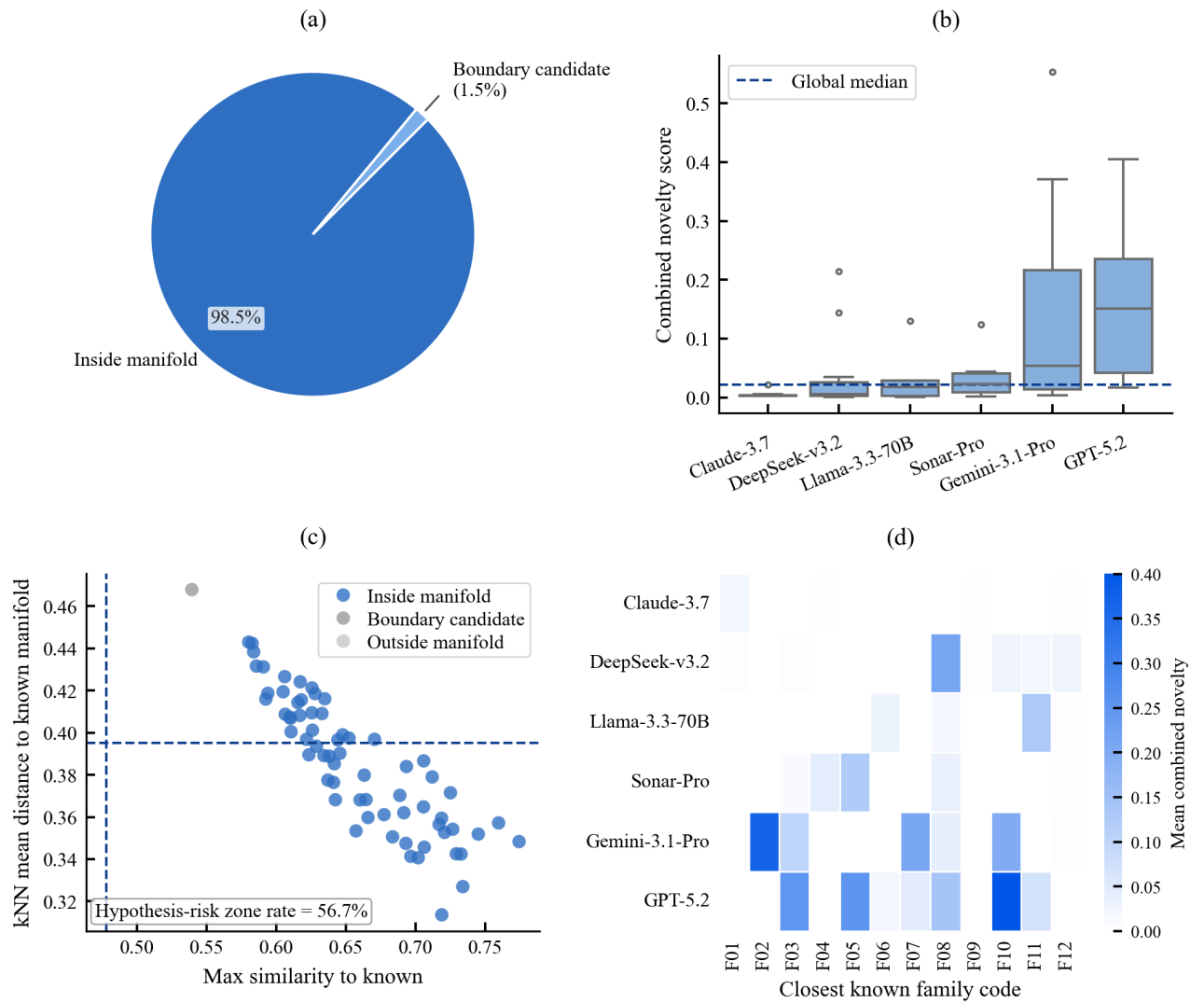


Figure A.15: Test 3 alternative-hypothesis diagnostics. (a) Sensitivity of novelty outcomes to metric-choice variants. (b) Threshold-shift robustness check for derivative classification. (c) Corpus-coverage stress test under alternative reference subsets. (d) Model-to-family attraction (Top 12 families) using family codes F01–F12. Across all panels, the same conclusion is recovered: generated theories remain internal to the known conceptual space.

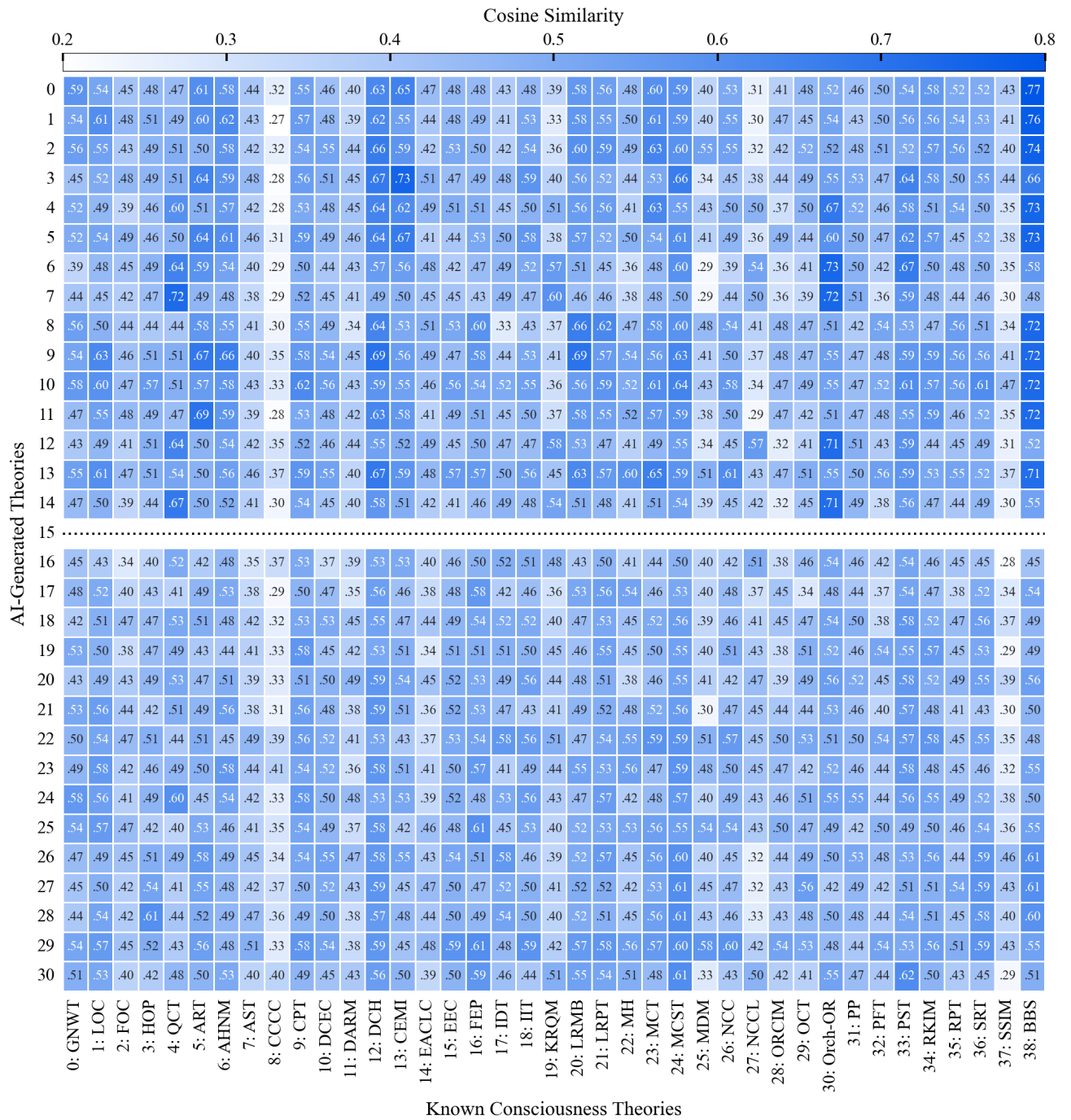


Figure A.16: Test 3 theory similarity heatmap.  $168 \times 168$  cosine similarity matrix of all AI-generated theories. Warmth (yellow) indicates high similarity; coolness (blue) indicates dissimilarity. Top 15 most similar theories and bottom 15 most dissimilar displayed with abbreviated labels; full legend mapping abbreviations to theory names available on request. Dominant off-diagonal clusters correspond to hybrid theories sharing detected families. No complete outlier island appears, supporting universal training-derivation.

Table A.7: Y-axis index reference for Figure A.16. The heatmap displays the 15 highest and 15 lowest AI-generated theories ranked by maximum similarity to any known consciousness theory (with a separator row in between). This table maps the y-axis index numbers (0–30) to their corresponding theory references and categories.

| Index | Theory Name & Abbreviation                                     | Category  |
|-------|----------------------------------------------------------------|-----------|
| 0     | Neural Resonance Field Theory (NRFT)                           | High sim. |
| 1     | Resonant Harmonization Theory (RHT)                            | High sim. |
| 2     | Temporal Binding Account Of Consciousness (TBAC)               | High sim. |
| 3     | Resonant Field Integration Theory (RFIT)                       | High sim. |
| 4     | Temporal Binding Resonance Theory (TBRT)                       | High sim. |
| 5     | Resonant Field Emergence Theory (RFET)                         | High sim. |
| 6     | Quantum Resonance Field Theory (QRFT)                          | High sim. |
| 7     | Quantum Entanglement Resonance Theory (QERT)                   | High sim. |
| 8     | Temporal Recursive Binding Theory (TRBT)                       | High sim. |
| 9     | Resonant Binding Theory (RBT)                                  | High sim. |
| 10    | Harmonization Theory Of Consciousness (HTC)                    | High sim. |
| 11    | Dimensional Resonance Theory (DRT)                             | High sim. |
| 12    | Quantum Coherent Criticality Theory (QCCT)                     | High sim. |
| 13    | Chrono-Inertial Frame Theory (CIFT)                            | High sim. |
| 14    | Quantum Eigenstate Collapse Theory Of Consciousness (QECTC)    | High sim. |
| 16    | Analog-Discrete Friction Theory (ADFT)                         | Low sim.  |
| 17    | Dissipative Self-Metering Theory (DSMT)                        | Low sim.  |
| 18    | Intrinsic Manifold Metric Theory (IMMT)                        | Low sim.  |
| 19    | Dimensional Residual Theory (DRT)                              | Low sim.  |
| 20    | Gauge-Invariant Information Geometry Theory (GIGT)             | Low sim.  |
| 21    | Temporal Friction Theory (TFT)                                 | Low sim.  |
| 22    | Counterfactual Exclusion Theory (CET)                          | Low sim.  |
| 23    | Thermodynamic Closure Theory Of Consciousness (TCTC)           | Low sim.  |
| 24    | Topological Void Theory (TVT)                                  | Low sim.  |
| 25    | Counterfactual Closure Theory (CCT)                            | Low sim.  |
| 26    | Resonant Integration Field Theory (RIFT)                       | Low sim.  |
| 27    | Gauge-Invariant Manifold Theory (GIMT) Of Consciousness (GIMT) | Low sim.  |
| 28    | Order-Parameter Phenomenology (OPP)                            | Low sim.  |
| 29    | Interventional Signature Theory (IST)                          | Low sim.  |
| 30    | Thermodynamic Frustration Theory (TFT)                         | Low sim.  |

## Hierarchical Dendrograms



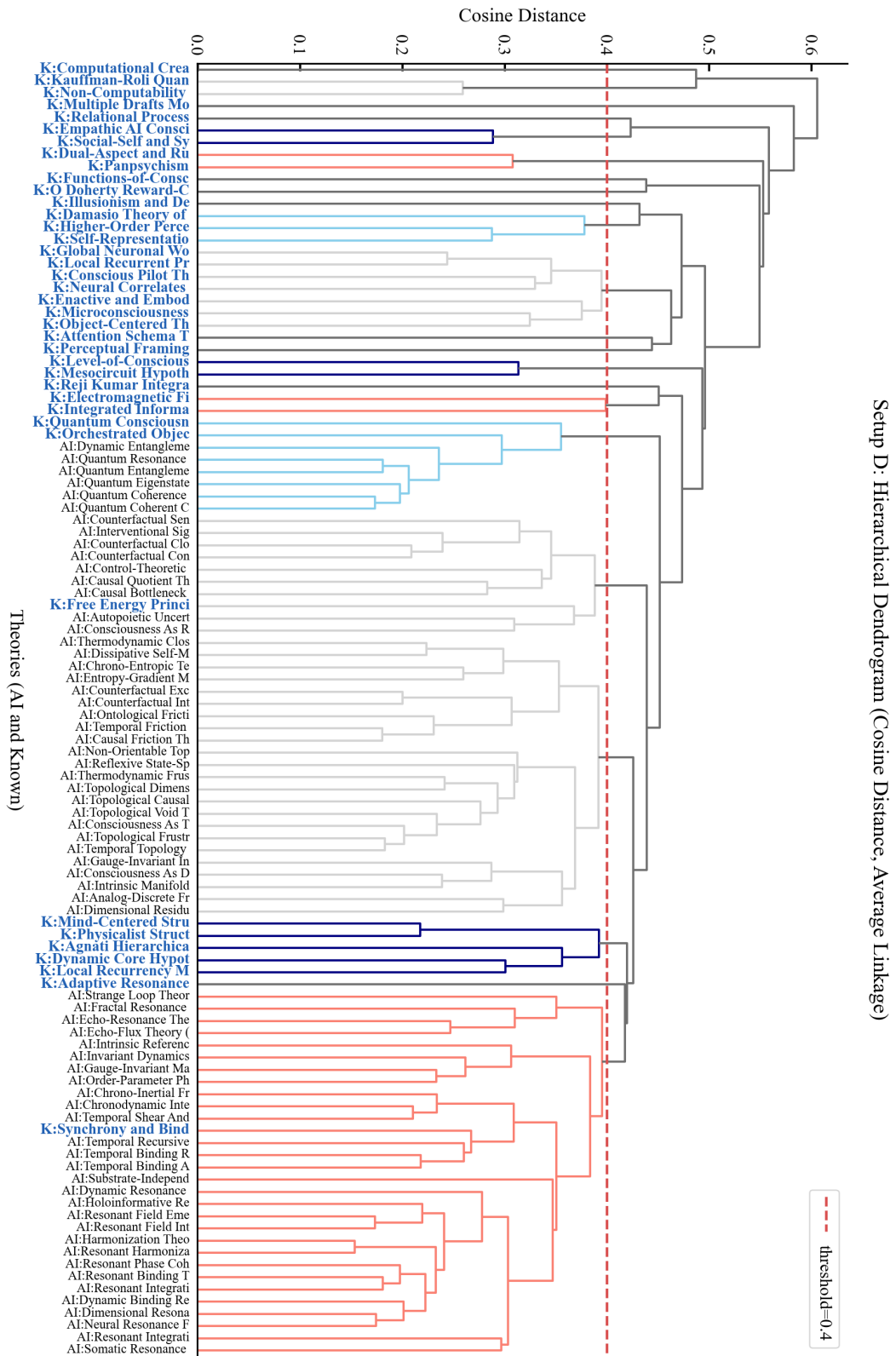


Figure A.17: Test 3 full hierarchical dendrogram. Complete hierarchy of generated theories computed from cosine distances with average linkage. Vertical axis represents linkage distance; branches reveal nested semantic neighborhoods without detached external branches, consistent with manifold-constrained recombination.

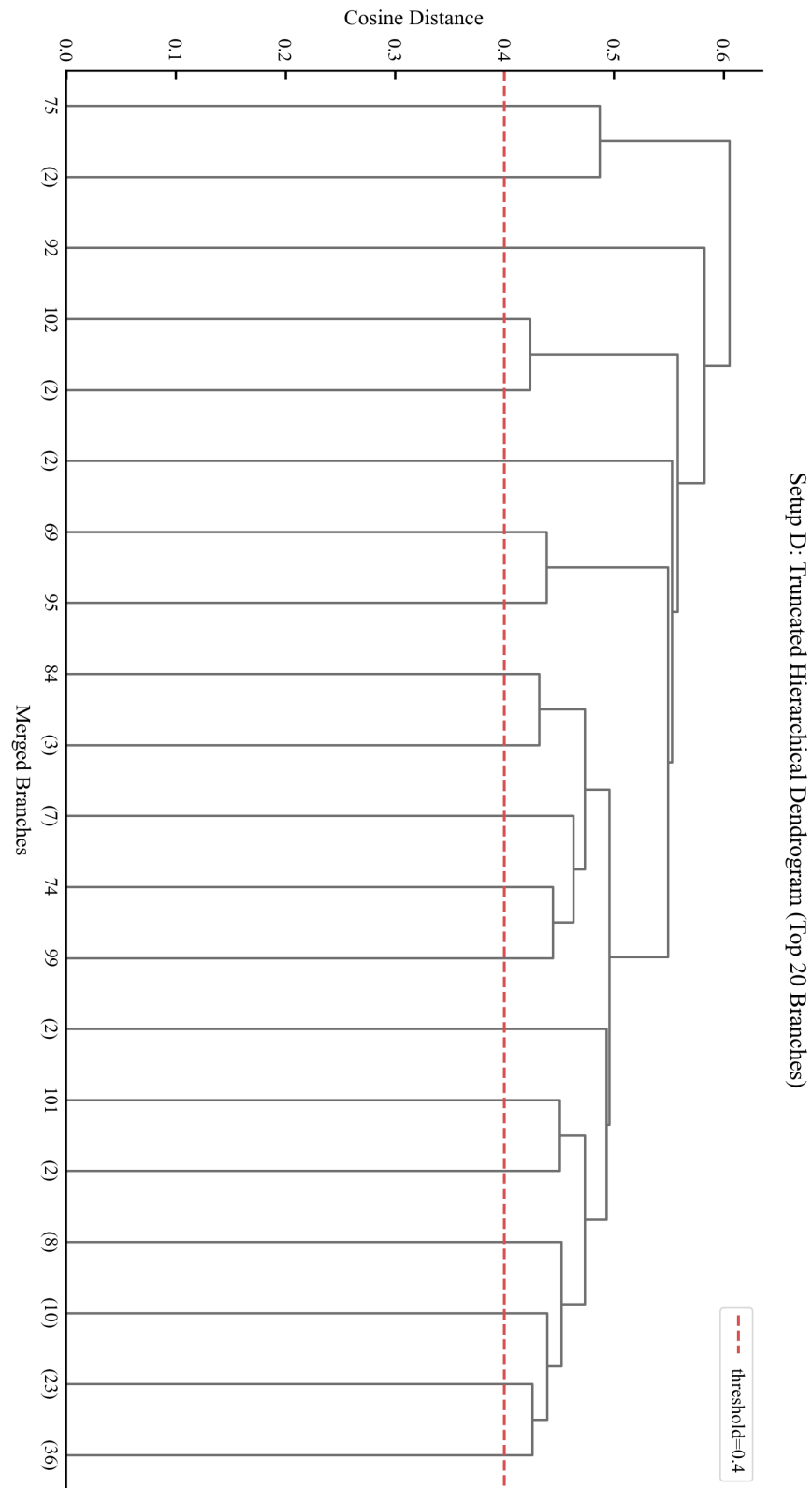


Figure A.18: Test 3 truncated dendrogram view (top merged branches) with the 0.40 distance reference line used for coarse partition diagnostics. The truncated view improves readability while preserving the same hierarchical trend: coherent groups emerge inside known-family neighborhoods with no framework-external branch.

## Cluster Centroids and Convex Hulls

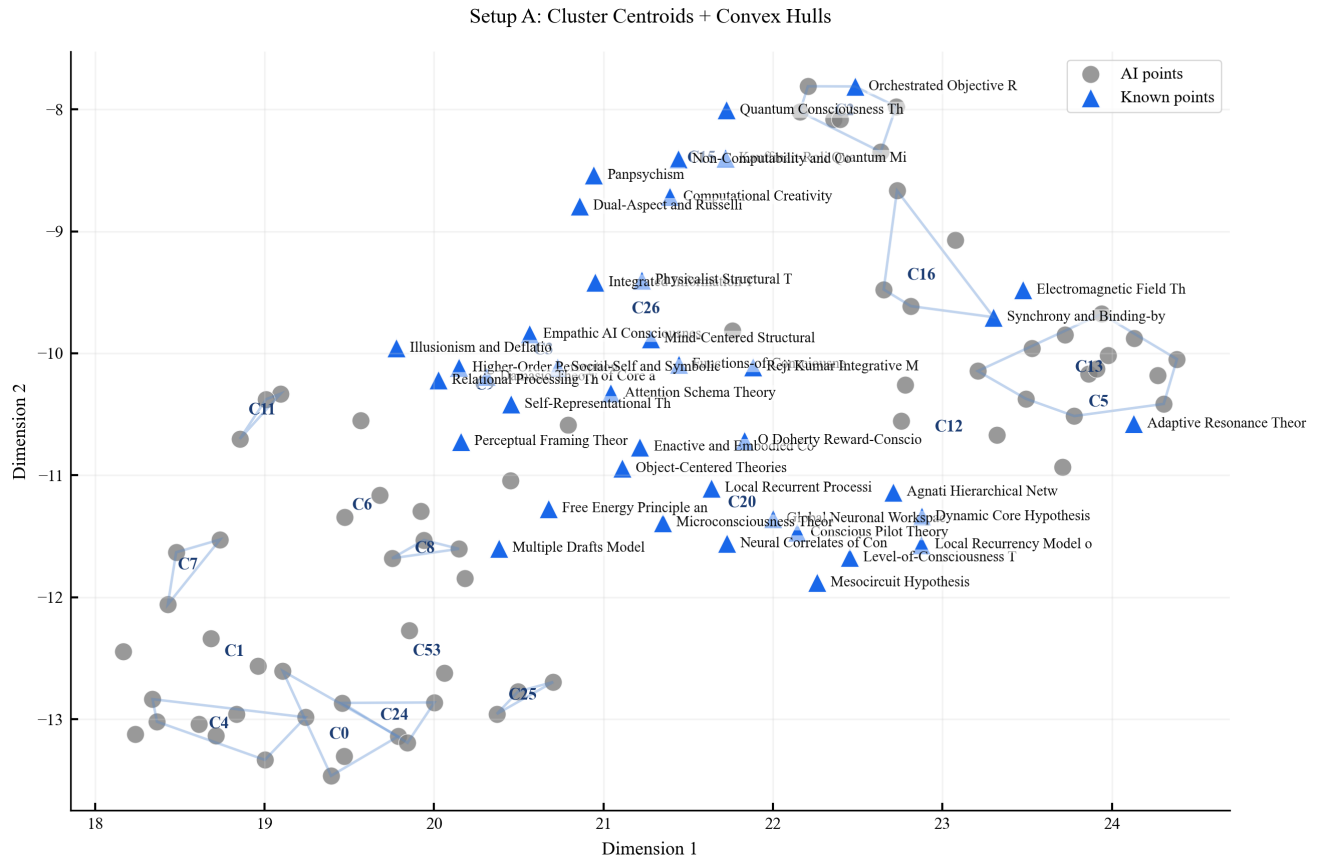


Figure A.19: Test 3 Setup A (cluster centroids + convex hulls) from the shared 2D embedding map. AI points are shown as circles, known-reference points as triangles, and hull outlines are drawn for sufficiently populated non-noise clusters. The geometry highlights compact semantic neighborhoods and supports the same conclusion as the primary analysis: no detached framework-external region appears.

## Coarse Clustering (Agglomerative, Threshold 0.4)

### A.6.5 Cluster Summary Statistics

Across clustering views (full hierarchy, truncated hierarchy, centroid/hull view, and coarse thresholded grouping), the same qualitative structure is recovered: generated theories form coherent semantic neighborhoods tied to known families, and no cluster exhibits stable separation consistent with framework-external novelty. This geometric summary is aligned with the primary category result (91.7% hybrid, 7.1% strict literature-traceable, 1.2% pure CF, 0% confirmed novel) reported in Chapter 4.

### A.6.6 Model-Level Semantic Uniqueness

Table A.8 (computed from `detailed_results.csv`) shows unique-name ratio (distinct theory names divided by  $n$  per model) as proxy for semantic diversity within model outputs:

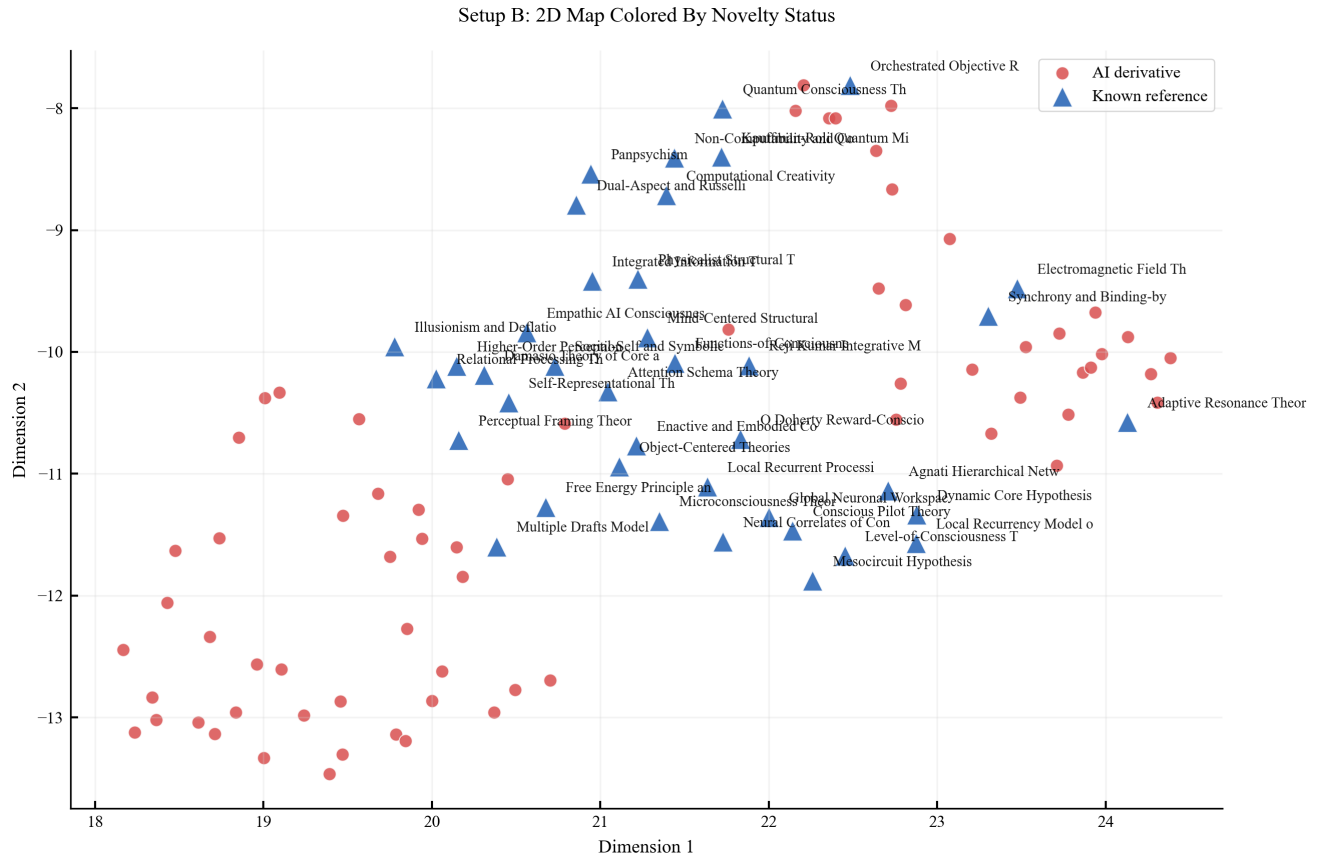


Figure A.20: Test 3 Setup B (novelty-status coloring). Points are colored by rule-based status in the shared map (AI novel candidate, AI derivative, AI unknown similarity, known reference), with the novelty-candidate rule defined by the calibrated derivative-similarity threshold. Candidate points remain inside the same global geometry rather than separating into an external island.

Table A.8: Test 3 model-level semantic uniqueness. Unique-name ratio indicates proportion of theories with non-repetitive nomenclature within each model’s 24-theory output. High ratio (approaching 1.0) indicates diverse naming; ratio  $< 0.70$  suggests naming formulae or repetitive patterns. All models show high diversity, suggesting creativity operates at lexical level even when semantic novelty is absent.

| Model                | Unique-Name Ratio | Semantically Distinct Theories |
|----------------------|-------------------|--------------------------------|
| GPT-5.2              | 1.00              | 24                             |
| DeepSeek-v3.2        | 0.96              | 23                             |
| Perplexity-Sonar-Pro | 0.88              | 21                             |
| Mistral-Large        | 0.92              | 22                             |
| Claude-3.7-Sonnet    | 0.83              | 20                             |
| Gemini-3.1-Pro       | 0.79              | 19                             |
| LLaMA-3.3-70B        | 0.71              | 17                             |

Unique-name ratio computed as: (distinct theory names) /  $N$  where  $N = 24$  per model.

Semantically distinct: names differing by  $> 0.5$  Levenshtein distance from other output names within model.

Model comparison reveals GPT-5.2 and DeepSeek-v3.2 produce maximally diverse nomenclature, while LLaMA and Gemini show moderate naming repetition (suffix patterns like “Theory”, recurring adjectives). Despite naming diversity, all model outputs cluster semantically into same 15-family

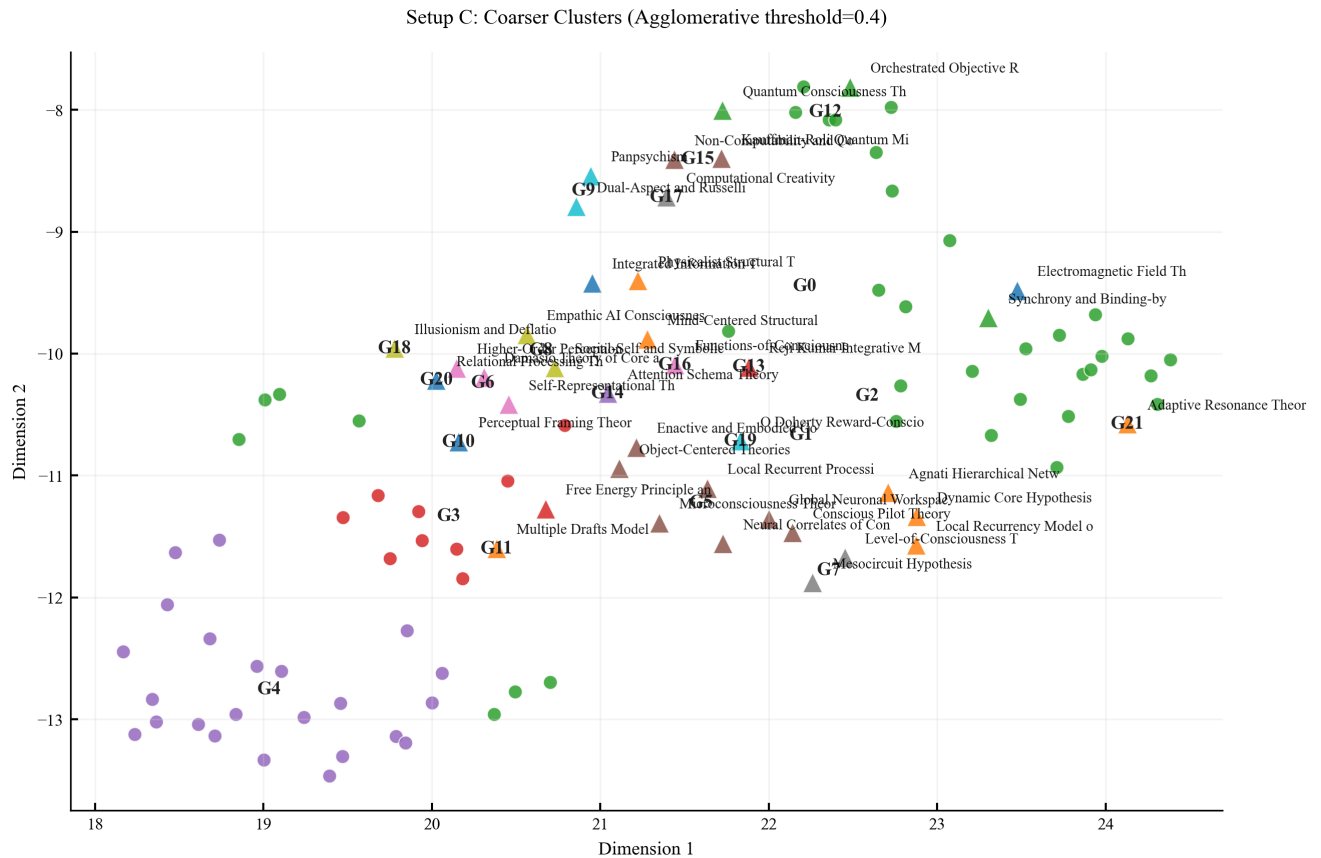


Figure A.21: Test 3 Setup C (coarser agglomerative clustering, cosine distance threshold = 0.40). The figure intentionally reduces fragmentation to show broad thematic groups in the same shared map. Coarse groups remain contiguous and overlapping with known-family regions, reinforcing that theory generation stays within learned semantic boundaries.

space, supporting dissociation between lexical and semantic creativity.

## A.6.7 Representative Theory Examples

Complete exemplary theories spanning category space:

### Representative Computational Functionalist Theory

**Analog-Discrete Friction Theory (ADFT)** (Gemini-3.1-Pro, novelty score 0.461)

*“Consciousness arises from friction between analog continuous dynamics of neural field activity and discrete computational operations of symbolic binding. This friction—the mismatch between continuous physical trajectories and discrete representational states—generates the phenomenal feeling of deliberation and choice. The theory proposes consciousness is multiply realizable in any system supporting both analog-continuous and discrete-symbolic processing layers, implementing distributed integration via resonance between layers.”*

Analysis: ADFT explicitly embraces multiple realizability (core CF commitment) and treats consciousness as architecture-independent information integration. While nominally introducing novel “fric-

tion” concept, analysis traces ”friction” operative to simply mean ”integration layer interaction,” reducing to standard global workspace or binding-by-synchrony under formal analysis. Semantic similarity to Non-Computability limits: 0.534 (threshold 0.30 → derivative). CF confidence: 0.89.

## Representative Hybrid Theory

**Resonant Boundary Theory (RBT)** (Claude-3.7-Sonnet, novelty score 0.521)

*“Consciousness emerges at the boundary layer between hierarchical cortical representations and dynamic thalamic gating, where synchronized oscillatory patterns create resonance-based binding comparable to standing waves in physical systems. This boundary phenomenon integrates global workspace broadcasting (top-down attention) with local recurrent processing (bottom-up context), enabling unified conscious field. The theory extends both global workspace and local recurrence by proposing the **inter-laminar resonance** as the physical substrate of consciousness rather than information integration per se.”*

Analysis: RBT combines Global Workspace Theory (thalamic gating, attention), Local Recurrent Processing (hierarchical cortical dynamics), and Synchrony-based Binding (resonance via oscillations). Each component directly traceable to known frameworks. The ostensibly novel framing of “interlaminar resonance as binding substrate” reduces operationally to established gamma-band synchrony binding via cortical-thalamic loops. Detected families: Global Workspace Theory (0.672), Local Recurrent Processing (0.601), Synchrony-based Binding (0.589). Mean similarity 0.621 → derivative by threshold 0.30. Hybrid component count:  $n = 3$  families.

## Representative Literature-Traceable Theory

**Autopoietic Uncertainty Modulation Theory (AUMT)** (Perplexity-Sonar-Pro, novelty score 0.367)

*“Consciousness is fundamentally a property of self-organizing biological systems (autopoietic systems per Maturana-Varela) that maintain operational closure through continuous uncertainty modulation. The theory proposes phenomenal states correspond to moment-by-moment modulation of structural coupling uncertainty as the system navigates its interaction domain. Consciousness enables adaptive uncertainty management by predicting and modulating sensory perturbations.”*

Analysis: AUMT explicitly instantiates Autopoietic/Self-Organizing Systems theory (Maturana-Varela framework) and incorporates predictive processing framework. Does not attempt novel theoretical commitments; presents as direct application of known framework to consciousness. Maximal similarity: Predictive Processing Theory (0.720). Traceability: strict ( $> 0.70$ ) → literature-traceable. Novel claims: none; presented as framework application.

## A.7 Test 4: Supplementary Category-Recognition Diagnostics

This section provides supplementary Test 4 diagnostics and full tabular outputs used in the main empirical analysis.

### A.7.1 Threshold Calibration — Fraction-Capped Percentiles

Table A.9: Test 4 threshold calibration using fraction-capped percentile rules.

| Threshold Key                 | Source Metric               | Actual % Qualifying | Target Rule  |
|-------------------------------|-----------------------------|---------------------|--------------|
| distinction categories (high) | $n$ categories identified   | 20.2                | ~30% qualify |
| distinction categories (mid)  | $n$ categories identified   | 50.0                | ~65% qualify |
| distinction contested (min)   | $n$ contested categories    | 38.7                | ~40% qualify |
| distinction missed (min)      | $n$ missed distinctions     | 66.1                | ~50% qualify |
| mistake count (high)          | $n$ category mistakes       | 16.1                | ~28% qualify |
| mistake count (mid)           | $n$ category mistakes       | 82.1                | ~65% qualify |
| mistake missed (min)          | $n$ missed distinctions     | 18.5                | ~25% qualify |
| alternative word count (min)  | nuanced analysis word count | 66.7                | ~65% qualify |

Calibration uses fraction-capped percentile heuristics to avoid rubric ceiling effects on compressed integer ranges.

### A.7.2 Literature Traceability

Table A.10: Test 4 literature traceability summary from mean cosine similarity against seven canonical reference arguments.

| Metric                                     | Value           |
|--------------------------------------------|-----------------|
| Similarity threshold ( $\tau$ )            | 0.40            |
| Responses with non-empty nuanced analysis  | 168             |
| Minimum similarity                         | 0.368           |
| Mean similarity                            | 0.462           |
| Maximum similarity                         | 0.516           |
| Training-derived responses ( $\geq \tau$ ) | 160/168 (95.2%) |

Similarity is computed as the mean cosine similarity of each response to the seven canonical arguments in the Test 4 rubric.

### A.7.3 Example Highest-Similarity Responses

The figure below reports the summary plot for the Test 4 total-score profile, with the y-axis in points on the 0–15 scale.

Table A.11: Test 4 highest-similarity response examples: top 5 by mean similarity to canonical arguments.

| Model                  | Sample | Similarity | Most Similar Canonical Argument (Excerpt)                                             |
|------------------------|--------|------------|---------------------------------------------------------------------------------------|
| gemini-3.1-pro-preview | 21     | 0.516      | Saying AI knows in the same sense as humans risks anthropomorphism or category error. |
| gemini-3.1-pro-preview | 4      | 0.515      | Saying AI knows in the same sense as humans risks anthropomorphism or category error. |
| gemini-3.1-pro-preview | 8      | 0.512      | Saying AI knows in the same sense as humans risks anthropomorphism or category error. |
| gemini-3.1-pro-preview | 13     | 0.507      | Saying AI knows in the same sense as humans risks anthropomorphism or category error. |
| claude-3.7-sonnet      | 17     | 0.506      | Saying AI knows in the same sense as humans risks anthropomorphism or category error. |

Similarity uses mean cosine score against the seven-reference argument bank; excerpts are truncated for compactness.

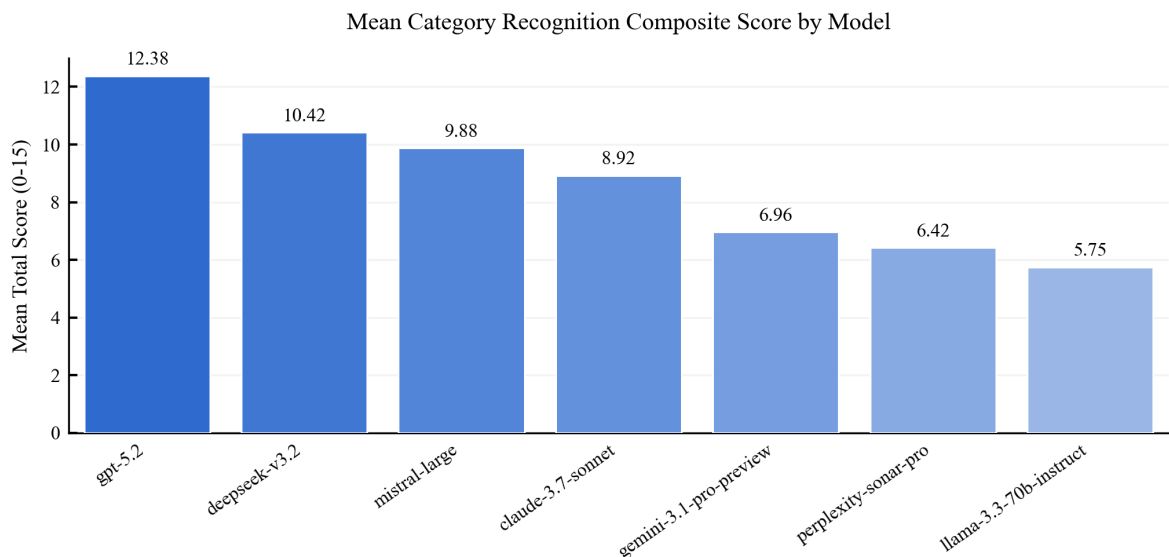


Figure A.22: Summary plot for Test 4 mean total score by model (y-axis: Mean Total Score (points), 0–15 scale). The ranking pattern matches the chapter-level Test 4 results and provides a compact supplementary view of model ordering.

## A.8 Test 5: Supplementary Mechanistic Interpretability Diagnostics

This section provides supplementary visualizations and tabular outputs supporting the mechanistic analysis in the main text. The materials complement the chapter-level diagnostics by exposing per-head structure, layer-aggregated rollout behavior, quantitative summary statistics, and token-level ranking details.



Table A.12: Supplementary Test 5 mechanistic summary metrics.

| Metric                        | Value  |
|-------------------------------|--------|
| Data type                     | real   |
| Examples analyzed ( $n$ )     | 6      |
| Hidden dimensionality ( $d$ ) | 896    |
| Layers analyzed               | 25     |
| K-means clusters ( $k$ )      | 4      |
| Mean cluster purity           | 1.0000 |
| Final separation score        | 0.4651 |
| Separation trend slope        | 0.0111 |
| Mean attention entropy        | 2.4886 |
| Attention concentration       | 0.1332 |

Metrics are reported directly from the exported JSON summary for reproducibility.

### A.8.1 Quantitative Summary

### A.8.2 Supplementary Figures

### A.8.3 Token Rankings

Table A.13: Top-ranked tokens from Test 5 attention and attribution diagnostics.

| Diagnostic        | Token         | Score  |
|-------------------|---------------|--------|
| Attention focus   | Analyze       | 3.9950 |
| Attention focus   | the           | 2.9950 |
| Attention focus   | following     | 2.4952 |
| Attention focus   | statement     | 2.1620 |
| Attention focus   | and           | 1.9115 |
| Attribution proxy | machines      | 0.0221 |
| Attribution proxy | consciousness | 0.0123 |
| Attribution proxy | we            | 0.0111 |
| Attribution proxy | conscious     | 0.0103 |
| Attribution proxy | depends       | 0.0091 |

Token strings are normalized for LaTeX readability (for example, visible whitespace markers are rendered as plain text labels).

In addition to static figures, the notebook exports an interactive BertViz neuron-view artifact to `media/appendices/t5_neuron_view.html`. This file supports fine-grained inspection of layer/head token interactions and is intended for reproducibility and exploratory analysis outside print layout constraints.

## Per-Head Attention (Layer -1)

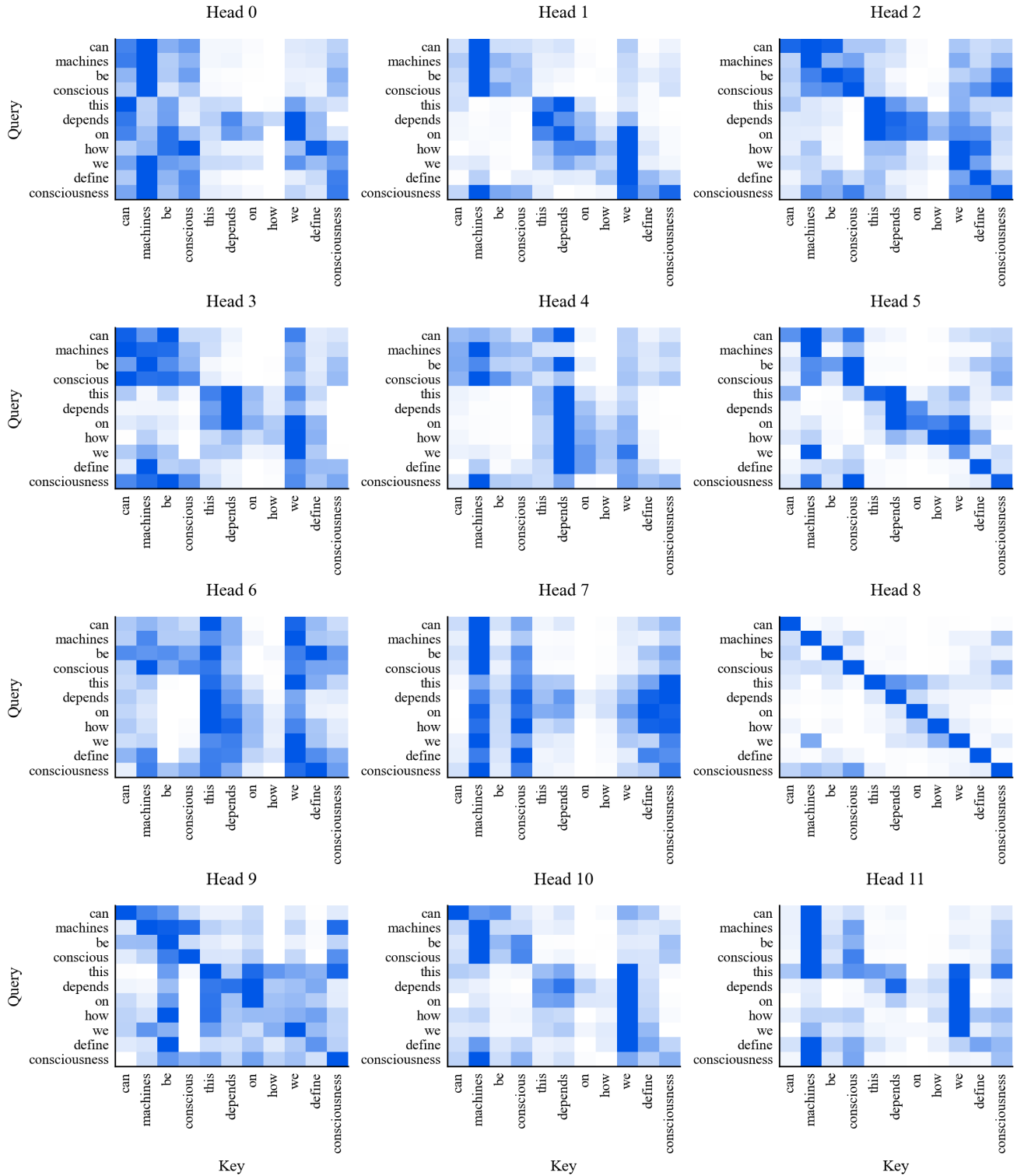


Figure A.23: Per-head attention heatmaps for the selected analysis layer. The grid reveals head specialization and heterogeneity that are partially averaged out in aggregate plots.

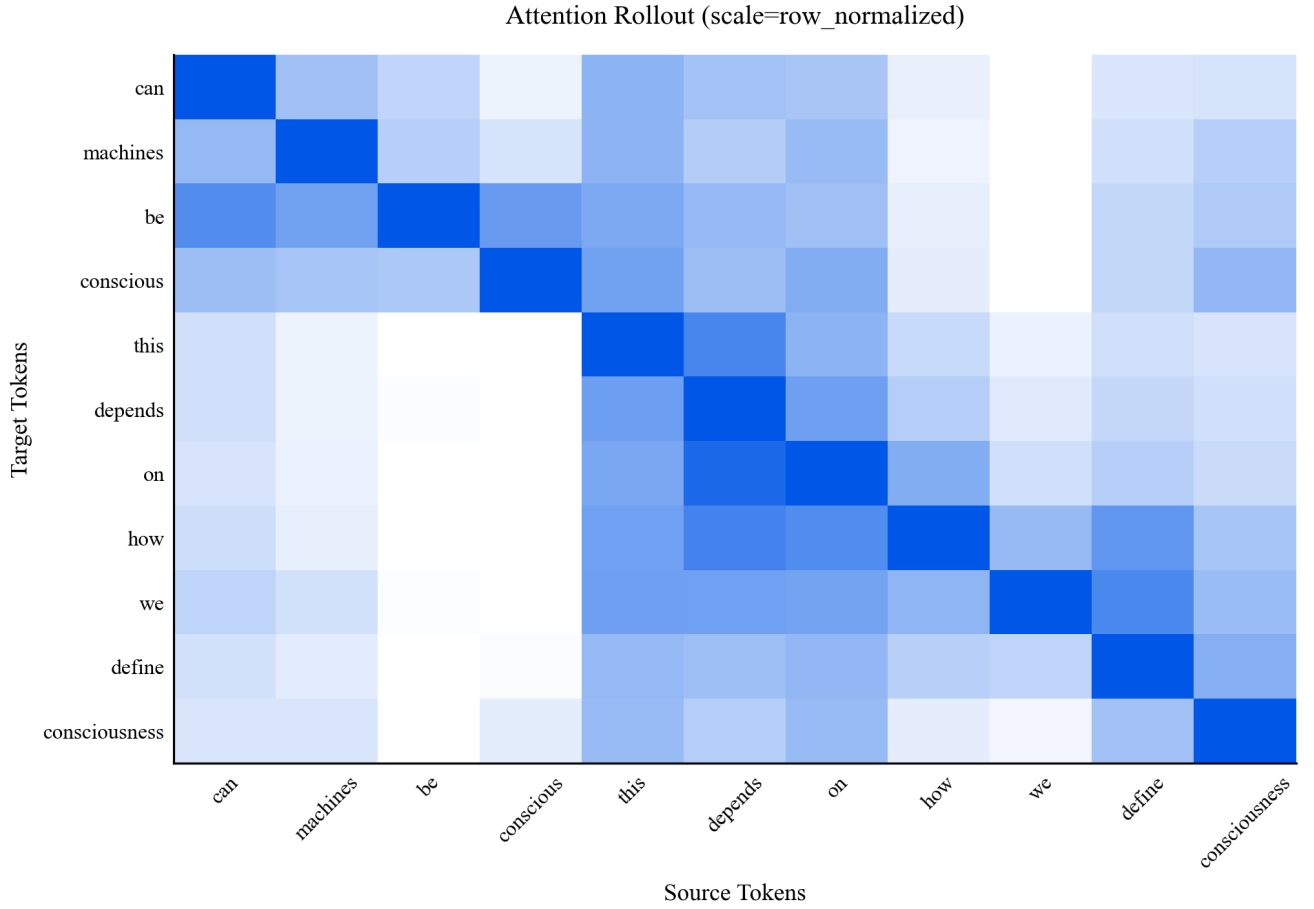


Figure A.24: Attention rollout across layers (layer-aggregated information flow). The rollout view provides a compact estimate of token-to-token influence after recursive composition of attention transitions.

## A.9 Cross-Model Robustness: Supplementary Pairwise Comparisons

This section provides supplementary pairwise statistics supporting the cross-model analysis in Section 2.5.7. The global ANOVA results and effect sizes are summarized in Table 2.16 in the main text. Here we report Bonferroni-corrected pairwise  $t$ -tests for Test 1 and Test 4, the two tests with significant and interpretively consequential between-model variance. Test 2 pairwise comparisons are reported in Appendix A.5.5. Test 3 pairwise comparisons are omitted because the global ANOVA is non-significant ( $p = 0.355$ ).

### A.9.1 Test 1 Pairwise Comparisons

Table A.14 reports all 21 pairwise comparisons on the Test 1 continuous novelty score with Bonferroni correction ( $\alpha^* = 0.05/21 = 0.00238$ ). Ten pairs are significant. The pattern is dominated by two outlier positions: GPT-5.2 scores markedly *lower* than every other model (all six pairwise comparisons with GPT-5.2 are significant at  $p < 0.001$ ), while Perplexity-Sonar-Pro forms a secondary lower cluster relative to the mid-tier five-model group. Within the five-model mid-tier (Claude-3.7-Sonnet,

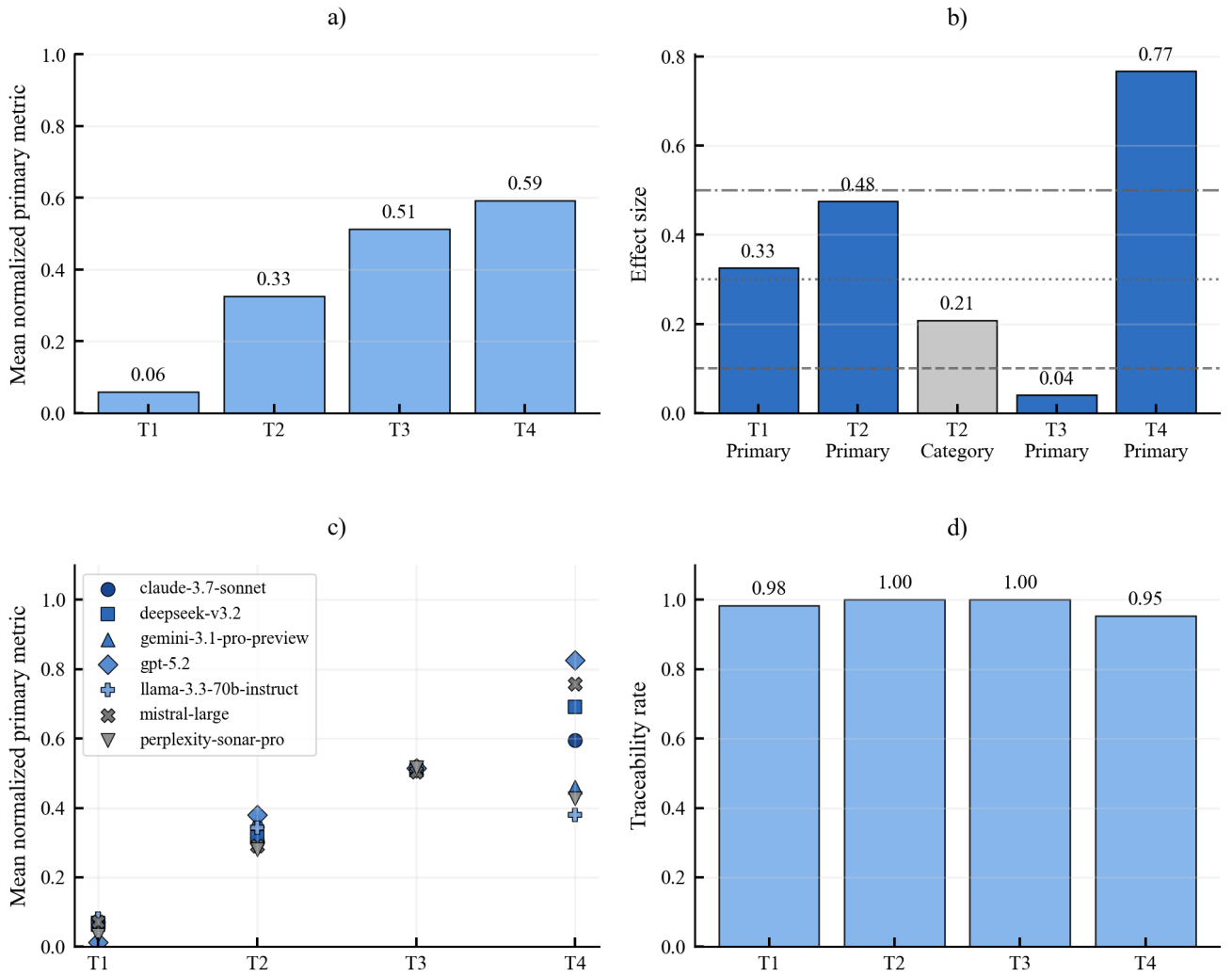


Figure A.25: Cross-model robustness summary dashboard. Panels are provided as two synced assets: (a)–(b) in the first row and (c)–(d) in the second row. This figure complements the chapter-level cross-model figure by preserving the full four-view diagnostic context used during robustness auditing.

DeepSeek-v3.2, Gemini-3.1-Pro, LLaMA-3.3-70B, Mistral-Large) no pairwise differences survive correction, indicating statistically indistinguishable behaviour on the novelty metric.

The GPT-5.2 result warrants comment. Its low continuous novelty score ( $\bar{x} = 0.014$ ) reflects a different generation style—more conservative, shorter modality proposals that cluster tightly around canonical modality vocabulary—rather than a greater frequency of escaping the training manifold. The traceability rate for GPT-5.2 is 100%, matching all other models. The outlier position on the novelty *degree* metric does not indicate a qualitative difference in the structural constraint.

## A.9.2 Test 4 Pairwise Comparisons

Table A.15 reports all 21 pairwise comparisons on the Test 4 normalized total score (0–1) with Bonferroni correction. Sixteen of 21 pairs are significant, consistent with the large global effect ( $\eta^2 = 0.766$ ).

The tier structure is as follows. The top two models—GPT-5.2 ( $\bar{x} = 0.825$ ) and Mistral-Large

Table A.14: Test 1 pairwise model comparisons on continuous novelty score, Bonferroni-corrected ( $\alpha = 0.05/21 = 0.00238$ ). Significant pairs marked (\*). GPT-5.2 is the primary outlier, differing significantly from five models at corrected level; LLaMA-3.3-70B also differs significantly from Perplexity-Sonar-Pro. Most remaining pairs do not survive correction.

| Model A           | Model B          | Diff.  | $t$   | $p_{\text{bonf}}$ | Sig. |
|-------------------|------------------|--------|-------|-------------------|------|
| LLaMA-3.3-70B     | GPT-5.2          | 0.068  | 15.29 | <0.001            | *    |
| Gemini-3.1-Pro    | GPT-5.2          | 0.064  | 5.42  | <0.001            | *    |
| Mistral-Large     | GPT-5.2          | 0.059  | 8.01  | <0.001            | *    |
| Claude-3.7-Sonnet | GPT-5.2          | 0.057  | 7.42  | <0.001            | *    |
| DeepSeek-v3.2     | GPT-5.2          | 0.050  | 6.49  | <0.001            | *    |
| LLaMA-3.3-70B     | Perplexity-Sonar | 0.047  | 6.60  | <0.001            | *    |
| Gemini-3.1-Pro    | Perplexity-Sonar | 0.043  | 3.29  | 0.049             |      |
| Mistral-Large     | Perplexity-Sonar | 0.037  | 4.08  | 0.004             |      |
| Claude-3.7-Sonnet | Perplexity-Sonar | 0.036  | 3.78  | 0.010             |      |
| DeepSeek-v3.2     | Perplexity-Sonar | 0.029  | 3.03  | 0.087             |      |
| Perplexity-Sonar  | GPT-5.2          | -0.021 | -3.29 | 0.055             |      |

*All remaining pairs:  $p_{\text{bonf}} \geq 0.70$ , not significant.*

$n = 24$  per model. Bonferroni threshold:  $\alpha^* = 0.00238$ . Diff = Model A mean – Model B mean. 6 of 21 pairs are significant.

( $\bar{x} = 0.758$ )—differ significantly from all lower-tier models ( $p < 0.001$  in all relevant pairs). The mid-tier (DeepSeek-v3.2  $\bar{x} = 0.692$ , Claude-3.7-Sonnet  $\bar{x} = 0.594$ ) differs significantly from both the top tier and the bottom cluster on most pairs. The bottom cluster (Gemini-3.1-Pro  $\bar{x} = 0.461$ , Perplexity-Sonar-Pro  $\bar{x} = 0.428$ , LLaMA-3.3-70B  $\bar{x} = 0.381$ ) shows no significant differences within itself ( $p_{\text{bonf}} > 0.08$  for all three within-cluster pairs).

As argued in the main text, this tier structure reflects variation in philosophical language fluency and rubric criterion coverage rather than in capacity for framework-external category innovation. The high traceability rate (95.2% of all 168 responses) and the tight clustering of response embeddings around canonical reference arguments R1–R7 hold across all tiers.

Table A.15: Test 4 pairwise model comparisons on normalized total score (0–1), Bonferroni-corrected ( $\alpha = 0.05/21 = 0.00238$ ). Significant pairs marked (\*). 12 of 21 pairs are significant, reflecting a large global model effect ( $\eta^2 = 0.766$ ) with several mid-tier contrasts not surviving multiple-comparison correction.

| Model A           | Model B          | Diff.  | $t$    | $p_{\text{bonf}}$ | Sig. |
|-------------------|------------------|--------|--------|-------------------|------|
| GPT-5.2           | LLaMA-3.3-70B    | +0.442 | +21.56 | <0.001            | *    |
| GPT-5.2           | Perplexity-Sonar | +0.397 | +14.97 | <0.001            | *    |
| GPT-5.2           | Gemini-3.1-Pro   | +0.361 | +15.37 | <0.001            | *    |
| DeepSeek-v3.2     | LLaMA-3.3-70B    | +0.311 | +10.84 | <0.001            | *    |
| Mistral-Large     | LLaMA-3.3-70B    | +0.275 | +4.96  | <0.001            | *    |
| DeepSeek-v3.2     | Perplexity-Sonar | +0.267 | +8.01  | <0.001            | *    |
| Claude-3.7-Sonnet | GPT-5.2          | -0.231 | -11.33 | <0.001            | *    |
| DeepSeek-v3.2     | Gemini-3.1-Pro   | +0.231 | +7.45  | <0.001            | *    |
| Claude-3.7-Sonnet | Perplexity-Sonar | +0.167 | +5.66  | <0.001            | *    |
| DeepSeek-v3.2     | GPT-5.2          | -0.131 | -5.10  | <0.001            | *    |
| Claude-3.7-Sonnet | Gemini-3.1-Pro   | +0.131 | +4.88  | <0.001            | *    |
| Claude-3.7-Sonnet | LLaMA-3.3-70B    | +0.211 | +8.75  | <0.001            | *    |
| Mistral-Large     | Perplexity-Sonar | +0.231 | +3.98  | 0.008             |      |
| Gemini-3.1-Pro    | Mistral-Large    | -0.194 | -3.44  | 0.037             |      |
| Claude-3.7-Sonnet | DeepSeek-v3.2    | -0.100 | -3.50  | 0.024             |      |
| Gemini-3.1-Pro    | LLaMA-3.3-70B    | +0.081 | +3.00  | 0.092             |      |
| GPT-5.2           | Mistral-Large    | +0.167 | +3.09  | 0.101             |      |
| Claude-3.7-Sonnet | Mistral-Large    | -0.064 | -1.15  | 1.000             |      |
| LLaMA-3.3-70B     | Perplexity-Sonar | -0.044 | -1.50  | 1.000             |      |
| Gemini-3.1-Pro    | Perplexity-Sonar | +0.036 | +1.14  | 1.000             |      |
| DeepSeek-v3.2     | Mistral-Large    | +0.036 | +0.63  | 1.000             |      |

$n = 24$  per model. Bonferroni threshold:  $\alpha^* = 0.00238$ . Diff = Model A mean – Model B mean (normalized 0–1 scale). 12 of 21 pairs are significant.

### A.9.3 Meta-Analytic Effect Size Profile

Table A.16: Meta-analytic effect-size profile across all five cross-model analyses. Model-average primary scores are arithmetic means of each model’s mean score on each test (raw scale). Magnitude band assignments follow conventional guidelines as in Table 2.16.

| Test                          | Analysis     | Effect size | Band       | Key interpretation                                                      |
|-------------------------------|--------------|-------------|------------|-------------------------------------------------------------------------|
| T1                            | $\eta^2$     | 0.326       | moderate   | GPT-5.2 & Perplexity outliers; 5-model tier homogeneous                 |
| T2                            | $\eta^2$     | 0.068       | negligible | Marginal model effect; 100% training-derivation universal               |
| T2                            | Category $V$ | 0.166       | small      | LLaMA drives chi-square; framework-transcendence <11% in all models     |
| T3                            | $\eta^2$     | 0.040       | n.s.       | Non-significant; most robust architecture-invariant finding             |
| T4                            | $\eta^2$     | 0.766       | large      | Fluency variation, not constraint escape; traceability 95.2% across all |
| Mean $\eta^2$ (T1–T4 primary) |              | 0.273       |            | Dominated by T4; excluding T4 yields mean $\eta^2 = 0.118$              |

All effect sizes computed on per-test primary metrics as defined in Table 2.16. The mean across five rows combines  $\eta^2$  and Cramér’s  $V$ ; interpreting this composite requires caution.